Classical Field Theory — Exercise Sheet 1

Local actions

Solutions

Use Lecture 1's sign $E_a = \partial L/\partial\phi^a - \partial_\mu p_a^\mu$. All fields and prescribed sources in this sheet are smooth. The first three problems use only differentiation, integration by parts and the fundamental lemma. The final problem also uses finite-dimensional linear ODEs.

Exercise 1.1 (A complex source and two real variations). On $\mathbb{R}^{1,1}$ let $\Phi = \phi_1 + \mathrm{i}\phi_2$ be a complex field, $J = J_1 + \mathrm{i}J_2$ a prescribed complex source, and $m \geq 0$. The density is

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\Phi_x|^2 - m^2|\Phi|^2\big) + \mathrm{Re}(\overline{J}\Phi).$$

For the action, take any compact region $D \subset \mathbb{R}^{1,1}$ and variations H whose support lies in the interior of D.

(a) Compute both real momentum currents and Euler expressions. Express the stationarity condition as one complex equation and derive the first variation for a compactly supported complex variation H.

(b) Find the source for which $\Phi(t,x) = (t^2+1) + \mathrm{i}(t^3 - x^2)$ is stationary.

(c) If only real-valued variations H are tested, must the complex equation hold? Give a counterexample with $m = 0$ and $J = 0$.

Solution. For the first part, the density in real components is

$$L = \frac{1}{2}\sum_{a=1}^{2}\big((\partial_t\phi_a)^2 - (\partial_x\phi_a)^2 - m^2\phi_a^2\big) + J_1\phi_1 + J_2\phi_2.$$

The definition of the momentum current and then the Euler expression give

$$p_a^0 = \partial_t\phi_a, \qquad p_a^1 = -\partial_x\phi_a, \qquad E_a = J_a - \big(\partial_t^2 - \partial_x^2 + m^2\big)\phi_a.$$

Let $E = E_1 + \mathrm{i}E_2$ and $H = H_1 + \mathrm{i}H_2$. The real first-variation formula is

$$\delta S_D[\Phi]H = \int_D (E_1H_1 + E_2H_2)\,\mathrm{d}t\mathrm{d}x = \mathrm{Re}\int_D \overline{E}H\,\mathrm{d}t\mathrm{d}x.$$

Tests with $H_2 = 0$ imply $E_1 = 0$, and tests with $H_1 = 0$ imply $E_2 = 0$. Together these are exactly the stated complex equation. No independent variation of two conjugate complex variables was assumed.

For the second part, apply the same differential operator to the actual field:

$$(\Box + m^2)(t^2+1) = 2 + m^2(t^2+1), \qquad (\Box + m^2)(t^3 - x^2) = 6t + 2 + m^2(t^3 - x^2).$$

These are the required two source components.

For the final part, the field $\Phi = \mathrm{i}t^2$ has $E = -2\mathrm{i}$. Its first variation against every real compact H vanishes because $\mathrm{Re}(\overline{E}H) = 0$. Its imaginary Euler expression is nonzero, so it is not stationary against all complex variations.

Exercise 1.2 (Natural conditions at the two spatial ends). Let $T, \ell > 0$, $m \geq 0$ and $\beta_0, \beta_\ell \geq 0$. On the rectangle $D = [0,T] \times [0,\ell]$, a real field has action

$$S_D[\phi] = \int_0^T \int_0^\ell \left(\frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - \frac{1}{2}m^2\phi^2 + J\phi \right) \mathrm{d}x\mathrm{d}t \\ - \frac{1}{2}\int_0^T \left(\beta_0 \phi(t,0)^2 + \beta_\ell \phi(t,\ell)^2\right) \mathrm{d}t,$$

where J is prescribed. Variations vanish near $t = 0, T$ and are unrestricted at $x = 0, \ell$.

(a) Integrate coordinate by coordinate and prove that stationarity is equivalent to the bulk equation and two Robin boundary conditions. Include the case $\beta_0 = \beta_\ell = 0$.

(b) For $\ell = 1$, $m = 0$, $J = 2$ and $\beta_0 = \beta_\ell = 1$, find the unique time-independent stationary field.

Solution. The variation of the bulk density is $\phi_t H_t - \phi_x H_x - m^2\phi H + JH$. Time integration has no endpoint term by the hypothesis on H. Spatial integration and the two boundary-action variations therefore give

$$\delta S_D[\phi]H = \int_0^T \int_0^\ell \left(J - \phi_{tt} + \phi_{xx} - m^2\phi\right) H \, \mathrm{d}x\mathrm{d}t \\ + \int_0^T \left(\phi_x(t,0) - \beta_0\phi(t,0)\right) H(t,0) \mathrm{d}t \\ + \int_0^T \left(-\phi_x(t,\ell) - \beta_\ell\phi(t,\ell)\right) H(t,\ell) \mathrm{d}t.$$

This derives the rectangle formula directly, rather than importing the lecture's torus-graph formula on a different region. Interior tests force the bulk coefficient to vanish. Given any compactly supported smooth time function g, the tests $H(t,x) = g(t)(1 - x/\ell)$ and $H(t,x) = g(t)x/\ell$ select the two traces independently. The one-dimensional fundamental lemma forces each boundary coefficient to vanish. Conversely, the bulk equation and both conditions make the displayed variation zero for every allowed H. If both coefficients β are zero, the conditions reduce to Neumann conditions $\phi_x = 0$ at both ends.

For the second part, write the static field as $u(x)$. The equation $-u'' = 2$ gives

$$u(x) = -x^2 + Ax + B.$$

The left condition gives $A = B$. The right condition gives $-2 + A = -(-1 + A + B)$, so $2A + B = 3$. Thus $A = B = 1$, uniquely. Both endpoint values are one and the derivatives are one and minus one, respectively, as the Robin conditions require.

Exercise 1.3 (Lorentz changes of coordinates). The Lorentz group in two coordinates is the matrix group

$$O(1,1) = \{\Lambda \in GL(2,\mathbb{R}) : \Lambda^{\mathsf{T}} \eta \Lambda = \eta\}, \qquad \eta = \operatorname{diag}(1,-1).$$

For $|v| < 1$, set

$$\gamma_v = (1-v^2)^{-1/2}, \qquad \begin{pmatrix} t' \\ x' \end{pmatrix} = \Lambda_v \begin{pmatrix} t \\ x \end{pmatrix}, \qquad \Lambda_v = \gamma_v \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}.$$

(a) Verify that $\Lambda_v \in O(1,1)$ and $\det \Lambda_v = 1$.

(b) For a scalar field, set $\phi'(t',x') = \phi(t,x)$ at corresponding points. Use the chain rule to prove invariance of $\partial_t^2 - \partial_x^2$ and of $\frac{1}{2}(\phi_t^2 - \phi_x^2) - V(\phi)$, where V is any smooth function. Conclude equality of the scalar actions on corresponding compact regions.

Solution. Matrix multiplication gives

$$\Lambda_v^{\mathsf{T}} \eta \Lambda_v = \gamma_v^2 \begin{pmatrix} 1-v^2 & 0 \\ 0 & -(1-v^2) \end{pmatrix} = \eta, \qquad \det \Lambda_v = \gamma_v^2 (1-v^2) = 1.$$

The inverse is Λ_{-v}, so the chain rule gives

$$\partial_{t'} = \gamma_v(\partial_t + v\partial_x), \qquad \partial_{x'} = \gamma_v(v\partial_t + \partial_x).$$

Mixed derivatives commute for a smooth scalar. Expanding the operator difference gives

$$\partial_{t'}^2 - \partial_{x'}^2 = \gamma_v^2\big((1-v^2)\partial_t^2 - (1-v^2)\partial_x^2\big) = \partial_t^2 - \partial_x^2.$$

The same algebra applied to the first derivatives gives

$$(\phi'_{t'})^2 - (\phi'_{x'})^2 = \phi_t^2 - \phi_x^2.$$

At corresponding points $V(\phi') = V(\phi)$. Thus the densities agree. Since $\mathrm{d}t'\mathrm{d}x' = \mathrm{d}t\mathrm{d}x$ and $D' = \Lambda_v D$, change of variables yields

$$\int_{D'} L[\phi']\,\mathrm{d}t'\mathrm{d}x' = \int_D L[\phi]\,\mathrm{d}t\mathrm{d}x.$$

This statement compares the transformed field and region together.

Exercise 1.4 ⋆. Let $N \geq 3$, $h_{\mathrm{lat}} = 2\pi/N$, and let $q_j(t)$, $j = 0, \dots, N-1$, be real coordinates with periodic indexing. The chain Lagrangian is

$$L_h = h_{\mathrm{lat}} \sum_{j=0}^{N-1} \left[\frac{1}{2}\dot{q}_j^2 - \frac{1}{2h_{\mathrm{lat}}^2}(q_{j+1} - q_j)^2 - \frac{1}{2}m^2 q_j^2 \right], \qquad m \geq 0.$$

(a) Derive its equations by varying every q_j.

(b) Let $T > 0$ and let ϕ be a given smooth periodic solution of $\phi_{tt} - \phi_{xx} + m^2\phi = 0$ on $[0,T] \times \mathbb{T}$. Assume $M_4 \geq 0$ and $|\partial_x^4 \phi| \leq M_4$ there. Prove that sampling this field in the chain equation produces a residual bounded by $h_{\mathrm{lat}}^2 M_4/12$ at every node.

(c) Give the chain the exact sampled initial position and velocity. Set $e_j = q_j - \phi(t, jh_{\mathrm{lat}})$, $D_h e_j = (e_{j+1} - e_j)/h_{\mathrm{lat}}$ and $\|w\|_h^2 = h_{\mathrm{lat}} \sum_j w_j^2$. With $A_h = \sqrt{2\pi} M_4 h_{\mathrm{lat}}^2/12$, prove, for $0 \leq t \leq T$,

$$\left(\|\dot{e}(t)\|_h^2 + \|D_h e(t)\|_h^2 + m^2 \|e(t)\|_h^2 \right)^{1/2} \leq A_h t, \qquad \|e(t)\|_h \leq \frac{1}{2} A_h t^2.$$

The finite linear chain has a global solution by the linear ODE theorem. No existence theorem for the continuum equation is assumed beyond the given ϕ.

Solution. For the first part, variation of the two springs meeting at node j gives

$$\frac{\partial L_h}{\partial q_j} = \frac{q_{j+1} - 2q_j + q_{j-1}}{h_{\mathrm{lat}}} - h_{\mathrm{lat}} m^2 q_j, \qquad \frac{\partial L_h}{\partial \dot{q}_j} = h_{\mathrm{lat}} \dot{q}_j.$$

Subtracting the time derivative of the second expression from the first and dividing by $-h_{\mathrm{lat}}$ gives the chain equation.

For consistency, define $\Delta_h w_j = (w_{j+1} - 2w_j + w_{j-1})/h_{\mathrm{lat}}^2$. Taylor's theorem at a fixed time gives

$$\phi(x \pm h_{\mathrm{lat}}) = \phi(x) \pm h_{\mathrm{lat}} \phi_x(x) + \frac{h_{\mathrm{lat}}^2}{2}\phi_{xx}(x) \pm \frac{h_{\mathrm{lat}}^3}{6}\phi_{xxx}(x) + r_{\pm}, \qquad |r_{\pm}| \leq \frac{h_{\mathrm{lat}}^4 M_4}{24}.$$

The odd terms cancel on addition. Therefore

$$|\Delta_h \phi(t,x) - \phi_{xx}(t,x)| \leq \frac{h_{\mathrm{lat}}^2 M_4}{12}.$$

The sampled residual is $R_j = \phi_{tt}(t, jh_{\mathrm{lat}}) - \Delta_h \phi(t, jh_{\mathrm{lat}}) + m^2 \phi(t, jh_{\mathrm{lat}}) = \phi_{xx}(t, jh_{\mathrm{lat}}) - \Delta_h \phi(t, jh_{\mathrm{lat}})$. Since $Nh_{\mathrm{lat}} = 2\pi$, the bound gives $\|R(t)\|_h \leq A_h$.

For the error estimate, subtraction gives $\ddot{e} - \Delta_h e + m^2 e = -R$, with $e(0) = \dot{e}(0) = 0$. Periodic reindexing proves the summation identity

$$h_{\mathrm{lat}} \sum_j v_j \Delta_h e_j = -h_{\mathrm{lat}} \sum_j (D_h v_j)(D_h e_j).$$

Define the discrete error energy by

$$\mathcal{E}_h(t) = \frac{1}{2}\left(\|\dot{e}\|_h^2 + \|D_h e\|_h^2 + m^2 \|e\|_h^2 \right).$$

Its derivative, using the identity and the error equation, is

$$\mathcal{E}_h' = -h_{\mathrm{lat}} \sum_j \dot{e}_j R_j \leq \sqrt{2\mathcal{E}_h}\, \|R\|_h.$$

For $\epsilon > 0$, differentiation gives $\frac{\mathrm{d}}{\mathrm{d}t}\sqrt{2\mathcal{E}_h + \epsilon^2} \le \|R\|_h$. Integrate from zero and let $\epsilon \downarrow 0$ to obtain $\sqrt{2\mathcal{E}_h(t)} \le A_h t$. Finally, integrate $\dot{e}$ in the finite-dimensional norm:

$$\|e(t)\|_h \le \int_0^t \|\dot{e}(s)\|_h \mathrm{d}s \le \int_0^t A_h s \mathrm{d}s = \frac{1}{2} A_h t^2.$$

The last step supplies the full position bound even when the mass is zero. The estimate compares a given smooth continuum solution to its sampled chain and proves the stated finite-time convergence.

Classical Field Theory — Exercise Sheet 2

Noether currents and their limits

Solutions

Use Lecture 2's signs $j^\mu = p_a^\mu Q^a - B^\mu$ and $\partial_\mu j^\mu = -E_a Q^a$. All fields and coefficients below are smooth. The unindexed $Q(t)$ denotes an integrated charge, and Q^a denotes a characteristic. The last problem is a different field theory from Klein–Gordon.

Exercise 2.1 (An improvement can change a boundary charge). Let j^μ be a current in $1+d$ dimensions and let $K^{\mu\nu} = -K^{\nu\mu}$ be smooth. Define $\widetilde{j}^\mu = j^\mu + \partial_\nu K^{\mu\nu}$. An added current with identically zero divergence is called an *identically conserved improvement* in this problem.

(a) Prove that the two currents have the same divergence for every field. Show that their charges agree on the spatial torus when K is periodic. On $\mathbb{R}^d$, assume the original slice charge exists and show equality when $K^{0i}(t,\cdot)$ is compactly supported.

(b) In $1+1$ dimensions take $K^{01} = t/(1+x^2)$ and $K^{10} = -K^{01}$. Compute the added density and flux. Find the change in total charge on $\mathbb{R}$ and the change in charge on the fixed interval $[0,1]$. Does an identically conserved improvement leave every regional charge unchanged?

Solution. The divergence of the improvement is $\partial_\mu \partial_\nu K^{\mu\nu}$. Interchange μ and ν, commute the derivatives and use antisymmetry. The same sum equals its negative, so it is zero identically. The density change is $\partial_i K^{0i}$ because $K^{00} = 0$. On a periodic cube its opposite face values cancel. For compactly supported K^{0i} on $\mathbb{R}^d$, extend the spatial integral over a containing cube; every face value is zero.

For the stated function, direct differentiation gives

$$\widetilde{j}^0 - j^0 = -\frac{2tx}{(1+x^2)^2}, \qquad \widetilde{j}^1 - j^1 = -\frac{1}{1+x^2}.$$

Their divergence is zero, even though neither component vanishes. On $[-R,R]$ the integrated density is $K^{01}(t,R) - K^{01}(t,-R) = 0$. The density is integrable, so the total charge change is zero. On the finite interval the same primitive gives

$$\widetilde{Q}_{[0,1]} - Q_{[0,1]} = K^{01}(t,1) - K^{01}(t,0) = \frac{t}{2} - t = -\frac{t}{2}.$$

Thus regional charges can change. Their changed endpoint flux accounts for that change; identical local divergences do not remove endpoint terms.

Exercise 2.2 (The converse with an off-shell identity supplied). For a first-order density, suppose a legitimate characteristic Q^a and specified local current j^μ satisfy $\partial_\mu j^\mu = -E_a Q^a$ for arbitrary fields, before imposing their equations. Prove that $B^\mu = p^\mu_a Q^a - j^\mu$ gives $\delta_Q L = \partial_\mu B^\mu$.

Apply this calculation on $\mathbb{R}^{1,1}$ to $L = \frac{1}{2}(\phi_t^2 - \phi_x^2)$, $\eta(t,x) = t + x$, and

$$j^0 = \eta\phi_t - \phi, \qquad j^1 = -\eta\phi_x + \phi.$$

Find its characteristic, the divergence term and an actual one-parameter point transformation realising that characteristic. Verify the off-shell identity directly.

For context, the following broader correspondence is a *cited input*, not a task to prove: in the local finite-derivative formulation for the scalar wave Euler–Lagrange equation, conservation-law classes have generalized variational characteristics. Currents are identified modulo currents vanishing on solutions and identically conserved improvements; characteristics are identified modulo ones vanishing on solutions. Here local means depending smoothly on the coordinates, field and finitely many of its derivatives. This is the scalar wave case of Olver, §5.3, Theorem 5.58, p. 334. Explain why the assertion that an arbitrary current is conserved on solutions is not the supplied off-shell identity used in the first calculation.

Solution. Lecture 1 gives the identity $\delta_Q L = E_a Q^a + \partial_\mu(p^\mu_a Q^a)$. The supplied hypothesis substitutes $E_a Q^a = -\partial_\mu j^\mu$ into that formula. Hence

$$\delta_Q L = \partial_\mu(p^\mu_a Q^a - j^\mu), \qquad B^\mu = p^\mu_a Q^a - j^\mu.$$

This proves the stated algebraic converse with its off-shell hypothesis.

For the massless density, $p^0 = \phi_t$, $p^1 = -\phi_x$ and $E = -(\phi_{tt} - \phi_{xx})$. Since $\eta_t = \eta_x = 1$, differentiation of the given current yields

$$\partial_t(\eta\phi_t - \phi) + \partial_x(-\eta\phi_x + \phi) = \eta(\phi_{tt} - \phi_{xx}).$$

Thus its characteristic is $Q^1 = \eta$. The point transformation fixes spacetime and sends the field value u to $u + \varepsilon(t + x)$. It is smooth and invertible for every parameter. The recovered divergence terms are

$$B^0 = \phi_t\eta - j^0 = \phi, \qquad B^1 = -\phi_x\eta - j^1 = -\phi.$$

Direct variation also gives $\delta_Q L = \phi_t\eta_t - \phi_x\eta_x = \phi_t - \phi_x = \partial_t B^0 + \partial_x B^1$.

An on-shell conservation law only says that a divergence vanishes after substitution of a solution. It does not specify a characteristic Q^a or a local identity valid on arbitrary fields. The cited converse supplies a local representation under its stated class and equivalences. The elementary argument above starts after such a representation has been supplied; it is not a proof of that general theorem. Neither assertion alone ensures convergence or conservation of a full-space charge.

Exercise 2.3 (A Noether charge escapes to infinity). Let p and q be real fields on $\mathbb{R}_t \times \mathbb{R}_x$, with density $L = p(q_t + x^2 q_x)$. Choose $p_0 \in C_c^\infty((1,2))$, nonnegative and with integral one, and extend it by zero to $\mathbb{R}$. Set $q = 0$ and

$$p(t,x) = \begin{cases} (1+tx)^{-2} p_0\big(x/(1+tx)\big), & 1+tx > 0, \\ 0, & 1+tx \le 0. \end{cases}$$

(a) Derive the two field equations. Find the Noether current for the constant shift of q. Prove that the displayed fields are smooth at every finite spacetime point and satisfy both equations.

(b) Compute $Q(t) = \int_{\mathbb{R}} p(t,x)\mathrm{d}x$. Find its derivative for $t > 0$ and the limiting flux as $x \to +\infty$. Show that the initial charge is one but the charge is zero for $t \ge 1$.

(c) Identify exactly which full-space charge hypothesis in Lecture 2 fails. Why is this neither an off-shell example nor a claim about Klein–Gordon propagation?

Solution. The density has no derivatives of p, so $E_p = q_t + x^2 q_x$. Variation of q gives $E_q = -p_t - \partial_x(x^2 p)$. For $q \mapsto q + \varepsilon$, the characteristic components are $Q^q = 1$ and $Q^p = 0$, and the density is invariant. Its momentum components for q are $p_q^0 = p$ and $p_q^1 = x^2 p$. Therefore the Noether current is $j^0 = p$, $j^1 = x^2 p$.

On the open branch $1 + tx > 0$, put $\xi = x/(1+tx)$. Differentiating the displayed formula verifies $p_t + \partial_x(x^2 p) = 0$. Equivalently, the trajectories and density Jacobian satisfy

$$x(t) = \frac{\xi}{1 - t\xi}, \qquad x_t = x^2, \qquad \frac{\partial x}{\partial \xi} = (1 - t\xi)^{-2}, \qquad p(t, x(t)) = (1 - t\xi)^2 p_0(\xi).$$

On the other branch $p = 0$, so the equation holds there too. At a finite point with $1 + tx = 0$, necessarily $x \neq 0$. In a small neighbourhood of that point, whenever the denominator is small, $|x/(1+tx)| > 2$. The profile is then zero. Thus p is identically zero near the join and is smooth across it. At $t = 0$ the denominator is one and $p(0,x) = p_0(x)$. The field $q = 0$ solves its equation everywhere.

The support lies on $x > 0$. If $t \le 0$, the change of variables above reaches every $\xi > 0$. If $t > 0$, it reaches exactly $0 < \xi < 1/t$. The density times its Jacobian is $p_0(\xi)$, so

$$Q(t) = 1 \quad (t \le 0), \qquad Q(t) = \int_0^{1/t} p_0(\xi)\mathrm{d}\xi \quad (t > 0).$$

These integrals are finite. For $0 < t \le 1/2$ the value is one, and for $t \ge 1$ it is zero, because the initial support lies in $(1,2)$. FTC now gives $Q'(t) = -p_0(1/t)/t^2$ for $t > 0$. For $t > 0$, at the right spatial end,

$$\lim_{x \to +\infty} x^2 p(t,x) = \frac{p_0(1/t)}{t^2},$$

while the left flux is zero. This is exactly the lost charge rate with the outward sign of Lecture 2.

The nonnegative profile is positive somewhere. Hence for an open interval of times in $(1/2, 1)$ the right flux limit is strictly positive. On each such slice $x^2 p$ is not integrable over $\mathbb{R}$. The spacetime-integrable flux hypothesis of Lecture 2 therefore fails. The current is locally conserved on an actual smooth solution, and its slice density is integrable. The failure concerns its spatial infinity. This model has transport speed x^2 and is not the Klein–Gordon equation.

Classical Field Theory — Exercise Sheet 3

Stress, improvements and angular momentum

Solutions

Use $T^\mu{}_\nu = p^\mu \partial_\nu \phi - \delta^\mu_\nu L$, $\eta = \operatorname{diag}(1, -1, \dots, -1)$ and the inverse-metric Hilbert convention of Lecture 3. All quantities below are smooth. The first problem has d spatial coordinates; the remaining problems have three. Problems 1–3 use Lectures 1–3. Problem 4 is a return problem after Lecture 9; its electromagnetic component conventions are supplied in the question.

Exercise 3.1 (Work by an external coupling). On $I \times \mathbb{T}^d$ consider

$$L = \frac{1}{2}\phi_t^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{\lambda(t)}{4}\phi^4,$$

where I is an open interval and λ is prescribed and smooth.

(a) Derive the field equation and energy density. For a classical solution, find the derivative of its total energy, including the term due to explicit time dependence. Decide whether its physical spatial momentum $P^i = \int_{\mathbb{T}^d} T^{0i} \mathrm{d}^d x$ remains constant.

(b) Set $I = (0, \pi/2)$, $\lambda(t) = 1/\sin^2 t$ and $\phi(t, x) = \sin t$. Verify the field equation, compute its energy and check the sign and value of the energy derivative by both methods.

Solution. The momentum is $p^0 = \phi_t$ and $p^i = -\partial_i \phi$. The Euler expression is $E = -\phi_{tt} + \Delta\phi - \lambda\phi^3$, so stationarity gives $\phi_{tt} - \Delta\phi + \lambda\phi^3 = 0$. The energy density and its spatial flux are

$$\rho_{\mathrm{en}} = \frac{1}{2}\phi_t^2 + \frac{1}{2}|\nabla\phi|^2 + \frac{\lambda}{4}\phi^4, \qquad f^i = -\phi_t \partial_i \phi.$$

The explicit derivative of the density is $\partial_t^{\mathrm{explicit}} L = -\lambda'\phi^4/4$. The general identity of Lecture 3 therefore gives, on solutions,

$$\partial_t \rho_{\mathrm{en}} + \partial_i f^i = \frac{\lambda'}{4}\phi^4, \qquad E'(t) = \frac{\lambda'(t)}{4}\int_{\mathbb{T}^d} \phi(t,x)^4 \mathrm{d}^d x.$$

The flux integral vanishes by periodicity. The density has no explicit spatial dependence, so $\partial_\mu T^\mu{}_i = 0$ on solutions. The integrated lower-index momentum is constant, and raising its spatial index changes only its sign. Thus $P^i = \int T^{0i}$ is also constant.

For the proposed homogeneous field, $\phi_{tt} = -\sin t$ and $\lambda\phi^3 = \sin t$. Their sum is zero. Its energy is

$$E(t) = (2\pi)^d \left(\frac{1}{2}\cos^2 t + \frac{1}{4}\sin^2 t\right).$$

Direct differentiation gives $E' = -(2\pi)^d \sin t \cos t/2$. The work formula gives the same value because $\lambda' = -2\cos t/\sin^3 t$ and $\phi^4 = \sin^4 t$. The positive potential does not force conservation when its coefficient is externally changed.

Exercise 3.2 (Angular momentum and weighted decay). For the autonomous scalar density $L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$ on $\mathbb{R}^{1+3}$, raise the second stress index with η. For fixed distinct spatial indices i, j, define

$$M^{\mu ij} = x^i T^{\mu j} - x^j T^{\mu i}.$$

(a) Prove its local conservation on solutions, keeping the derivatives of the explicit coordinate factors.

(b) Prove that $J^{ij}(t) = \int_{\mathbb{R}^3} M^{0ij}\mathrm{d}^3x$ is constant if, on every compact time interval, every component of $T^{\mu\nu}$ is bounded by $C(1+|x|)^{-4-\sigma}$ for some $C, \sigma > 0$. Explain why an unweighted integrability statement for stress alone does not automatically establish the angular charge hypotheses.

Solution. The scalar stress with both indices raised is $T^{\mu\nu} = \partial^\mu\phi\partial^\nu\phi - \eta^{\mu\nu}L$. It is symmetric. Its divergence vanishes on solutions because the density is autonomous. The product rule gives

$$\partial_\mu M^{\mu ij} = T^{ij} - T^{ji} + x^i\partial_\mu T^{\mu j} - x^j\partial_\mu T^{\mu i} = 0.$$

This keeps the coordinate-factor contribution rather than discarding it.

Under the stated bound, $|x^i|$ and $|x^j|$ are at most $1+|x|$, so $|M^{\mu ij}| \le 2C(1+|x|)^{-3-\sigma}$. The spatial integral of this bound is finite: its tail is bounded by a constant times $\int_1^\infty r^{-1-\sigma}\mathrm{d}r$. It is uniform on each compact time interval. Thus the density is integrable on each slice and the spatial flux is spacetime integrable there. Lecture 2's cutoff argument proves that J^{ij} is constant. Mere integrability of T supplies no weighted bound for $|x|T$; the angular current requires that extra control.

Exercise 3.3 (Belinfante improvement from a spin identity). Suppose a smooth tensor $T^{\mu\nu}$ on $\mathbb{R}^{1+3}$ is divergence-free, and a supplied smooth tensor $S^{\lambda\mu\nu}$ satisfies

$$S^{\lambda\mu\nu} = -S^{\lambda\nu\mu}, \qquad \partial_\lambda S^{\lambda\mu\nu} = T^{\nu\mu} - T^{\mu\nu}.$$

This spin identity is a given hypothesis; it is not inferred from scalar Lecture 3. Define

$$B^{\lambda\mu\nu} = \frac{1}{2}\big(S^{\lambda\mu\nu} + S^{\mu\nu\lambda} + S^{\nu\mu\lambda}\big), \qquad \Theta^{\mu\nu} = T^{\mu\nu} + \partial_\lambda B^{\lambda\mu\nu}.$$

Prove that Θ is symmetric and divergence-free. Whenever the slice integrals exist, find the difference between its integrated four-momentum and that of T. Give a sufficient boundary condition for equality, and explain why local improvement alone is insufficient.

Solution. Using antisymmetry in the last two slots of S gives

$$B^{\mu\lambda\nu} = \frac{1}{2}\big(-S^{\mu\nu\lambda} - S^{\lambda\mu\nu} - S^{\nu\mu\lambda}\big) = -B^{\lambda\mu\nu}.$$

Subtracting the two expressions with μ and ν interchanged gives $B^{\lambda\mu\nu} - B^{\lambda\nu\mu} = S^{\lambda\mu\nu}$. Consequently the supplied spin identity implies

$$\Theta^{\mu\nu} - \Theta^{\nu\mu} = T^{\mu\nu} - T^{\nu\mu} + \partial_\lambda S^{\lambda\mu\nu} = 0.$$

The added divergence has identically zero divergence because its first two slots are antisymmetric and mixed derivatives commute. Hence $\partial_\mu\Theta^{\mu\nu} = \partial_\mu T^{\mu\nu} = 0$.

On a spatial cube Q_R, coordinate FTC gives the momentum difference

$$\int_{Q_R} (\Theta^{0\nu} - T^{0\nu})\mathrm{d}^3x = \int_{\partial Q_R} B^{i0\nu} n_i \,\mathrm{d}S.$$

There is no $\partial_t B^{00\nu}$ term, since $B^{00\nu} = 0$. If $B^{i0\nu}$ is compactly supported in space the boundary integral vanishes for large R. Periodicity also makes opposite faces cancel. Under either condition, provided the original charges exist, the improved and original momenta agree. Without control of this boundary term, the local identities do not imply equality of integrated charges.

Exercise 3.4 (Maxwell Hilbert stress, after Lecture 9). Return to this problem after Lecture 9. Let $F_{\mu\nu} = -F_{\nu\mu}$ be held fixed with both indices down while a Lorentz metric is varied. Use the density and Hilbert convention

$$L = -\frac{1}{4} g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma}, \qquad \delta S = \frac{1}{2} \int T_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g, \quad k^{\mu\nu} = \delta g^{\mu\nu}.$$

The volume variation is $\delta \mathrm{vol}_g = -\frac{1}{2} g_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. Derive $T_{\mu\nu}$ by varying both inverse metrics. At $g = \eta$, set $F_{i0} = E_i$ and $F_{ij} = \epsilon_{ijk} B_k$, with $\epsilon_{123} = 1$. Compute T_{00} and T^{i0} in terms of the vectors $\mathbf{E}$ and $\mathbf{B}$. Evaluate them for $\mathbf{E} = (1, 2, 0)$ and $\mathbf{B} = (0, 0, 3)$.

Solution. Varying each inverse metric in the stated contraction and relabelling the summed indices gives two equal terms. Thus

$$\delta L = -\frac{1}{2} F_{\mu\alpha} F_{\nu}{}^{\alpha} k^{\mu\nu}.$$

Multiplying by the varied volume gives

$$\delta(L \mathrm{vol}_g) = \frac{1}{2} \left[-F_{\mu\alpha} F_{\nu}{}^{\alpha} + \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right] k^{\mu\nu} \mathrm{vol}_g.$$

Comparison with the plus Hilbert convention identifies the tensor inside the brackets as $T_{\mu\nu}$.

The supplied electromagnetic convention gives $F_{\alpha\beta} F^{\alpha\beta} = 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)$ and $F_{0\alpha} F_0{}^{\alpha} = -|\mathbf{E}|^2$. Therefore

$$T_{00} = \frac{1}{2} (|\mathbf{E}|^2 + |\mathbf{B}|^2).$$

For a spatial index the trace term is zero, and direct contraction gives $T_{i0} = -\epsilon_{ijk} E_j B_k$. Raising that spatial index gives $T^{i0} = \epsilon_{ijk} E_j B_k = (\mathbf{E} \times \mathbf{B})_i$. In the example the squared magnitudes are 5 and 9, so $T_{00} = 7$. The cross product is $(2 \cdot 3, -1 \cdot 3, 0) = (6, -3, 0)$. These signs follow from the inverse-metric convention, not a choice to make energy positive after the calculation.

Classical Field Theory — Exercise Sheet 4

Formula, stability and weak derivatives

Solutions

Use the unitary Fourier transform $\widehat{f}(k) = (2\pi)^{-1/2}\int_{\mathbb{R}} e^{-ikx}f(x)\mathrm{d}x$ and $\|f\|_{H^1}^2 = \int_{\mathbb{R}}(1+k^2)|\widehat{f}(k)|^2\mathrm{d}k$. All wave equations on this sheet have one spatial dimension and zero mass. The solution class is that of Theorem 4.3.

Exercise 4.1 (Two initial data and two travelling packets). Let $f, g \in C_c^\infty(\mathbb{R};\mathbb{R})$ be initial displacement and velocity.

(a) Derive d'Alembert's formula from Lecture 4's Fourier multipliers. Use an oriented integral when $t < 0$. Verify the equation and both data, and justify identifying the formula with the energy-class solution.

(b) Let F, G be smooth real functions supported in $[-1/4, 1/4]$ and strictly positive on its interior. Take

$$f(x) = F(x+3) + 2G(x-2), \qquad g(x) = -F'(x+3) + 2G'(x-2).$$

Find the solution. For $t \geq 0$, find the open time intervals on which an observer at $x = 0$ sees each contribution. Where and when do the packet centres meet? Find the corresponding visibility interval at that position.

Solution. Lecture 4 constructs the solution by $\widehat{v} = \cos(t|k|)\widehat{f} + S_t(|k|)\widehat{g}$. Since $\cos(t|k|) = \cos(tk)$, Fourier translation gives the displacement term $[f(x+t) + f(x-t)]/2$. The integral representation of the sine multiplier gives its velocity contribution as

$$\frac{1}{2}\int_0^t [g(x+s) + g(x-s)]\mathrm{d}s = \frac{1}{2}\int_{x-t}^{x+t} g(y)\mathrm{d}y.$$

The identity holds also for negative t with oriented integrals. Fourier inversion and the time integral can be interchanged here because the initial Fourier transforms are Schwartz functions.

Write the last contribution as w. The fundamental theorem of calculus gives

$$w_t = \frac{1}{2}[g(x+t)+g(x-t)], \qquad w_x = \frac{1}{2}[g(x+t)-g(x-t)], \qquad w_{tt} = w_{xx} = \frac{1}{2}[g'(x+t)-g'(x-t)].$$

The translated displacement term also solves $v_{tt} = v_{xx}$. At zero it equals f, while its time derivative vanishes; $w(0) = 0$ and $w_t(0) = g$. On each compact time interval all these functions have a common compact spatial support and are smooth. Dominated convergence therefore gives $v \in C(H^1) \cap C^1(L^2)$. The classical equation implies the weak equation by integration against compact tests, so Lecture 4's uniqueness applies.

For the specified data, the travelling field

$$v(t,x) = F(x+3-t) + 2G(x-2+t)$$

has exactly the required two initial values, and hence is that solution. At $x = 0$ its arguments lie in $(-1/4, 1/4)$ precisely for $11/4 < t < 13/4$ and $7/4 < t < 9/4$, respectively. Strict positivity gives nonzero contributions throughout those intervals, and smooth support gives

zero at their endpoints. The centres follow $x = t - 3$ and $x = 2 - t$. They meet at $t = 5/2$, $x = -1/2$. At this position the two arguments are $5/2 - t$ and $t - 5/2$, so both visibility intervals are $(9/4, 11/4)$. The linear sum remains valid while the packets overlap.

Exercise 4.2 (Hadamard instability for an elliptic Cauchy problem). For each positive integer n, seek a smooth function $u_n(x, y)$, periodic in $x \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, satisfying

$$\partial_y^2 u_n + \partial_x^2 u_n = 0, \qquad u_n(x, 0) = 0, \qquad \partial_y u_n(x, 0) = e^{-\sqrt{n}} \cos(nx).$$

Find such a solution by separation of variables. Prove that the initial velocity tends to zero in every fixed $C^r(\mathbb{T})$ seminorm, while $\|u_n(\cdot, y)\|_{L^2(\mathbb{T})} \to \infty$ for every fixed $y > 0$. What continuity claim does this disprove? Why does it not contradict Lecture 4?

Solution. For a mode $u_n = a_n(y) \cos(nx)$, the equation is $a_n'' - n^2 a_n = 0$. Its specified data give $a_n(y) = e^{-\sqrt{n}} \sinh(ny)/n$. Each derivative of order $j \le r$ of the initial velocity has supremum at most $n^j e^{-\sqrt{n}} \le n^r e^{-\sqrt{n}}$. For fixed r, its logarithm $r \log n - \sqrt{n}$ tends to $-\infty$. Thus all the fixed smooth seminorms tend to zero.

Since $\int_0^{2\pi} \cos^2(nx) \mathrm{d}x = \pi$ for $n \ge 1$,

$$\|u_n(\cdot, y)\|_2 = \frac{\sqrt{\pi}}{n} e^{-\sqrt{n}} \sinh(ny).$$

For fixed $y > 0$ and sufficiently large n, $\sinh(ny) \ge e^{ny}/4$. The logarithm of the resulting lower bound is $ny - \sqrt{n} - \log n + \log(\sqrt{\pi}/4)$, which tends to $+\infty$. Consequently a Cauchy solution assignment agreeing with these modes cannot be continuous at zero from the smooth-data topology to L^2 on any positive slice. This is instability even among data possessing explicit smooth solutions; no assertion of existence for all smooth data is needed. The equation here is elliptic: it has $+\partial_x^2$, whereas Lecture 4's wave equation is $\partial_t^2 - \partial_x^2 = 0$. Reversing wave time would not change that sign.

Exercise 4.3 (A corner and its weak derivatives ⋆). Let $f(x) = e^{-|x|}$. Compute its unitary Fourier transform and its H^1 norm. Identify its first weak derivative by testing against every $\zeta \in C_c^\infty(\mathbb{R})$, and calculate its second distributional derivative. Decide whether f belongs to the space

$$H^2(\mathbb{R}) = \left\{ h \in L^2(\mathbb{R}) : \int_{\mathbb{R}} (1 + k^2)^2 |\widehat{h}(k)|^2 \mathrm{d}k < \infty \right\}.$$

Distinguish the value assigned to a weak derivative at the single point zero from the existence of a classical derivative there.

Solution. The two half-line integrals give

$$\widehat{f}(k) = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{1 - ik} + \frac{1}{1 + ik} \right) = \frac{\sqrt{2/\pi}}{1 + k^2}.$$

Hence the Fourier definition and $\int_{\mathbb{R}} (1 + k^2)^{-1} \mathrm{d}k = \pi$ give $\|f\|_{H^1}^2 = 2$. Direct integration also gives $\|f\|_2^2 = 1$.

For a compact smooth test ζ, integration by parts on both half-lines gives

$$\int_{\mathbb{R}} f\zeta' = \zeta(0) - \int_{-\infty}^{0} e^x \zeta \mathrm{d}x - \zeta(0) + \int_0^{\infty} e^{-x} \zeta \mathrm{d}x = -\int_{\mathbb{R}} h\zeta,$$

where $h = e^x$ for $x < 0$ and $h = -e^{-x}$ for $x > 0$. Thus h is the weak derivative. It belongs to L^2 and has norm squared 1, agreeing with the Fourier norm identity. Its value at zero may be chosen arbitrarily without changing this L^2 element.

Differentiating h distributionally retains its jump. More explicitly, the same half-line calculation gives

$$\langle h', \zeta \rangle = -\int_{\mathbb{R}} h\zeta' = \int_{\mathbb{R}} f\zeta - 2\zeta(0).$$

Therefore $f'' = f - 2\delta_0$. The Fourier H^2 integrand is the nonzero constant $2/\pi$, so its integral diverges. Finally the classical one-sided derivatives of f at zero are 1 and -1; changing a weak derivative's value at one point cannot make these agree.

Exercise 4.4 (The sharp massless sine bound). Choose a real even $\rho \in C_c^\infty((-1,1))$ with $\|\rho\|_2 = 1$. For $0 < \varepsilon \le 1$, let g_ε satisfy $\widehat{g}_\varepsilon(k) = \varepsilon^{-1/2}\rho(k/\varepsilon)$, and let v_ε have initial data $(0, g_\varepsilon)$.

(a) Prove that g_ε is real and normalized in L^2, and that for every $T > 0$,

$$\sup_{|t|\le T} \|v_\varepsilon(t) - tg_\varepsilon\|_2 \le \frac{T^3\varepsilon^2}{6}.$$

(b) For fixed real t, determine the operator norm on real $L^2(\mathbb{R})$ of the Fourier multiplier $S_t(|k|)$, where $S_t(0) = t$. Explain why conservation of massless energy cannot give a bound for $\|v(t)\|_2$ uniform in both time and all initial data with energy $1/2$.

Solution. The transform is real and even, so it satisfies $\widehat{g}_\varepsilon(-k) = \overline{\widehat{g}_\varepsilon(k)}$. Fourier inversion preserves reality. Substitution $k = \varepsilon q$ and Plancherel give

$$\|g_\varepsilon\|_2^2 = \int_{\mathbb{R}} \varepsilon^{-1}|\rho(k/\varepsilon)|^2 \mathrm{d}k = 1.$$

The inverse transform is Schwartz, so all these data are permitted by Lecture 4. Integrating $|1 - \cos s| \le s^2/2$ gives $|\sin z - z| \le |z|^3/6$ for real z. Thus

$$|S_t(|k|) - t| \le \frac{|t|^3k^2}{6}, \qquad \|(S_t(|k|) - t)\widehat{g}_\varepsilon\|_2 \le \frac{|t|^3\varepsilon^2}{6}\|\widehat{g}_\varepsilon\|_2.$$

This proves the compact-time estimate, including $k = 0$ by continuity. In particular the reverse triangle inequality gives $\big|\|v_\varepsilon(t)\|_2 - |t|\big| \le |t|^3\varepsilon^2/6$.

Lecture 4's bound $|S_t| \le |t|$ supplies an operator norm at most $|t|$. The normalized real test family just constructed gives the reverse bound by taking $\varepsilon \to 0$. Hence the norm is $|t|$ for every real t. The initial energy is $\|g_\varepsilon\|_2^2/2 = 1/2$. Given a proposed common displacement bound M, choose a time with $|t| > M$ and then choose ε small enough that the displacement norm exceeds M. This disproves a bound uniform over this energy level and all time. It does not assert unbounded growth for each fixed datum; the finite-time estimate of Lecture 4 remains valid.

Classical Field Theory — Exercise Sheet 5

Cones with forcing and interior tails

Solutions

Write $\Box = \partial_t^2 - \Delta$. Every cone argument below concerns real C^2 fields on an open neighbourhood of the closed cone $\{0 \le t \le R,\ |x-a| \le R-t\}$, with $R > 0$. No existence theorem for a nonlinear equation is assumed or proved here.

Exercise 5.1 (Source work on a shrinking cone). Suppose $(\Box + m^2)w = F$, with $m \ge 0$ and continuous F. Let $E_m(t)$ be the integral over $B_{R-t}(a)$ of $e_m = (w_t^2 + |\nabla w|^2 + m^2 w^2)/2$.

(a) Derive the exact shrinking-ball balance, including source work and all lateral flux terms. Prove, for $0 \le t < R$,

$$E_m(t) \le e^t \left[E_m(0) + \frac{1}{2} \int_0^t e^{-s} \|F(s)\|^2_{L^2(B_{R-s}(a))} \mathrm{d}s \right].$$

(b) If $F = 0$ on the cone and both initial data vanish on its base, prove that $w = 0$ on the closed cone, including when $m = 0$. Explain why this is a local statement even if total energy is infinite.

Solution. The energy density and spatial flux are $e_m = (w_t^2 + |\nabla w|^2 + m^2 w^2)/2$ and $f = -w_t \nabla w$. Differentiating and using the forced equation gives

$$\partial_t e_m + \operatorname{div} f = w_t (w_{tt} - \Delta w + m^2 w) = w_t F.$$

Lecture 5's moving-ball and ball-divergence formulas apply to these C^1 quantities. With n the outward spatial normal they give

$$E_m'(t) = \int_{B_{R-t}(a)} w_t F \, \mathrm{d}^d x - \int_{\partial B_{R-t}(a)} (e_m + f \cdot n) \mathrm{d}S.$$

The boundary density is

$$e_m + f \cdot n = \frac{1}{2} \left[(w_t - \partial_n w)^2 + |\nabla_{\tan} w|^2 + m^2 w^2 \right] \ge 0.$$

The inequality $|w_t F| \le (w_t^2 + F^2)/2$ therefore yields $E_m' \le E_m + \|F\|_2^2/2$, with the norm on the current shrinking ball. Multiplying by e^{-t} and integrating from zero proves the stated estimate.

If the source and both data vanish, the estimate gives $E_m(t) = 0$. Continuity implies $w_t = 0$ on each open shrinking ball. For an interior cone point (t, x), the segment (s, x), $0 \le s \le t$, stays in the cone. Hence $w(t,x) = w(0,x) + \int_0^t w_t(s,x) \mathrm{d}s = 0$. This retains the initial displacement, which energy alone would miss for a massless constant. Continuity gives zero on the lateral boundary and vertex. Every integral used a compact cone slice; no all-space integrability was needed.

Exercise 5.2 (Finite speed for a semilinear equation, conditional on solutions). Let $V \in C^2(\mathbb{R})$ and suppose two real C^2 fields ϕ_1, ϕ_2 solve $\Box\phi + V'(\phi) = 0$ on a neighbourhood of the same closed cone. Assume their displacements and velocities agree on $B_R(a)$ at time zero.

(a) Prove that the fields agree on the closed cone. Use the mass-one energy of their difference, allowing V'' to have either sign. Give an explicit finite coefficient in the energy inequality in terms of a bound for $|V''|$ on the field values occurring in the cone.

(b) Suppose instead that the fields are defined on an open slab $I \times \mathbb{R}^d$, with $0 \in I$, and their two initial data agree outside a closed set K. Prove agreement at every (t, x) with $\operatorname{dist}(x, K) > |t|$. What extra condition makes the zero field an available comparison solution?

Solution. Put $w = \phi_1 - \phi_2$. The ordinary fundamental theorem gives

$$b(t,x) = \int_0^1 V''(\phi_2 + s(\phi_1 - \phi_2))\mathrm{d}s, \qquad \Box w + bw = 0.$$

The fields are bounded on the compact cone. Their values and every segment between them lie in some compact interval $[-M, M]$. Thus $L = \max_{|z|\le M} |V''(z)|$ is finite and $|b| \le L$. No positivity of b is required. Write the equation as $(\Box + 1)w = (1 - b)w$. The exact balance from the first problem gives

$$E_1'(t) \le \int_{B_{R-t}} |1 - b|\,|w_t w|\mathrm{d}^d x \le \frac{1+L}{2} \int_{B_{R-t}} (w_t^2 + w^2)\mathrm{d}^d x \le (1+L)E_1(t).$$

Matching smooth initial displacement also gives matching initial spatial gradients on the open ball, so $E_1(0) = 0$. The integrating factor $e^{-(1+L)t}$ yields $E_1 = 0$. Its mass term gives $w = 0$ on the interior directly; continuity includes the cone boundary and vertex.

For a later event outside the asserted influence set, the two data coincide on a neighbourhood of $\overline{B_t(x)}$. Apply the preceding argument to the compact cone of height t with this base and the event as vertex. It lies in the open slab. Thus the fields agree at the event. Reverse time for earlier events. No bound uniform over all of space is needed: a separate compact cone supplies a finite L for each event. The constant field zero solves the equation exactly when $V'(0) = 0$. Under that condition the result also gives finite propagation of the support of any existing solution with compactly supported data. None of these arguments constructs such a solution.

Exercise 5.3 (Huygens' principle and the dimension). Consider the massless wave with zero displacement and real $g \in C_c^\infty(\mathbb{R}^d)$ as initial velocity. The following classical representation formulas may be used as given facts for $t > 0$:

$$u_1(t,x) = \frac{1}{2}\int_{x-t}^{x+t} g(y)\,\mathrm{d}y, \qquad u_2(t,x) = \frac{1}{2\pi}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,\mathrm{d}^2y,$$

$$u_3(t,x) = \frac{1}{4\pi t}\int_{|y-x|=t} g(y)\,\mathrm{d}S_y.$$

For the higher-dimensional classification one may use these general forms. Let α_d be the volume of the unit ball, $\fint$ the integral divided by the region's volume or area, and $D_t = t^{-1}\partial_t$. For odd $d \geq 3$ and even $d \geq 2$, respectively,

$$u_d = \frac{1}{1\cdot 3\cdots(d-2)} D_t^{(d-3)/2}\left[t^{d-2}\fint_{|y-x|=t} g(y)\,\mathrm{d}S_y\right],$$

$$u_d = \frac{1}{2\cdot 4\cdots d} D_t^{(d-2)/2}\left[\frac{1}{\alpha_d}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,\mathrm{d}^dy\right].$$

These facts hold for the smooth compact data specified here; see Evans, *Partial Differential Equations*, 2nd ed., §2.4.1, formulas (31), (38), Theorems 2–3, pp. 73–77. Nonzero displacement contributes the time derivative of the corresponding velocity formula with g replaced by that displacement.

(a) For $d = 1, 2, 3$, let $g \geq 0$, $g \not\equiv 0$, with support inside $B_1(0)$. Decide whether $u_d(t,0)$ vanishes for $t > 1$. Explain the difference between finite speed and dependence only on data arbitrarily near the base sphere.

(b) Use the general formulas to classify all positive spatial dimensions. For even d, test data supported strictly inside the base sphere so that every time derivative acts on a smooth kernel. Explain why a statement that all even-dimensional kernels are positive would be too strong.

Solution. In one dimension the interval $[-t,t]$ contains the support for $t > 1$. Thus $u_1(t,0) = \frac{1}{2}\int g > 0$ even though g is zero near its two endpoints. In two dimensions the kernel is strictly positive on the fixed support, so $u_2(t,0) > 0$. In three dimensions the sphere of radius $t > 1$ misses the support, hence $u_3(t,0) = 0$. All three respect finite speed: their values use only data in the closed base ball. Sphere dependence is the stronger statement that data vanishing near the base sphere have zero effect at the event, even when they are nonzero in its interior.

For odd $d \geq 3$, the supplied formula uses a finite number of time derivatives of spherical averages. If both data vanish on a neighbourhood of the sphere at the specified time, these averages vanish on a time neighbourhood there, and all their derivatives vanish. This proves the sphere property, also including the given displacement term.

For even $d = 2\ell + 2$, put the nonnegative nonzero data inside $B_1(0)$ and fix $t > 1$. The support stays strictly away from the moving boundary for nearby times, so differentiation under the fixed-support integral is valid. Induction gives, with the empty product equal to one,

$$D_t^\ell (t^2 - r^2)^{-1/2} = (-1)^\ell (1\cdot 3\cdots(2\ell-1))(t^2-r^2)^{-\ell-1/2}.$$

The coefficient is nonzero and the remaining kernel is strictly positive on the support. Its integral against g cannot vanish. Thus every even dimension has an interior-tail counterexample to the sphere property. The sign alternates, becoming negative already for $d = 4$; positivity of the two-dimensional kernel must not be asserted in all even dimensions.

Classical Field Theory — Exercise Sheet 6

Causal kernels and frequency boundary values

Solutions

Use $\Box = \partial_t^2 - \Delta$ and Lecture 6's normalized kernel $G_{\mathrm{ret}} = \delta(t-|x|)/(4\pi|x|)$ in three spatial dimensions. Its meaning at the vertex is its test pairing, not a product of distributions. All distribution pairings are bilinear. Write H for the Heaviside function; its value at zero does not affect the distributions below.

Exercise 6.1 (An entire radial source response). Let $q \in C_c^\infty(\mathbb{R})$, and let h be a smooth even compactly supported function such that $x \mapsto h(|x|)$ is smooth on $\mathbb{R}^3$. Take $f(t,x) = q(t)h(|x|)$ and $u = G_{\mathrm{ret}} * f$.

(a) Perform the angular integration to obtain a one-dimensional radial integral for $u(t,r)$, $r = |x| > 0$, with an inner integral over travel distance. Find its limit at $r = 0$ and explain why no singularity remains there.

(b) Set $v(t,r) = ru(t,r)$ for $r \geq 0$ and extend it oddly to the real line. Reduce the equation for v to a forced one-dimensional wave and derive its causal double-integral formula. Check its agreement with part (a).

(c) Let $a < b$ and $R > 0$. If $\operatorname{supp} q \subset [a,b]$ and $\operatorname{supp} h \subset [-R,R]$, find the possible response interval at an observer $r > R$. If $q > 0$ on (a,b) and $h > 0$ for $|s| < R$, prove that the response is positive throughout the interior of that interval.

Solution. Lecture 6 gives a smooth rotation-invariant response by integrating $q(t-|x-z|)h(|z|)/(4\pi|x-z|)$ over z. For $r,s > 0$, put $\mu = \cos\theta$. Then $\ell^2 = r^2 + s^2 - 2rs\mu$ and $\mathrm{d}\mu = -\ell\,\mathrm{d}\ell/(rs)$. The azimuthal integral is 2π. Consequently

$$u(t,r) = \frac{1}{2r}\int_0^\infty sh(s)\int_{|r-s|}^{r+s} q(t-\ell)\,\mathrm{d}\ell\,\mathrm{d}s.$$

For fixed $s > 0$, the inner integral divided by $2r$ tends to $q(t-s)$ as $r \downarrow 0$. Its absolute value is at most $\|q\|_\infty$, since its interval has length $2\min(r,s) \leq 2r$. Dominated convergence with the integrable majorant $s|h(s)|\|q\|_\infty$ proves

$$u(t,0) = \int_0^\infty sh(s)q(t-s)\,\mathrm{d}s.$$

This also follows directly from the original three-dimensional integral at $x = 0$. Smoothness at the centre is supplied by Lecture 6's convolution theorem, not merely by finiteness of this radial limit. Radial smoothness implies $u_r(t,0) = 0$.

For the second part, the radial Laplacian identity gives $v_{tt} - v_{rr} = rq(t)h(r)$ for $r > 0$. The smooth even radial profile of u makes v a smooth odd function of r, and the odd forcing is $rq(t)h(|r|)$. The causal one-dimensional formula is

$$v(t,r) = \frac{1}{2}\int_{-\infty}^{t} q(\tau)\int_{r-(t-\tau)}^{r+(t-\tau)} sh(|s|)\,\mathrm{d}s\,\mathrm{d}\tau.$$

To verify it without assuming a Duhamel theorem, differentiate the inner endpoint integral twice. Its zero-length interval at $\tau = t$ contributes $rh(|r|)q(t)$ to v_{tt}; the remaining terms equal v_{rr}. All integrals and derivatives are justified on compact sets by compact support

and smoothness. The formula vanishes before q starts, so Lecture 5's one-dimensional local uniqueness identifies it with ru.

There is also a direct equality of the two integrals. Splitting the odd source into positive and negative s and interchanging integrals yields, for $r \geq 0$,

$$v(t,r) = \frac{1}{2}\int_0^\infty sh(s)\left[\int_{-\infty}^{t-|r-s|} q(\tau)\mathrm{d}\tau - \int_{-\infty}^{t-r-s} q(\tau)\mathrm{d}\tau\right]\mathrm{d}s.$$

The bracket is $\int_{|r-s|}^{r+s} q(t-\ell)\mathrm{d}\ell$, proving agreement.

For the final part, a source position in $B_R(0)$ is at distance between $r-R$ and $r+R$ from an exterior observer. The retarded argument must be in $[a,b]$, so the field vanishes outside $[a+r-R, b+r+R]$. Under the positivity hypotheses, an interior time satisfies

$$(r-R, r+R) \cap (t-b, t-a) \neq \varnothing.$$

Choose a travel distance in this open intersection. The corresponding source points include an open set inside $B_R(0)$ on which both factors of the original integrand are positive. Its integral is positive. At the endpoints the field is zero by smoothness and its vanishing on the adjacent exterior time intervals. The strict positivity assertion requires these extra hypotheses; a support bound alone would allow cancellation.

Exercise 6.2 (Which side of the frequency poles?). The spacetime Fourier convention is

$$\widetilde{F}(\nu,k) = (2\pi)^{-2}\int e^{i\nu t - ik\cdot x}F(t,x)\mathrm{d}t\mathrm{d}^3x.$$

For a tempered distribution it is defined by transposition on Schwartz tests, with this same kernel. Fourier transformation preserves Schwartz space and its topology, so this definition preserves convergence of tempered distributions. The Schwartz Fourier automorphism is the one from *Analysis* I used in Lecture 4; transposition here specifies its distributional extension. Let $G_\varepsilon = e^{-\varepsilon t}G_{\mathrm{ret}}$, $\varepsilon > 0$; the product is understood on the future cone, where the factor is bounded.

(a) Compute its spatial transform, then its full spacetime transform. Prove convergence as $\varepsilon \downarrow 0$ on Schwartz tests and express $\widetilde{G}_{\mathrm{ret}}$ as a frequency boundary value. Check the Fourier coefficient of the equation $\Box G_{\mathrm{ret}} = \delta^4$.

(b) For fixed $\kappa = |k| > 0$, derive the principal-value and delta terms of that boundary value. Here $\langle \mathrm{pv}(1/s), \zeta\rangle = \lim_{a\downarrow 0}\int_{|s|>a}\zeta(s)/s\,\mathrm{d}s$. Derive the identity for $(s \pm i0)^{-1}$ by separating real and imaginary parts. Compare the advanced and half-retarded plus half-advanced kernels. State why the fixed-κ decomposition cannot simply be evaluated at $\kappa = 0$.

Solution. The spatial transform includes the factor $(2\pi)^{-3/2}$. Angular integration on the shell of radius t therefore gives

$$\widehat{G}_\varepsilon(t,k) = (2\pi)^{-3/2}H(t)e^{-\varepsilon t}\frac{\sin(t\kappa)}{\kappa}, \qquad \kappa = |k|,$$

with continuous value t for the quotient at $\kappa = 0$. Splitting the sine into exponentials and integrating on $t \geq 0$ gives

$$\int_0^\infty e^{-(\varepsilon - i\nu)t}\frac{\sin(t\kappa)}{\kappa}\mathrm{d}t = \frac{1}{(\varepsilon - i\nu)^2 + \kappa^2} = \frac{1}{\kappa^2 - (\nu + i\varepsilon)^2}.$$

The remaining unitary time factor is $(2\pi)^{-1/2}$. Thus the full transform is

$$\widetilde{G}_\varepsilon = (2\pi)^{-2}\frac{1}{|k|^2-(\nu+i\varepsilon)^2},$$

also at $\kappa = 0$ by its integral limit.

The cone pairing with a Schwartz test is bounded by a constant times $\int_0^\infty r(1+r)^{-N}\mathrm{d}r$ for $N > 2$. The same majorant works with $e^{-\varepsilon r} \le 1$, so dominated convergence gives $G_\varepsilon \to G_{\mathrm{ret}}$ as tempered distributions. Transposing the continuous Schwartz Fourier map passes this convergence to the transforms. Thus

$$\widetilde{G}_{\mathrm{ret}} = (2\pi)^{-2}\lim_{\varepsilon\downarrow 0}\frac{1}{|k|^2-(\nu+i\varepsilon)^2},$$

where the limit is a distribution on all four frequency variables. Transforming the already proved fundamental-solution equation gives $(|k|^2-\nu^2)\widetilde{G}_{\mathrm{ret}} = (2\pi)^{-2}$, since the transform of δ^4 is the constant $(2\pi)^{-2}$. It is not the constant one in the unitary convention.

For the second part, write

$$\frac{1}{s\pm i\varepsilon} = \frac{s}{s^2+\varepsilon^2} \mp i\frac{\varepsilon}{s^2+\varepsilon^2}.$$

For a Schwartz test, substitute $s = \varepsilon z$ in the second term. Dominated convergence gives its limit $\pi\zeta(0)$. For the first term subtract $\zeta(0)$ times an even cutoff equal to one near zero; the subtracted part has zero odd integral. The remaining test vanishes at zero, so dominated convergence near zero and at infinity gives the principal value. Consequently $(s\pm i0)^{-1} = \mathrm{pv}(1/s) \mp i\pi\delta(s)$. At a fixed $\kappa > 0$, the partial fractions are

$$\frac{1}{\kappa^2-(\nu+i\varepsilon)^2} = \frac{1}{2\kappa}\left[-\frac{1}{\nu-\kappa+i\varepsilon}+\frac{1}{\nu+\kappa+i\varepsilon}\right].$$

Applying the identity gives

$$\lim_{\varepsilon\downarrow 0}\frac{1}{\kappa^2-(\nu+i\varepsilon)^2} = \mathrm{pv}(\kappa^2-\nu^2)^{-1} + \frac{i\pi}{2\kappa}[\delta(\nu-\kappa)-\delta(\nu+\kappa)].$$

The principal value at the two distinct roots is understood by the same partial fractions. Time reflection gives the advanced denominator $\kappa^2-(\nu-i0)^2$. Its imaginary term has the opposite sign, so the half-sum cancels it. The factor $1/(2\kappa)$ and the two-root separation require $\kappa > 0$. The full boundary-value limit above already includes zero spatial frequency; one must not replace it by an undefined substitution into the separated terms.

Exercise 6.3 (Descending the spatial dimension). Starting with the three-dimensional retarded pairing, integrate out one auxiliary spatial coordinate to construct a kernel G_2 on $\mathbb{R}^{1+2}$, and two auxiliary coordinates to construct G_1 on $\mathbb{R}^{1+1}$. Make the construction legitimate on compact smooth tests: a test independent of the auxiliary coordinates is not itself compactly supported. Derive explicit locally integrable formulas, prove $(\partial_t^2-\Delta_d)G_d = \delta^{1+d}$ for $d = 1, 2$, and identify their support and their relation to Sheet 5's interior tails.

Solution. Let $\psi(t,x)$ be a compact test in the retained coordinates, with $|t| \le T$ on its support. Lift it to $\psi(t,x)\zeta(z)$, where the auxiliary cutoff is one for $|z| \le T+1$. On the three-dimensional cone $t = \sqrt{|x|^2+|z|^2}$, a nonzero lifted test necessarily has $|z| \le T$. The resulting pairing is independent of any larger cutoff and defines G_d. Its local continuity follows from the three-dimensional pairing with this fixed cutoff for tests of fixed support.

For one removed coordinate, the two roots of $t = \sqrt{r^2 + z^2}$, for $t > r = |x|$, are $z = \pm\sqrt{t^2 - r^2}$. Changing variables on the two half-lines gives

$$G_2(t, x) = \frac{H(t - r)}{2\pi\sqrt{t^2 - r^2}}.$$

This fraction is evaluated for $t > r$ and the function is zero outside that open cone; values on the cone itself do not affect its distribution. This is a locally integrable function: for $0 < t < T$, its spatial radial integral contains $\int_0^t r(t^2 - r^2)^{-1/2}\mathrm{d}r = t$. The change of variables in the original pairing also proves that no additional cone-supported atom was omitted.

For two removed coordinates, let their radius be s and set $\ell = \sqrt{x^2 + s^2}$. The polar Jacobian gives

$$\int_0^\infty \frac{2\pi s}{4\pi\sqrt{x^2 + s^2}}\delta(t - \sqrt{x^2 + s^2})\mathrm{d}s = \frac{1}{2}\int_{|x|}^\infty \delta(t - \ell)\mathrm{d}\ell = \frac{1}{2}H(t - |x|).$$

These delta expressions abbreviate changes of variables on tests, so there is no multiplication of singular distributions.

Finally apply the three-dimensional identity to the lifted test. The retained wave operator acts on ψ; the extra term differentiating ζ is supported where $|z| > T + 1$ and hence has zero cone pairing. The vertex has $\zeta(0) = 1$. Therefore

$$\langle G_d, (\partial_t^2 - \Delta_d)\psi\rangle = \psi(0, 0), \qquad d = 1, 2.$$

The formulas vanish outside the future cone and are strictly positive in its interior. Their supports are its whole closure. Their convolution with initial velocity gives exactly the one- and two-dimensional formulas used on Sheet 5, explaining why an interior tail is compatible with finite speed.

Classical Field Theory — Exercise Sheet 7

Modes, complex pairs and admissibility

Solutions

Use $\{F,G\} = \int (F_\phi G_\pi - F_\pi G_\phi)$ and $\Omega = \int \delta\phi \wedge \delta\pi$. Except for the explicitly finite torus model in Problem 1, functionals have Lecture 7's smooth Schwartz-valued gradients. A formal distributional derivative does not automatically belong to that class.

Exercise 7.1 (Real oscillators in a finite torus model). On $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, fix an integer $N \geq 2$ and use the real basis

$$e_0 = (2\pi)^{-1/2}, \qquad e_{n,c} = \pi^{-1/2}\cos(nx), \qquad e_{n,s} = \pi^{-1/2}\sin(nx), \quad 1 \leq n \leq N.$$

Let $\phi = q_0 e_0 + \sum_{n=1}^{N}(q_{n,c}e_{n,c} + q_{n,s}e_{n,s})$ and let π have the same expansion with coefficients p.

(a) Verify orthonormality and compute the pulled-back canonical form and the Hamiltonian $H = \frac{1}{2}\int_{\mathbb{T}}(\pi^2 + \phi_x^2 + m^2\phi^2)\mathrm{d}x$, where $m \geq 0$. Derive the oscillator equations and treat the zero mode at $m = 0$.

(b) Define $\widehat{\phi}_n = (2\pi)^{-1/2}\int e^{-inx}\phi\mathrm{d}x$, and similarly $\widehat{\pi}_n$, for $|n| \leq N$. Express them in the real coordinates and compute their complex bilinear brackets, including $n = 0$.

(c) Compute all nonzero real coefficients and the energy of the data $\phi = 3\cos(2x) - 2\sin x$, $\pi = 5\sin(2x) + \cos x$.

Solution. The elementary integrals of $\cos(nx)\cos(lx)$ and $\sin(nx)\sin(lx)$ are $\pi\delta_{nl}$ for positive indices, while every sine-cosine integral and every nonconstant mean vanish. The constant has squared integral one with the stated normalization. Substituting both expansions into the form therefore gives

$$\Omega_N = \delta q_0 \wedge \delta p_0 + \sum_{n=1}^{N}\sum_{\alpha=c,s}\delta q_{n,\alpha} \wedge \delta p_{n,\alpha}.$$

Spatial differentiation interchanges the sine/cosine directions with coefficients n and $-n$. Thus

$$H_N = \frac{1}{2}(p_0^2 + m^2 q_0^2) + \frac{1}{2}\sum_{n=1}^{N}\sum_{\alpha=c,s}\left[p_{n,\alpha}^2 + (n^2+m^2)q_{n,\alpha}^2\right].$$

Hamilton's equations are $\dot{q} = p$, $\dot{p} = -(n^2+m^2)q$ for each nonzero harmonic, and $\dot{q}_0 = p_0$, $\dot{p}_0 = -m^2 q_0$. At $m = 0$ the constant mode is a free particle: $q_0(t) = q_0(0) + tp_0(0)$, $p_0(t) = p_0(0)$.

For positive n, direct integration gives

$$\widehat{\phi}_n = \frac{q_{n,c} - iq_{n,s}}{\sqrt{2}}, \qquad \widehat{\phi}_{-n} = \frac{q_{n,c} + iq_{n,s}}{\sqrt{2}}, \qquad \widehat{\phi}_0 = q_0,$$

and the same formulas for $\widehat{\pi}$ in terms of p. The real canonical bracket has $\{q_{n,\alpha}, p_{l,\beta}\} = \delta_{nl}\delta_{\alpha\beta}$. Complex bilinear extension gives zero for two equal positive modes, since $1 + (-i)^2 = 0$, and one for opposite modes, since $1 + (-i)i = 2$. Distinct absolute indices give zero, and the constant bracket is one. Hence

$$\{\widehat{\phi}_n, \widehat{\pi}_l\} = \delta_{n+l,0}, \qquad \{\widehat{\phi}_n, \widehat{\phi}_l\} = \{\widehat{\pi}_n, \widehat{\pi}_l\} = 0.$$

No conjugation is inserted into this bilinear bracket.

For the last part the nonzero coefficients are $q_{2,c} = 3\sqrt{\pi}$, $q_{1,s} = -2\sqrt{\pi}$, $p_{2,s} = 5\sqrt{\pi}$ and $p_{1,c} = \sqrt{\pi}$. The kinetic contribution is $26\pi/2$, and the displacement contribution is $[9(4+m^2)+4(1+m^2)]\pi/2$. Their sum is $\pi(33+13m^2/2)$.

Exercise 7.2 (A complex scalar is two real canonical pairs). Write $\Phi = \phi_1 + i\phi_2$, with both real fields Schwartz, and take

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\nabla\Phi|^2 - m^2|\Phi|^2\big), \qquad m \geq 0.$$

Derive the two momenta and Hamiltonian by real variation. Compute all functional derivatives of $Q = \int(\phi_1\pi_2 - \phi_2\pi_1)\mathrm{d}^d x$ and its action on both fields and both momenta. Identify its finite flow, prove $\{Q, H\} = 0$, and verify admissibility in the product of the two real Schwartz phase spaces. Evaluate Q on $(\phi_1, \phi_2; \pi_1, \pi_2) = (f, 2f; -f, 3f)$, where $f \in \mathcal{S}(\mathbb{R}^d; \mathbb{R})$ and $\|f\|_2 = 1$.

Solution. The density is the sum of two real Klein–Gordon densities, each with factor $1/2$. Thus the Legendre maps give $\pi_a = \partial_t\phi_a$, and subtraction gives

$$H = \frac{1}{2}\sum_{a=1}^{2}\int_{\mathbb{R}^d}(\pi_a^2 + |\nabla\phi_a|^2 + m^2\phi_a^2)\mathrm{d}^d x.$$

For the charge, variation gives the four derivatives

$$Q_{\phi_1} = \pi_2, \quad Q_{\phi_2} = -\pi_1, \quad Q_{\pi_1} = -\phi_2, \quad Q_{\pi_2} = \phi_1.$$

Taking brackets in the order $\delta F = \{F, Q\}$ therefore gives

$$\delta\phi_1 = -\phi_2, \quad \delta\phi_2 = \phi_1, \qquad \delta\pi_1 = -\pi_2, \quad \delta\pi_2 = \pi_1.$$

Solving these two rotation systems gives simultaneous multiplication by $e^{i\alpha}$ of the complex field and complex momentum. The charge derivatives are continuous linear Schwartz maps. The Hamiltonian derivatives are $(-\Delta\phi_a + m^2\phi_a, \pi_a)$ as in Lecture 7. Both functionals are admissible in this product class.

For conservation, substitute the derivatives into the bracket to obtain

$$\{Q, H\} = \int\big[\pi_2\pi_1 - \pi_1\pi_2 + \phi_2(-\Delta\phi_1 + m^2\phi_1) - \phi_1(-\Delta\phi_2 + m^2\phi_2)\big]\mathrm{d}^d x = 0.$$

The kinetic and mass terms cancel directly. Integrating each Laplacian term by parts leaves opposite copies of $\nabla\phi_1 \cdot \nabla\phi_2$; Schwartz decay removes the boundary terms. For the supplied data, the charge density is $3f^2 + 2f^2 = 5f^2$, so $Q = 5$. The real normalization $\Phi = \phi_1 + i\phi_2$ was used throughout; no additional $1/\sqrt{2}$ or Wirtinger derivative is needed.

Exercise 7.3 (A smooth window and a sharp interval). In one dimension let $w \in C_c^\infty(\mathbb{R}; \mathbb{R})$ be a fixed window and set

$$F_w[\phi, \pi] = \frac{1}{2}\int_{\mathbb{R}} w(x)\phi_x(x)^2\mathrm{d}x, \qquad F_\#[\phi, \pi] = \frac{1}{2}\int_0^1 \phi_x(x)^2\mathrm{d}x.$$

Compute both derivatives against arbitrary real Schwartz variations. Prove F_w is admissible. Is $F_\#$ an admissible functional on the whole Schwartz phase space? Test the field $\phi(x) = (x + x^2)\eta(x)$, where $\eta \in C_c^\infty(\mathbb{R})$ equals one on a neighbourhood of $[0, 1]$. Display its endpoint distribution terms.

Solution. For the smooth window, differentiation and integration by parts give

$$DF_w[\phi,\pi](h_\phi,h_\pi)=\int_{\mathbb{R}} w\phi_x\partial_x h_\phi \mathrm{d}x=-\int_{\mathbb{R}}\partial_x(w\phi_x)h_\phi \mathrm{d}x.$$

Compact support of w removes the boundary term. Multiplication by a fixed smooth compact function and spatial differentiation are continuous on Schwartz space. Thus the displayed gradient is a smooth linear Schwartz-valued map, proving admissibility.

For the sharp functional the same step instead gives

$$DF_\#[\phi,\pi]h=\phi_x(1)h_\phi(1)-\phi_x(0)h_\phi(0)-\int_0^1\phi_{xx}h_\phi \mathrm{d}x.$$

Consequently its distributional field derivative is $-\mathbf{1}_{(0,1)}\phi_{xx}+\phi_x(1)\delta_1-\phi_x(0)\delta_0$; its momentum derivative is zero. At the chosen field, $\phi_x(0)=1$, $\phi_x(1)=3$ and $\phi_{xx}=2$ on the interval. Take smooth test bumps of height one at 1, supported in intervals of radius ε around 1, with uniformly bounded magnitude and zero value at 0. The derivative tends to 3 as $\varepsilon\downarrow 0$. Pairing any fixed smooth candidate gradient with these bumps tends to zero because their supports have length 2ε. This contradiction excludes a Schwartz gradient there. The quadratic functional itself is smooth on Schwartz space, but smoothness alone is weaker than Lecture 7's admissibility requirement.

Exercise 7.4 (The sign of the Hamiltonian commutator). For admissible F,G, let $H_F=D(\operatorname{grad}F)$ and $\mathbb{J}(a,b)=(b,-a)$ as in Lecture 7. Define the vector-field bracket by

$$[X,Y](z)=DY(z)X(z)-DX(z)Y(z),$$

the sign for which it acts on functionals as $X(YK)-Y(XK)$. Use the actual bracket-gradient formula of Lecture 7 to prove $X_{\{F,G\}}=-[X_F,X_G]$. For a nonzero example, choose real $f\in\mathcal{S}$ with $\|f\|_2=1$, put $q=\int f\phi$, $p=\int f\pi$, and take $F=q^2/2$, $G=p^2/2$. Compute both vector fields, their commutator, and $X_{\{F,G\}}$.

Solution. Write $a=\operatorname{grad}F$ and $b=\operatorname{grad}G$. Lecture 7's closure formula, proved using the symmetric Hessians, is

$$\operatorname{grad}\{F,G\}=H_F\mathbb{J}b-H_G\mathbb{J}a.$$

Since $\mathbb{J}$ is constant, $DX_F=\mathbb{J}H_F$ and $DX_G=\mathbb{J}H_G$. Therefore

$$[X_F,X_G]=\mathbb{J}H_G\mathbb{J}a-\mathbb{J}H_F\mathbb{J}b=-\mathbb{J}\operatorname{grad}\{F,G\}=-X_{\{F,G\}}.$$

Every direction and every output remains Schwartz by admissibility; no extension to arbitrary smooth functionals is being assumed.

For the example, the normalization gives $\{q,p\}=\int f^2=1$. The gradients and vectors are

$$\operatorname{grad}F=(qf,0),\quad X_F=(0,-qf),\qquad \operatorname{grad}G=(0,pf),\quad X_G=(pf,0).$$

The bracket is $\{F,G\}=qp$, whose vector is $(qf,-pf)$. Differentiating the displayed vectors gives $DX_G[X_F]=(-qf,0)$ and $DX_F[X_G]=(0,-pf)$. Their difference is $(-qf,pf)$, as claimed. At the allowed datum $(\phi,\pi)=(f,2f)$ it is $(-f,2f)\neq 0$, so this example detects an incorrect plus sign. The sign agrees with $X_FK=\{K,F\}$.

Classical Field Theory — Exercise Sheet 8

Slices, boundary conditions and causal brackets

Solutions

For real Klein–Gordon variations with compact smooth Cauchy data, use $\omega^\mu(h_1, h_2) = h_1\partial^\mu h_2 - h_2\partial^\mu h_1$. The future-oriented graph element is $(1, -\nabla\tau)\mathrm{d}^d x$. All fields below are smooth. The last problem explicitly restricts the field equation to the massless wave in $3+1$ dimensions.

Exercise 8.1 (An unbounded Cauchy graph). Let $\tau(x) = t_* + v \cdot x$ on $\mathbb{R}^d$, where $t_* \in \mathbb{R}$ and $v \in \mathbb{R}^d$, $|v| < 1$. Let $h_1, h_2 \in \mathcal{S}_m$ be the actual compact-Cauchy-data solutions of Lecture 8, $m \geq 0$.

(a) Prove that the graph is spacelike and meets each inextendible timelike curve exactly once. A future timelike curve can be parametrized by coordinate time with $|\dot{x}(t)| < 1$; justify that its time range is all of $\mathbb{R}$. Here an inextendible curve has no past or future endpoint in Minkowski spacetime.

(b) Although τ is unbounded, prove that the current is supported in a compact subset of the region between the graph and $t = 0$. Give an explicit spatial bound in terms of t_*, $|v|$, and a radius R containing both sets of initial data. Use signed time integration to prove $\Omega_{\Sigma_\tau}(h_1, h_2) = \Omega_0(h_1, h_2)$, including where the graph crosses zero.

(c) Choose nonzero real $f \in C_c^\infty(\mathbb{R}^d)$ and solutions with data $(f, 0)$ and $(0, f)$. Determine their pairing on this graph.

For comparison, the following general result is supplied: in full Minkowski spacetime, the same compact-data solutions have finite, equal symplectic integrals on every smooth spacelike Cauchy surface with future orientation. It uses spacelike compact support, not just current closure. See Khavkine, §3.3.3, Lemma 3.11, p. 41, arXiv:1402.1282. Your proof here should establish the affine case directly.

Solution. A graph tangent $(v \cdot a, a)$ has Minkowski square $(v \cdot a)^2 - |a|^2 < 0$ for $a \neq 0$, so the graph is spacelike. On a future timelike curve parametrized by t,

$$\frac{\mathrm{d}}{\mathrm{d}t}(t - t_* - v \cdot x(t)) = 1 - v \cdot \dot{x}(t) \geq 1 - |v| > 0.$$

If the time range had a finite upper endpoint, the speed bound would make $x(t)$ Cauchy there. Its finite limit would allow a continuation by a timelike curve, contradicting inextendibility. The same argument applies at the lower endpoint. Thus the time range is $\mathbb{R}$ and the displayed function runs from $-\infty$ to $+\infty$, with exactly one zero.

Finite propagation gives support of both solutions and their derivatives inside $|x| \leq R + |t|$. Between 0 and $\tau(x)$, one also has $|t| \leq |t_*| + |v||x|$. Where the current can be nonzero this implies

$$(1 - |v|)|x| \leq R + |t_*|.$$

The spatial bound also bounds t, so the relevant closed support in the comparison region is compact. In particular both graph and flat integrals are finite.

Define $A^i(x) = \int_0^{\tau(x)} \omega^i(t, x)\mathrm{d}t$ with its signed orientation. The support bound makes A^i vanish outside a fixed ball. The product rule, FTC and $\partial_\mu \omega^\mu = 0$ give

$$\omega^0(\tau(x), x) - \omega^i(\tau(x), x)v_i - \omega^0(0, x) = -\partial_i A^i(x).$$

Integrating in a large rectangle gives zero on the right by coordinate FTC. The left uses the future graph element $(1, -v)$, proving the equality even when τ changes sign. For the final data the flat pairing is $\int [f \cdot f - 0 \cdot 0] \mathrm{d}^d x = \|f\|_2^2 > 0$, so the same value holds on the affine graph. This direct proof did not assume the supplied general theorem.

Exercise 8.2 (Common boundary conditions remove side flux). Let two real solutions of $(\partial_t^2 - \partial_x^2 + m^2)h = 0$ be smooth on a neighbourhood of $I \times [0, L]$, where I is an open interval, $L > 0$ and $m \geq 0$. Set $\Omega_{[0,L]}(h_1, h_2) = \int_0^L (h_1 h_{2,t} - h_2 h_{1,t}) \mathrm{d}x$. At each endpoint, consider the following common conditions on both solutions: Dirichlet $h = 0$, Neumann $\partial_n h = 0$, or real Robin $\partial_n h + \kappa h = 0$, with the same prescribed real κ for both solutions at that endpoint. The two endpoints may have different such conditions.

(a) Prove that each choice makes the side symplectic flux zero and conserves $\Omega_{[0,L]}$. Specify the outward derivative at each endpoint.

(b) For $m = 0$, $L = \pi$, compare $h_D(t, x) = \sin x \cos t$ and $h_N(t, x) = 1$. Each obeys one of the listed conditions, but they obey different conditions. Compute their current, pairing and net outward side flux. Is individual satisfaction of some listed condition sufficient?

Solution. The outward derivatives are $\partial_n = -\partial_x$ at zero and $\partial_n = +\partial_x$ at L. Since $\omega^1 = -h_1 h_{2,x} + h_2 h_{1,x}$, its outward contraction at either endpoint is

$$n\omega^1 = -(h_1 \partial_n h_2 - h_2 \partial_n h_1).$$

Common Dirichlet conditions kill both undifferentiated factors. Common Neumann conditions kill both normal derivatives. For common Robin conditions, substituting $\partial_n h_j = -\kappa h_j$ makes the two terms equal and opposite. Thus each side flux is zero. Integrating the current divergence over the interval gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \Omega_{[0,L]} = -[\omega^1(t, L) - \omega^1(t, 0)] = 0.$$

For the second part, h_D solves the wave equation and vanishes at both endpoints, while h_N solves it and has zero normal derivative. Their ordered current is

$$\omega^0 = \sin x \sin t, \qquad \omega^1 = \cos x \cos t.$$

Integration gives $\Omega = 2 \sin t$. The net outward flux is $\omega^1(\pi) - \omega^1(0) = -2 \cos t$, whose negative is the derivative of the pairing. A Dirichlet solution paired with a Neumann solution need not have zero boundary current. Common boundary conditions, rather than separate choices for each solution, supply the cancellation.

Exercise 8.3 (A causal source bracket and its sign). In $3+1$ dimensions let G_{ret} be Lecture 6's massless kernel, let $G_{\mathrm{adv}}(t,x) = G_{\mathrm{ret}}(-t,x)$, and put $E = G_{\mathrm{adv}} - G_{\mathrm{ret}}$. For real $f, g \in C_c^\infty(\mathbb{R}^{1+3})$, define on the massless compact-data solution space

$$F_f(\phi) = \int_{\mathbb{R}^{1+3}} f\phi \, \mathrm{d}t\mathrm{d}^3x, \qquad Ef = (G_{\mathrm{adv}} - G_{\mathrm{ret}}) * f.$$

The identification of the resulting causal bracket with the linear *Peierls bracket* may be taken as given; see Khavkine, §§3.3.4–3.3.5, Theorem 3.2, and the scalar case §4.1, arXiv:1402.1282. The signs in this problem must be fixed by the following calculation, using the course convention $\iota_{X_F}\Omega = DF$.

(a) Show that Ef is a smooth homogeneous solution with compact smooth data on every constant slice. Apply the sourced current balance to $G_{\mathrm{ret}} * f$ and a homogeneous compact-data solution h to prove $\Omega(Ef, h) = \int fh$. Identify X_{F_f} and derive the bracket $\{F_f, F_g\} = \int f(Eg)$. Prove its antisymmetry and its vanishing when the two test supports are spacelike separated.

(b) As a separate finite-dimensional sign test, take one oscillator $\ddot{q} + \nu^2 q = 0$, $\nu \geq 0$, with $\{q(0), p(0)\} = 1$ and $p = \dot{q}$. Compute $\{q(t), q(s)\}$ and $\{q(t), p(s)\}$, using continuous limits at $\nu = 0$. Recover the equal-time canonical sign. Explain why this finite model does not make point evaluation of a field admissible in Lecture 7.

Solution. Both convolutions are smooth sourced solutions by Lecture 6 and time reflection. Their difference is homogeneous. If the source lies in $[a,b] \times B_R$, the retarded contribution on a fixed slice is spatially supported in a ball of radius at most $R + \max(t-a, 0)$ and is zero for $t < a$. The advanced counterpart has the analogous radius $R + \max(b-t, 0)$ and is zero for $t > b$. The same is true of all derivatives. Thus Ef has compact smooth Cauchy data and belongs to the actual tangent space.

Let $r_f = G_{\mathrm{ret}} * f$ and let h be homogeneous. The massless current satisfies

$$\partial_\mu \omega^\mu(r_f, h) = r_f \Box h - h \Box r_f = -hf.$$

Choose constant slices $t_- < a$ and $t_+ > b$. Compact support on this bounded time interval removes every side term in a sufficiently large rectangle. The initial current is zero. Its integrated balance is

$$\Omega_{t_+}(r_f, h) = -\int fh \, \mathrm{d}t\mathrm{d}^3x.$$

At the late slice the advanced field is zero, so $Ef = -r_f$. Lecture 8's slice independence for the homogeneous pair then gives $\Omega(Ef, h) = \int fh = DF_f(h)$ on every constant slice. Nondegeneracy identifies $X_{F_f} = Ef$. The bracket order is therefore

$$\{F_f, F_g\} = DF_f(X_{F_g}) = \int f(Eg) = \Omega(Ef, Eg) = -\Omega(Eg, Ef) = -\{F_g, F_f\}.$$

If every event in the first compact support is spacelike separated from every event in the second, the retarded and advanced support properties make Eg zero on the support of f. The bracket vanishes. This proves the displayed linear bracket identity; the supplied Peierls identification is not a theorem here about arbitrary nonlinear theories.

For the finite oscillator, write

$$q(t) = \cos(\nu t) q(0) + \frac{\sin(\nu t)}{\nu} p(0).$$

The canonical two-variable bracket gives

$$\{q(t), q(s)\} = \frac{\cos(\nu t)\sin(\nu s) - \sin(\nu t)\cos(\nu s)}{\nu} = \frac{\sin(\nu(s-t))}{\nu}.$$

Differentiating with respect to s gives $\{q(t), p(s)\} = \cos(\nu(s-t))$. For $\nu = 0$ the solution is $q(0)+tp(0)$ and the same brackets are $s-t$ and 1. This agrees with $E(t-s) = -\sin(\nu(t-s))/\nu$ for the advanced-minus-retarded oscillator response. Here $q(t)$ is a smooth function of two real coordinates. Field point evaluation instead has a delta derivative, excluded from the Schwartz-gradient class of Lecture 7.

Classical Field Theory — Exercise Sheet 9

Gauge identities and source conditions

Solutions

Use $A_{\mathrm{em}} = -\phi_{\mathrm{em}}\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}x$, $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t\mathbf{A}$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$. Unless a weaker class is explicitly discussed, fields and coefficients are smooth. Compact support is stated separately in each problem.

Exercise 9.1 (The adjoint differentiates coefficients too). Let a smooth first-order density $L(x,\phi,\partial\phi)$ have Euler expressions E_a on an open coordinate region $U \subset \mathbb{R}^{1+d}$. Vary its action on a compact region $D \subset U$, with every parameter supported in the interior of D. For smooth parameters ϵ^α, consider the finite-order variation

$$\delta_\epsilon\phi^a = \sum_{|I|\le r} R^{aI}_\alpha(x, j^s\phi)\partial_I\epsilon^\alpha.$$

Here I is a derivative multi-index, each multi-index occurs once in the sum, and $j^s\phi$ denotes the field and all its derivatives through finite order s. The coefficients are smooth functions of these arguments. Assume the action is invariant under these variations for every compact parameter and every smooth field, without imposing the field equations.

(a) Derive the identity satisfied by the Euler expressions, including the variable coefficients. For the one-dimensional test operator $R\epsilon = (1+x^2)\epsilon_x + x\epsilon$, display R^*E explicitly. This test asks for its adjoint, not for a claim that every density has that symmetry.

(b) For Maxwell's transformation $\delta\phi_{\mathrm{em}} = -\epsilon_t$, $\delta\mathbf{A} = \nabla\epsilon$, derive the identity for the scalar and vector Euler expressions. With $\mathcal{M}_0 = \operatorname{div}\mathbf{E}$ and $\boldsymbol{\mathcal{M}} = \partial_t\mathbf{E} - \operatorname{curl}\mathbf{B}$, include a prescribed source $(\rho,\mathbf{J})$ and identify its necessary conservation condition. Explain the distributional interpretation for only C^2 potentials.

Solution. The parameter and all its derivatives have compact support, so the field variation does too. Lecture 1's compact-variation formula gives

$$0 = \delta_\epsilon S = \int \sum_{a,I} E_a R^{aI}_\alpha \partial_I\epsilon^\alpha \mathrm{vol} = \int \sum_{a,I} (-1)^{|I|}\partial_I(E_a R^{aI}_\alpha)\epsilon^\alpha \mathrm{vol}.$$

Every boundary term vanishes by compact support. The fundamental lemma gives one identity for each α. These derivatives act after the coefficient has been evaluated on the field, including its implicit coordinate dependence through the field derivatives. For the test operator, ordinary integration by parts gives

$$R^*E = -\partial_x((1+x^2)E) + xE = -(1+x^2)E_x - xE.$$

Dropping the derivative of $1+x^2$ would leave the incorrect coefficient $+xE$ instead of $-xE$.

For Maxwell the compact variation pairs as $\int[-E_\phi\epsilon_t + E_A\cdot\nabla\epsilon]\mathrm{vol}$. Its adjoint is $\partial_t E_\phi - \operatorname{div}E_A$. Lecture 9's density gives

$$E_\phi = \mathcal{M}_0 - \rho, \qquad E_A = \boldsymbol{\mathcal{M}} + \mathbf{J}.$$

The source-free identity follows directly from commuting derivatives: $\partial_t\mathcal{M}_0 - \operatorname{div}\boldsymbol{\mathcal{M}} = 0$. Consequently

$$\partial_t E_\phi - \operatorname{div}E_A = -\rho_t - \operatorname{div}\mathbf{J}.$$

This vanishes identically exactly for a conserved source. For a C^2 potential the Euler expressions are continuous but need not have classical first derivatives. Pair the identity with compact smooth tests and move derivatives onto those tests. Distributional mixed derivatives commute, which proves the same identity without assuming third derivatives of the potential.

Exercise 9.2 (Polarization and magnetization). Let $P, M \in C_c^\infty(\mathbb{R}^{1+3}; \mathbb{R}^3)$ and define

$$\rho = -\operatorname{div} P, \qquad \mathbf{J} = \partial_t P + \operatorname{curl} M.$$

Prove conservation, zero total charge on each slice and $\int y\rho(t,y)\mathrm{d}^3y = \int P(t,y)\mathrm{d}^3y$. For any smooth potential, rewrite the integrated source coupling $\int(-\rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A})\mathrm{vol}$ in terms of P, M and the fields, displaying why all discarded terms vanish. Does this construction assert that the potential of this source is unique?

Solution. Differentiation and commutation of spatial derivatives give

$$\rho_t + \operatorname{div}\mathbf{J} = -\operatorname{div} P_t + \operatorname{div} P_t + \operatorname{div}\operatorname{curl} M = 0.$$

Compact spatial support removes every face integral, so $\int \rho = 0$. For its first moment, integrating each coordinate derivative gives

$$\int y_i\rho(t,y)\mathrm{d}^3y = -\int y_i\partial_{y_j}P_j(t,y)\mathrm{d}^3y = \int P_i(t,y)\mathrm{d}^3y.$$

Substitution in the coupling and integration by parts give

$$\begin{aligned}\int(-\rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A})\mathrm{vol} &= \int[(\operatorname{div}P)\phi_{\mathrm{em}} + P_t\cdot\mathbf{A} + (\operatorname{curl}M)\cdot\mathbf{A}]\mathrm{vol}\\ &= \int[-P\cdot\nabla\phi_{\mathrm{em}} - P\cdot\partial_t\mathbf{A} + M\cdot\operatorname{curl}\mathbf{A}]\mathrm{vol}\\ &= \int(P\cdot\mathbf{E} + M\cdot\mathbf{B})\mathrm{vol}.\end{aligned}$$

The spatial face terms contain P or M; the time-face term contains P. All vanish outside a fixed compact set, regardless of the smooth potential's behaviour at infinity. The curl sign also follows from $\operatorname{div}(M\times\mathbf{A}) = \mathbf{A}\cdot\operatorname{curl}M - M\cdot\operatorname{curl}\mathbf{A}$. The final expression depends only on the fields and is gauge invariant. It supplies a family of compatible sources, not a uniqueness theorem for potentials: adding $\mathrm{d}\chi$ still leaves their fields unchanged.

Exercise 9.3 (Vacuum duality with the Lorentzian star). For a vacuum Maxwell field, put $F' = \star F_{\mathrm{em}}$, using exactly the Hodge table of Lecture 9 and $\star^2 = -1$ on two-forms. Find the transformed electric and magnetic vectors, and verify all four vacuum component equations. Determine the effect of two and four successive transformations. Check the energy and Poynting densities for the constant vacuum fields $\mathbf{E} = (1,0,2)$, $\mathbf{B} = (0,3,0)$ before and after the map. These are densities, not finite total energies on $\mathbb{R}^3$. Does the same transformation preserve Maxwell's equations with a prescribed nonzero electric current three-form $\mathcal{J}$ and no magnetic source?

Solution. The Hodge table gives $\star(\mathrm{d}x^i\wedge\mathrm{d}t)$ equal to the corresponding spatial magnetic wedge, and the star of that wedge is $\mathrm{d}t\wedge\mathrm{d}x^i = -\mathrm{d}x^i\wedge\mathrm{d}t$. Thus $E'_i = -B_i$ and $B'_i = E_i$. The transformed divergences vanish because $\operatorname{div}\mathbf{E}' = -\operatorname{div}\mathbf{B} = 0$ and $\operatorname{div}\mathbf{B}' = \operatorname{div}\mathbf{E} = 0$. The two evolution equations become

$$\partial_t\mathbf{B}' + \operatorname{curl}\mathbf{E}' = \partial_t\mathbf{E} - \operatorname{curl}\mathbf{B} = 0, \qquad \operatorname{curl}\mathbf{B}' - \partial_t\mathbf{E}' = \operatorname{curl}\mathbf{E} + \partial_t\mathbf{B} = 0.$$

Applying the map twice changes the sign of both fields and four times restores them.

In the example the transformed fields are $(0,-3,0)$ and $(1,0,2)$. Both squared-magnitude sums are 14, and direct cross products give

$$\frac{1}{2}(|\mathbf{E}|^2+|\mathbf{B}|^2)=7, \qquad \mathbf{E}\times\mathbf{B}=(-6,0,3)=\mathbf{E}'\times\mathbf{B}'.$$

For a prescribed electric source, $\mathrm{d}F_{\mathrm{em}}=0$ but $\mathrm{d}{\star}F_{\mathrm{em}}=\mathcal{J}$. Hence $\mathrm{d}F'=\mathcal{J}$ and $\mathrm{d}{\star}F'=-\mathrm{d}F_{\mathrm{em}}=0$. These are not the original equations with the same nonzero electric source; the transformed Bianchi equation contains a magnetic-source term.

Exercise 9.4 (Conservation does not remove a time-face term). Let $D=[t_0,t_1]\times\mathbb{T}^3$, $t_1>t_0$, and use the Maxwell density $L=\frac{1}{2}(|\mathbf{E}|^2-|\mathbf{B}|^2)-\rho\phi_{\mathrm{em}}+\mathbf{J}\cdot\mathbf{A}$. Take smooth periodic fields and a prescribed smooth conserved source. Allow a smooth periodic gauge function $\chi(t,x)$ with nonzero time-face values.

(a) Find the exact action change, retaining the time faces and explaining the cancellation on opposite spatial faces.

(b) Set $\rho=\sin x^1$, $\mathbf{J}=0$, $\phi_{\mathrm{em}}=\sin x^1$, $\mathbf{A}=0$, and $\chi=\gamma t\sin x^1$, where real $\gamma\neq 0$. Verify Maxwell's equations before and after the transformation and compute the nonzero action change. Why is this consistent with Lecture 9's compact-parameter gauge invariance?

Solution. A gauge change leaves the two field strengths fixed, so only the source coupling changes. Its exact difference is

$$\Delta S=\int_D(\rho\chi_t+\mathbf{J}\cdot\nabla\chi)\mathrm{vol}=\int_D[\partial_t(\rho\chi)+\mathrm{div}(\chi\mathbf{J})]\mathrm{vol},$$

where conservation removed the remaining bulk term. Periodicity cancels opposite spatial faces, leaving

$$\Delta S=\left[\int_{\mathbb{T}^3}\rho(t,x)\chi(t,x)\mathrm{d}^3x\right]_{t_0}^{t_1}.$$

In the example, the static electric field is $(-\cos x^1,0,0)$ and its divergence is $\sin x^1=\rho$. The magnetic field and spatial current are zero; all remaining Maxwell equations hold. The source is conserved and neutral. After the gauge change the potentials are

$$\phi'_{\mathrm{em}}=(1-\gamma)\sin x^1, \qquad \mathbf{A}'=(\gamma t\cos x^1,0,0).$$

Their electric contributions add to the same $-\cos x^1$, and their curl is zero. Thus the same Maxwell equations hold. Finally,

$$\Delta S=\gamma(t_1-t_0)\int_{\mathbb{T}^3}\sin^2 x^1\mathrm{d}^3x=4\pi^3\gamma(t_1-t_0)\neq 0.$$

Lecture 9 imposed compact support of the gauge parameter away from every boundary. The present parameter has different nonzero face data, so that hypothesis is absent. Local field invariance and source conservation alone do not remove this boundary charge pairing.

Classical Field Theory — Exercise Sheet 10

Residual gauges and charged tails

Solutions

Write $\Box = \partial_t^2 - \Delta$. Use $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t\mathbf{A}$, $\mathbf{B} = \operatorname{curl}\mathbf{A}$ and $\ell = -(\partial_t\phi_{\mathrm{em}} + \operatorname{div}\mathbf{A})$. Lecture 10's theorem assumes a smooth conserved source of compact *spacetime* support and compatible compact smooth potential data. Problems 2–3 explicitly separate a charged background outside that class.

Exercise 10.1 (All residual shifts of Cauchy data). Let a smooth Lorenz-gauge Maxwell potential have initial data $(a_\phi, b_\phi, a_A, b_A)$ as in Lecture 10, with a fixed source ρ. Let χ be the smooth wave with arbitrary real compact smooth data $\chi(0) = p$, $\chi_t(0) = q$. Set $\phi'_{\mathrm{em}} = \phi_{\mathrm{em}} - \chi_t$, $\mathbf{A}' = \mathbf{A} + \nabla\chi$.

(a) Find the increments of all four potential data. Verify directly that both compatibility expressions $b_\phi + \operatorname{div} a_A$ and $-\Delta a_\phi - \operatorname{div} b_A - \rho(0)$ are unchanged. Compare the two initial electromagnetic fields.

(b) Prove that fixing all four potential data rules out every nonzero residual gauge wave in this compact-data class. Contrast this with fixing only the electromagnetic fields. If compactness of the gauge data is dropped, explain why a constant χ is physically trivial.

Solution. The wave equation gives $\chi_{tt}(0) = \Delta p$. Evaluating the potential transformation and its first time derivative gives

$$a'_\phi = a_\phi - q, \quad b'_\phi = b_\phi - \Delta p, \qquad a'_A = a_A + \nabla p, \quad b'_A = b_A + \nabla q.$$

The first compatibility increment is $-\Delta p + \operatorname{div}\nabla p = 0$. The second increment is $\Delta q - \operatorname{div}\nabla q = 0$; the prescribed source is fixed. The electromagnetic data satisfy

$$\mathbf{E}'_0 = -\nabla(a_\phi - q) - (b_A + \nabla q) = \mathbf{E}_0, \qquad \mathbf{B}'_0 = \operatorname{curl}(a_A + \nabla p) = \mathbf{B}_0.$$

The full fields also agree by the gauge identity of Lecture 9, and Lorenz gauge persists because $\ell' = \ell + \Box\chi = 0$.

If all four potential data coincide, the first and third increments imply $q = 0$ and $\nabla p = 0$. A smooth function with zero gradient on connected $\mathbb{R}^3$ is constant; compact support forces that constant to be zero. Lecture 4 then gives the zero wave. By contrast, arbitrary compact p, q produce transformations with identical electromagnetic fields, many of which have nonzero potential increments. Without compactness, a constant χ changes no potential since every component of $\mathrm{d}\chi$ is zero. These distinctions concern uniqueness of potentials versus uniqueness of fields.

Exercise 10.2 (A smooth charge with a Coulomb tail). Choose a radial nonnegative $b \in C_c^\infty(\mathbb{R}^3)$ with $\int b \mathrm{d}^3x = 1$ and support in $\overline{B_R(0)}$, $R > 0$. Fix real $Q \neq 0$ and prescribe the static source $\rho_Q(x) = Qb(x)$, $\mathbf{J} = 0$ for all time. For $r > 0$ define, writing $\rho_Q(s)$ for its radial profile,

$$M_Q(r) = 4\pi \int_0^r s^2 \rho_Q(s) \mathrm{d}s, \quad E_r(r) = \frac{M_Q(r)}{4\pi r^2}, \quad \phi_Q(r) = \int_r^\infty E_r(s) \mathrm{d}s.$$

(a) Prove that $E_Q(x) = E_r(|x|)x/|x|$ and $\phi_Q(|x|)$ extend smoothly across zero. Prove Gauss's law there as well as away from zero. Show that $(\phi_{\mathrm{em}}, \mathbf{A}) = (\phi_Q, 0)$ solves Maxwell in Lorenz gauge.

(b) Determine the exact potential and field for $r \geq R$. Prove $\phi_Q \notin L^2$, hence $\phi_Q \notin H^1$, while $\nabla\phi_Q \in L^2$ and the field energy is finite. Calculate the electric energy outside B_R.

(c) Identify the failed hypotheses of Lecture 10's theorem. Why does conservation alone not force $Q = 0$ for this source?

Solution. Changing variables $s = ar$ in M_Q gives, for $x \neq 0$,

$$E_Q(x) = x \int_0^1 a^2 \rho_Q(ax) \mathrm{d}a.$$

The right side is smooth for every x, by differentiating under a compact parameter integral, and is $O(|x|)$ near zero. It supplies the extension. Taking its divergence gives an identity valid also at zero:

$$\operatorname{div} E_Q = \int_0^1 [3a^2 \rho_Q(ax) + a^3 x \cdot \nabla\rho_Q(ax)] \mathrm{d}a = \int_0^1 \partial_a [a^3 \rho_Q(ax)] \mathrm{d}a = \rho_Q(x).$$

The potential integral converges at infinity since M_Q is constant there, and at zero since $E_r(r) = O(r)$. Its radial derivative is $-E_r$. More explicitly, its smooth extension can be written

$$\phi_Q(x) = \phi_Q(0) - \int_0^1 E_Q(\tau x) \cdot x \mathrm{d}\tau.$$

This equals the radial integral for $x \neq 0$, so it is a smooth extension and $\nabla\phi_Q = -E_Q$ there and, by continuity, at zero. The field is static and has zero curl; $\mathbf{A} = 0$ gives zero magnetic field. Gauss's law was proved above, Ampère's law has both sides zero, and the remaining two equations follow from the potential. Its Lorenz function is $-(\partial_t \phi_Q + \operatorname{div} 0) = 0$.

For $r \geq R$, normalization gives $M_Q(r) = Q$. Hence

$$E_r(r) = \frac{Q}{4\pi r^2}, \qquad \phi_Q(r) = \frac{Q}{4\pi r}.$$

The potential's exterior squared norm contains $Q^2 \int_R^\infty \mathrm{d}r/(4\pi) = \infty$. The gradient's exterior squared norm is instead

$$\int_{|x|>R} |\nabla\phi_Q|^2 \mathrm{d}^3x = \frac{Q^2}{4\pi} \int_R^\infty r^{-2} \mathrm{d}r = \frac{Q^2}{4\pi R}.$$

Smoothness on the interior ball makes its remaining gradient integral finite. The electric energy is half that norm squared, so its exterior part is $Q^2/(8\pi R)$ and its total field energy is finite.

The source persists for all time and the scalar potential data are neither compactly supported nor in H^1. Lecture 10's compact-source/data theorem therefore does not apply.

Conservation still makes the total charge constant, but there is no early time at which this static source vanishes. The step that forced neutrality for a compact spacetime source is absent.

Exercise 10.3 (An affine charged evolution class). Keep the background of Exercise 10.2. Let $(\rho_1, J_1) \in C_c^\infty(\mathbb{R}^{1+3}; \mathbb{R} \times \mathbb{R}^3)$ be conserved. Give compact smooth perturbation data $(a_{1\phi}, b_{1\phi}, a_{1A}, b_{1A})$ satisfying

$$b_{1\phi} + \operatorname{div} a_{1A} = 0, \qquad -\Delta a_{1\phi} - \operatorname{div} b_{1A} = \rho_1(0).$$

Construct a smooth Lorenz-gauge solution for total source $(\rho_Q + \rho_1, J_1)$ and total initial data obtained by adding the static background. Prove its total charge is Q, its electric and magnetic fields have finite energy on each slice, and $\phi_{\mathrm{em}}(t) \in \phi_Q + H^1(\mathbb{R}^3)$ while $\phi_{\mathrm{em}}(t) \notin H^1$. State the uniqueness conclusion actually supplied by Lecture 10. Explain why this construction does not put the charged background in a Schwartz canonical phase space.

Solution. The compact source and the displayed perturbation conditions are exactly the hypotheses of Theorem 10.2. It supplies a smooth global perturbation (ϕ_1, A_1) in Lorenz gauge. Its construction is a retarded compact-source field plus a homogeneous wave with compact corrected data. Lecture 6's support bound and Lecture 5's finite propagation put that perturbation and all its derivatives in a common spatial ball on each bounded time interval. In particular, on each slice $\phi_1 \in H^1$ and its electromagnetic fields are square integrable.

Maxwell's equations and the Lorenz condition are linear. Adding the fixed background gives

$$\phi_{\mathrm{em}} = \phi_Q + \phi_1, \qquad \mathbf{A} = A_1, \qquad \mathbf{E} = E_Q + E_1, \qquad \mathbf{B} = B_1.$$

These have the total source and the prescribed total data. The compact conserved perturbation current has zero total charge by Lecture 10's early-time argument. Thus $\int (\rho_Q + \rho_1) = Q$ at every time. Both electric summands and the magnetic field are in L^2, so the total field energy is finite, including their integrable electric cross term.

Since $\phi_1(t) \in H^1$, the scalar potential lies in $\phi_Q + H^1$. If it also lay in H^1, subtraction of ϕ_1 would put ϕ_Q in H^1, contradicting the preceding problem. Subtracting the fixed smooth background from any other C^2 Lorenz-gauge Maxwell solution with these total potential data gives a perturbation covered by Lecture 10's uniqueness conclusion. Thus the total potential is unique in that class with those full data. Residual transformations with different potential data remain possible as in Problem 1. This is a concrete construction around one fixed background, not a general existence theorem for arbitrary charged tails. Its potential has a $1/r$ tail and is not Schwartz; no Schwartz field bracket on that background has been established by adding the two solutions.

Exercise 10.4 (Local potentials with no global real one-form). On the oriented unit sphere, with area form $\sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi$, let $g_{\mathrm{mag}} \neq 0$ be real. On $U_N = S^2 \setminus \{\text{south pole}\}$ and $U_S = S^2 \setminus \{\text{north pole}\}$, respectively, take

$$A_N = g_{\mathrm{mag}}(1 - \cos\theta)\mathrm{d}\varphi, \qquad A_S = -g_{\mathrm{mag}}(1 + \cos\theta)\mathrm{d}\varphi.$$

These are spatial real one-form potentials; the task uses exterior calculus from *Topology and Geometry*, Part I, not bundle theory as a prerequisite.

(a) Verify smoothness at each included pole, compute the common curvature $F = \mathrm{d}A_N = \mathrm{d}A_S$, and find its total flux.

(b) Find a gauge function on a simply connected azimuthal sector in the overlap. Prove that there is no single-valued real gauge function for this shift on the whole overlap, and no globally smooth real one-form A on S^2 with $\mathrm{d}A = F$.

The interpretation by nontrivial bundles and single-valued charged transition phases, including the allowed flux quantization, is a later topic: *Topology and Geometry*, Lectures 20–22. No bundle construction or quantization theorem is assumed in this exercise.

Solution. Away from the coordinate poles, $\mathrm{d}\varphi = (X\mathrm{d}Y - Y\mathrm{d}X)/(X^2 + Y^2)$ and $X^2 + Y^2 = (1 - Z)(1 + Z)$. Thus

$$A_N = \frac{g_{\mathrm{mag}}}{1 + Z}(X\mathrm{d}Y - Y\mathrm{d}X), \qquad A_S = -\frac{g_{\mathrm{mag}}}{1 - Z}(X\mathrm{d}Y - Y\mathrm{d}X).$$

The first expression is smooth on U_N, including the north pole; the second is smooth on U_S, including the south pole. Their exterior derivatives in the angular region agree and extend to the smooth two-form

$$F = g_{\mathrm{mag}} \sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi, \qquad \int_{S^2} F = 4\pi g_{\mathrm{mag}}.$$

On a sector where φ is single-valued, $A_N - A_S = 2g_{\mathrm{mag}}\mathrm{d}\varphi = \mathrm{d}\chi_{NS}$ with $\chi_{NS} = 2g_{\mathrm{mag}}\varphi$. On the full overlap, integrate the difference around the positively oriented equator. Its integral is $4\pi g_{\mathrm{mag}} \neq 0$. An exact one-form of a single-valued real function has zero integral on a closed curve, so no such global overlap function exists. If a global smooth real A on S^2 had $\mathrm{d}A = F$, Stokes on the closed sphere would give $\int F = 0$, again a contradiction. Local smooth potentials can therefore describe a curvature which has no single global real one-form potential. Bundle transition phases require additional structure and consistency conditions, left to the stated later course.

Classical Field Theory — Exercise Sheet 11

Constraints and boundaries

Solutions

Use Lecture 11's canonical bracket, with $\mathbf{\Pi} = -\mathbf{E}$. A primary constraint is imposed only after forming the Hamilton equations. All field data below are real and smooth. The second problem uses Schwartz data on $\mathbb{R}^3$; the third explicitly changes the spatial region and its allowed boundary variations.

Exercise 11.1 (An entire constraint chain). Consider two mechanical coordinates $q(t)$ and $u(t)$ with Lagrangian

$$L(q, u, \dot{q}, \dot{u}) = \frac{1}{2}(\dot{q} - u)^2.$$

Use the following consistency procedure: find the image of the Legendre map, form the canonical Hamiltonian on that image, add an arbitrary multiplier for each primary constraint, and preserve the constraints until no new condition appears. The bracket on $(q, u; p_q, p_u)$ is the ordinary canonical bracket of mechanics.

(a) Carry out every step. State which conditions hold strongly and which hold only on the constraint surface. Classify the constraints.

(b) Find a time-dependent generator producing $\delta q = \epsilon(t)$ and $\delta u = \dot{\epsilon}(t)$ for an arbitrary smooth real function ϵ. Determine the transformation of the total-Hamiltonian multiplier needed to map solutions to solutions.

(c) At a fixed time, show that every two points on the constraint surface are gauge equivalent. How many canonical pairs remain on this quotient?

Solution. For the first part, the Legendre map gives

$$p_q = \dot{q} - u, \qquad p_u = 0.$$

Its velocity Hessian is $\mathrm{diag}(1, 0)$. The primary surface is $p_u = 0$, while $\dot{q} = p_q + u$ is determined. The subtraction gives

$$H_c = p_q(p_q + u) - \frac{1}{2}p_q^2 = \frac{1}{2}p_q^2 + up_q, \qquad H_T = H_c + \lambda p_u.$$

The Hamilton equations on the extended space are

$$\dot{q} = p_q + u, \qquad \dot{u} = \lambda, \qquad \dot{p}_q = 0, \qquad \dot{p}_u = -p_q.$$

Thus preserving $p_u = 0$ requires the secondary constraint $p_q = 0$. Its preservation gives no further condition and leaves λ arbitrary. The mutual constraint brackets vanish strongly. In contrast, $\{p_u, H_T\} = -p_q$ is only weakly zero. Both constraints are first class.

For the second part, define

$$G_\epsilon = \epsilon p_q + \dot{\epsilon} p_u.$$

Taking brackets in the order $\delta F = \{F, G_\epsilon\}$ gives

$$\delta q = \epsilon, \qquad \delta u = \dot{\epsilon}, \qquad \delta p_q = \delta p_u = 0.$$

The combination $\dot q - u$ is unchanged. To check Hamiltonian solutions, put $q' = q + \epsilon$ and $u' = u + \dot\epsilon$. Then

$$\dot q' = p_q + u', \qquad \dot u' = \lambda + \ddot\epsilon.$$

Hence $\lambda' = \lambda + \ddot\epsilon$ preserves every Hamilton equation. A generator containing only the secondary constraint would miss the shift of the auxiliary coordinate.

For the final part, the constraint surface has coordinates (q, u). At any fixed time, choose ϵ and $\dot\epsilon$ equal to the two differences between prescribed points. A smooth function with these two values exists, for example an affine function near that time. Its gauge transformation maps one point to the other. The quotient is a point and has no remaining canonical pair. This conclusion was established directly, without assuming a general constraint-counting theorem.

Exercise 11.2 (A mass term changes the constraint class). On $\mathbb{R}^{1+3}$ take the vacuum density

$$L_M = \frac{1}{2}|\mathbf{A}_t + \nabla\phi_{\mathrm{em}}|^2 - \frac{1}{2}|\operatorname{curl}\mathbf{A}|^2 + \frac{M^2}{2}(\phi_{\mathrm{em}}^2 - |\mathbf{A}|^2), \qquad M > 0.$$

At each time all canonical fields and the multiplier are Schwartz. Use the extended bracket of Lecture 11 and define $P[\eta] = \int \eta\pi_\phi \,\mathrm{d}^3x$ for real Schwartz η.

(a) Derive the momenta, canonical Hamiltonian and consistency chain. Does preserving the secondary constraint leave $\partial_t\phi_{\mathrm{em}}$ arbitrary?

(b) Put $\mathcal{D}_M = \operatorname{div}\mathbf{\Pi} - M^2\phi_{\mathrm{em}}$ and $C_M[\chi] = -\int \chi\mathcal{D}_M \,\mathrm{d}^3x$. Compute all primary/secondary brackets. For smearings with $\int \eta\chi = 1$, write the two-by-two off-diagonal bracket block and its inverse. Why does its inverse have no massless limit?

(c) Impose both constraints and express the scalar variables in terms of $\mathbf{A}$ and $\mathbf{\Pi}$. Pull back the canonical form and determine how many spatial component pairs remain. Contrast this with Maxwell's gauge quotient.

Solution. For the first part, differentiation in the velocities gives

$$\pi_\phi = 0, \qquad \mathbf{\Pi} = \mathbf{A}_t + \nabla\phi_{\mathrm{em}} = -\mathbf{E}.$$

The Legendre subtraction, with $\mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}$, is

$$H_c = \int \left[\frac{1}{2}(|\mathbf{\Pi}|^2 + |\operatorname{curl}\mathbf{A}|^2) - \mathbf{\Pi}\cdot\nabla\phi_{\mathrm{em}} - \frac{M^2}{2}\phi_{\mathrm{em}}^2 + \frac{M^2}{2}|\mathbf{A}|^2\right]\mathrm{d}^3x.$$

On a cube Q_R, the term containing ϕ_{em} has the boundary contribution

$$-\int_{Q_R} \mathbf{\Pi}\cdot\nabla\phi_{\mathrm{em}} = \int_{Q_R} \phi_{\mathrm{em}} \operatorname{div}\mathbf{\Pi} - \int_{\partial Q_R} \phi_{\mathrm{em}}\mathbf{\Pi}\cdot n \,\mathrm{d}S.$$

Schwartz decay removes the face integral as $R \to \infty$. The magnetic variation has the same vanishing boundary term as in Lecture 11. Adding $\int \lambda\pi_\phi$ gives

$$\partial_t\phi_{\mathrm{em}} = \lambda, \qquad (\pi_\phi)_t = -\operatorname{div}\mathbf{\Pi} + M^2\phi_{\mathrm{em}} = -\mathcal{D}_M,$$
$$\mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}, \qquad \mathbf{\Pi}_t = -\operatorname{curl}\operatorname{curl}\mathbf{A} - M^2\mathbf{A}.$$

The primary condition therefore requires $\mathcal{D}_M = 0$. Its derivative is

$$(\mathcal{D}_M)_t = -M^2 \operatorname{div}\mathbf{A} - M^2\lambda.$$

Since $M > 0$, preservation fixes $\lambda = -\operatorname{div}\mathbf{A}$. There is no tertiary constraint. The Lorenz condition is now forced by consistency, rather than selectable by residual Maxwell gauge freedom.

For the second part, the nonzero functional derivatives are

$$P_{\pi_\phi} = \eta, \qquad (C_M)_{\phi_{\mathrm{em}}} = M^2\chi, \qquad (C_M)_{\mathbf{\Pi}} = \nabla\chi.$$

They give

$$\{P[\eta], P[\zeta]\} = 0, \quad \{C_M[\chi], C_M[\vartheta]\} = 0, \quad \{P[\eta], C_M[\chi]\} = -M^2 \int \eta\chi \, \mathrm{d}^3x.$$

Thus the constraints do not form Maxwell's first-class algebra. The normalized scalar bracket block and inverse are

$$\begin{pmatrix} 0 & -M^2 \\ M^2 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & M^{-2} \\ -M^{-2} & 0 \end{pmatrix}.$$

This is an invertible second-class block for positive M; it is singular at $M = 0$. These displayed matrices describe the scalar constraint pairing, not an unqualified inverse of a pointwise field-distribution matrix.

For the final part, the constraints solve the scalar variables as

$$\pi_\phi = 0, \qquad \phi_{\mathrm{em}} = M^{-2} \operatorname{div}\mathbf{\Pi}.$$

Every Schwartz spatial pair gives such scalar data. On this constraint space, $\delta\pi_\phi = 0$, so the canonical form restricts to

$$\Omega_M = \int_{\mathbb{R}^3} \sum_{i=1}^{3} \delta A_i \wedge \delta\Pi_i \, \mathrm{d}^3x.$$

It is nondegenerate on the three spatial Schwartz pairs, by testing a vector (a, p) against $(-p, a)$. There is no spatial gradient null direction to quotient. Maxwell instead has a longitudinal constraint on $\mathbf{\Pi}$ and a spatial gauge identification of $\mathbf{A}$. The positive-mass theory therefore has three component pairs rather than two transverse pairs.

Exercise 11.3 (A boundary generator need not vanish). Let $\Omega = B_1(0) \subset \mathbb{R}^3$. For this problem the phase variables $\mathbf{A}, \mathbf{\Pi}$ are smooth on a neighbourhood of $\overline{\Omega}$, with no fixed normal trace for $\mathbf{\Pi}$. Their canonical form is $\int_\Omega \delta A_i \wedge \delta\Pi_i \, \mathrm{d}^3x$. A functional has a *smooth bulk derivative* if its variation is an integral of smooth coefficients against arbitrary smooth field variations, without a boundary evaluation term. For a fixed smooth real χ, set

$$C_\Omega[\chi] = -\int_\Omega \chi \operatorname{div}\mathbf{\Pi} \, \mathrm{d}^3x, \qquad G_\Omega[\chi] = \int_\Omega \mathbf{\Pi} \cdot \nabla\chi \, \mathrm{d}^3x.$$

(a) Derive their relation, including the surface integral, and vary it. Explain why $C_\Omega[\chi]$ generally lacks a smooth bulk derivative when χ is nonzero on the boundary, while $G_\Omega[\chi]$ has one.

(b) Find the Hamiltonian vector of $G_\Omega[\chi]$. On the Gauss surface $\operatorname{div}\mathbf{\Pi} = 0$, must its value and its contraction with the constrained canonical form vanish?

(c) Evaluate both functionals and their boundary difference for $\mathbf{\Pi} = e_1$ and $\chi = x^1$. Identify the boundary assumption that distinguishes this problem from the all-space integrations by parts in Lecture 11.

Solution. For the first part, the product rule and the ball divergence formula give

$$C_\Omega = G_\Omega - \int_{\partial\Omega} \chi\mathbf{\Pi} \cdot n \, \mathrm{d}S.$$

The domain, χ and normal are fixed during phase variation. Hence

$$\delta C_\Omega = \int_\Omega \nabla\chi \cdot \delta\mathbf{\Pi}\, \mathrm{d}^3x - \int_{\partial\Omega} \chi\delta\mathbf{\Pi}\cdot n\, \mathrm{d}S, \qquad \delta G_\Omega = \int_\Omega \nabla\chi\cdot\delta\mathbf{\Pi}\, \mathrm{d}^3x.$$

Interior tests force any proposed smooth momentum derivative of C_Ω to equal $\nabla\chi$. It cannot then account for the remaining boundary term. If χ has nonzero trace, choose a smooth variation with normal trace $\delta\mathbf{\Pi}\cdot n = \chi$ there. Its boundary term is $-\int_{\partial\Omega}\chi^2 \neq 0$. Thus no smooth bulk derivative exists. Adding the displayed boundary functional cancels this obstruction exactly.

For the second part, contraction with the canonical form shows

$$X_{G_\Omega} = (\nabla\chi, 0), \qquad \Omega(X_{G_\Omega}, (a,p)) = \int_\Omega \nabla\chi\cdot p\, \mathrm{d}^3x.$$

For a tangent with $\operatorname{div} p = 0$, this becomes $\int_{\partial\Omega}\chi p\cdot n\, \mathrm{d}S$. It need not be zero. On the Gauss surface $C_\Omega = 0$, but $G_\Omega = \int_{\partial\Omega}\chi\mathbf{\Pi}\cdot n\, \mathrm{d}S$ can remain nonzero.

For the last part, $\operatorname{div} e_1 = 0$ and $\nabla x^1 = e_1$, so

$$C_\Omega = 0, \qquad G_\Omega = \operatorname{vol}(B_1) = \frac{4\pi}{3}.$$

On the unit sphere $x^1 = n_1$. Rotation to its polar axis gives

$$\int_{\partial\Omega} x^1 n_1\, \mathrm{d}S = 2\pi\int_{-1}^{1} s^2 \mathrm{d}s = \frac{4\pi}{3}.$$

The same value is the pairing with the transverse tangent $(0, e_1)$. These boundary transformations are therefore not null directions of this constrained form. The all-space Schwartz integrations by parts in Lecture 11 have vanishing surface contributions. They do not justify dropping a free boundary trace on this ball. A gauge reduction must specify which boundary transformations it identifies.

Classical Field Theory — Exercise Sheet 12

Completing transverse reduction

Solutions

Use the unitary Fourier transform of Lecture 4. The real canonical form is

$$\Omega((a,p),(b,q)) = \int (a\cdot q - b\cdot p).$$

The projector is $P_{\mathrm{T}}(k) = I - kk^{\mathsf{T}}/|k|^2$ for $k \neq 0$. All claims about a fixed annulus below concern the data and smearings specified in Lecture 12, not pointwise Fourier observables.

Exercise 12.1 (A finite Dirac bracket and an oblique mode). For a fixed $k \in \mathbb{R}^3 \setminus \{0\}$, consider the finite canonical model $(a,p) \in \mathbb{R}^3 \times \mathbb{R}^3$ with $\{a_i, p_j\} = \delta_{ij}$. Set $c_1 = k\cdot p$ and $c_2 = k\cdot a$. Given an invertible constraint matrix $C_{ab} = \{c_a, c_b\}$, its *Dirac bracket* is

$$\{F,G\}_D = \{F,G\} - \{F,c_a\}(C^{-1})_{ab}\{c_b,G\}.$$

This formula is supplied for this finite system.

(a) Compute C, its inverse, and every coordinate Dirac bracket. Check that each constraint has zero Dirac bracket with every coordinate.

(b) At $k = (1,2,2)$, display the projector and project $(3,-1,2)$. Recover the annulus smeared bracket of Lecture 12 from this matrix symbol, using real vector smearings f, g.

(c) At $k = 0$, which step fails? Explain why periodic real gauge parameters on $\mathbb{T}^3$ leave three constant canonical pairs, and give their form.

Solution. For the first part, the two scalar constraints give

$$C = \begin{pmatrix} 0 & -|k|^2 \\ |k|^2 & 0 \end{pmatrix}, \qquad C^{-1} = \begin{pmatrix} 0 & |k|^{-2} \\ -|k|^{-2} & 0 \end{pmatrix}.$$

Only $\{a_i, c_1\} = k_i$ and $\{c_2, p_j\} = k_j$ contribute to the mixed coordinate correction. Thus

$$\{a_i,p_j\}_D = \delta_{ij} - \frac{k_i k_j}{|k|^2}, \qquad \{a_i,a_j\}_D = \{p_i,p_j\}_D = 0.$$

Contracting either index with k proves the claimed zero brackets with c_1 and c_2. In this finite model the Gauss direction and a spatial gauge-fixing condition have removed the longitudinal pair.

For the second part, $|k|^2 = 9$ gives

$$P_{\mathrm{T}}(k) = \frac{1}{9}\begin{pmatrix} 8 & -2 & -2 \\ -2 & 5 & -4 \\ -2 & -4 & 5 \end{pmatrix}, \qquad P_{\mathrm{T}}(k)\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 22 \\ -19 \\ 8 \end{pmatrix}.$$

The dot product of this result with $(1,2,2)$ is zero. For real annulus smears, the gradients of $A_T(f)$ and $\Pi_T(g)$ are $(P_{\mathrm{T}}f, 0)$ and $(0, P_{\mathrm{T}}g)$. Plancherel and orthogonality therefore give

$$\{A_T(f), \Pi_T(g)\}_{\mathrm{red}} = \int P_{\mathrm{T}}f \cdot P_{\mathrm{T}}g = \int f \cdot P_{\mathrm{T}}g.$$

This is the tested matrix symbol. The finite model does not turn a Fourier point value into an admissible observable, or establish an unrestricted full-frequency delta kernel from one annulus.

For the last part, both constraints become identically zero at $k = 0$. The matrix cannot be inverted there. A periodic real χ has $\int_{\mathbb{T}^3} \nabla\chi = 0$, so it cannot alter a constant potential. Gauss imposes no condition on constant momentum. Integrating the canonical form for constant pairs gives

$$\Omega_0 = (2\pi)^3 \sum_{i=1}^{3} \delta a_i \wedge \delta p_i.$$

All three harmonic pairs remain.

Exercise 12.2 (A union of annuli, rather than one annulus). For $n \geq 2$ let $\mathcal{U}_n = \{n^{-1} < |k| < n\}$ and let $\mathcal{Y}$ be the union of the real Schwartz classes with smooth compact Fourier support in $\mathcal{U}_n$.

(a) Prove that $\mathcal{Y}$ is dense in the real spaces $L^2(\mathbb{R}^3;\mathbb{R})$ and $H^1(\mathbb{R}^3;\mathbb{R})$, using the appropriate norm in each case. You may use the usual L^2 density of compact smooth functions, proved by truncation and smoothing.

(b) Prove the corresponding density for transverse vector fields, with approximants $P_{\mathrm{T}}v_n$. Keep the real-field Fourier condition.

(c) Show that the analogous class in one fixed annulus is not dense in either whole space. Explain why a bounded extension from a dense union must not be justified by density of just one annulus.

Solution. For the first part, write $w_s(k) = (1 + |k|^2)^s$ with $s = 0$ or $s = 1$. Choose real even smooth cutoffs η_n supported in $\mathcal{U}_n$, between zero and one, and tending to one at every $k \neq 0$. Dominated convergence gives

$$\int w_s |\eta_n \widehat{f} - \widehat{f}|^2 \mathrm{d}^3k \longrightarrow 0.$$

At fixed n, approximate $\eta_n \widehat{f}$ by smooth compact profiles inside $\mathcal{U}_n$. This can be done by first cutting away any remaining boundary strip and then convolving with a sufficiently small mollifier. On that bounded set the weighted and ordinary L^2 norms are equivalent. Choose the approximation error at most $1/n$ in the weighted norm. Replace its profile b_n by

$$\frac{1}{2}\big(b_n(k) + \overline{b_n(-k)}\big).$$

This preserves its annulus support, gives Fourier reality, and does not increase its error from the real target. Inverse transformation proves the stated density in the corresponding spatial norm.

Here H^0 means L^2. For the second part, apply that construction componentwise to a transverse v. The multiplier is real and even, so $P_{\mathrm{T}}v_n$ remains real with the same smooth annulus support. Its contraction estimate with either weight is

$$\|P_{\mathrm{T}}(v_n - v)\|_{H^s}^2 = \int w_s |P_{\mathrm{T}}(k)(\widehat{v}_n - \widehat{v})|^2 \mathrm{d}^3k \leq \|v_n - v\|_{H^s}^2.$$

Since $P_{\mathrm{T}}v = v$, the projected approximants converge to v.

For the final part, choose a nonzero real even smooth Fourier profile supported in a shell disjoint from the fixed annulus. Normalize its inverse transform to have norm one in the

space being tested. Every fixed-annulus profile is orthogonal to it in the weighted Fourier inner product. Consequently its distance from every such profile is at least one. One annulus is not dense. The direct global multiplier in Lecture 12, or extension from the dense union just proved, avoids this error.

Exercise 12.3 (A gauge parameter outside L^2). Choose a real radial $\eta \in C_c^\infty(\mathbb{R}^3)$ equal to one on $|k| \le 1$. Define a tempered scalar distribution and a vector by their Fourier profiles

$$\widehat{\chi}_0(k) = \frac{\eta(k)}{|k|^2}, \qquad \widehat{A}(k) = ik\,\widehat{\chi}_0(k) \quad (k \neq 0).$$

A locally integrable function of at most polynomial growth defines a tempered distribution. You may also use that a distribution with all first derivatives zero is constant.

(a) Show that χ_0 is well-defined and real, that $A \in H^1$, and that $\chi_0 \notin L^2$. Can any L^2 scalar have gradient A?

(b) For a general real $A \in H^1(\mathbb{R}^3;\mathbb{R}^3)$ compare the two profiles $\widehat{\chi}_L = -i(k\cdot\widehat{A})/|k|^2$ and $-\widehat{\chi}_L$. Which one has gradient $(I-P_{\mathrm{T}})A$, and which gauges A to $P_{\mathrm{T}}A$? Justify that χ_L is tempered even near zero frequency.

Solution. For the first part, r^{-2} is locally integrable in three dimensions because $\int_0^1 r^2 r^{-2}\mathrm{d}r < \infty$. Its compact support makes the resulting distribution tempered. The profile is real and even, so the distribution is real. The vector profile is imaginary and odd, which is the Fourier reality condition for A. The squared parameter profile has radial integral $\int_0^1 r^{-2}\mathrm{d}r = \infty$. Plancherel therefore rules out $\chi_0 \in L^2$. For its gradient, the weighted radial integral near zero is instead

$$\int_0^1 (1+r^2) r^2 r^{-2}\mathrm{d}r < \infty.$$

There is no large-frequency issue because η is compactly supported. Thus $A \in H^1$. If an L^2 parameter had this gradient, it would differ from χ_0 by a constant distribution. In Fourier space a nonzero constant adds a point delta at zero, which cannot be an L^2 function; a zero constant leaves the non-L^2 profile η/r^2. Neither gives an L^2 parameter.

For the second part, direct multiplication gives

$$ik\widehat{\chi}_L = \frac{k(k\cdot\widehat{A})}{|k|^2} = (I-P_{\mathrm{T}})\widehat{A}, \qquad \widehat{A} + ik(-\widehat{\chi}_L) = P_{\mathrm{T}}\widehat{A}.$$

The two signs serve different purposes. Fourier reality follows by replacing k by $-k$ and conjugating $\widehat{A}$. Near zero the bound $|\widehat{\chi}_L| \le |\widehat{A}|/|k|$ gives

$$\int_{|k|<1} |\widehat{\chi}_L|\mathrm{d}^3k \le \|\widehat{A}\|_2 \left(\int_{|k|<1} |k|^{-2}\mathrm{d}^3k\right)^{1/2} < \infty.$$

At infinity its pairing with any Schwartz test is finite by Cauchy–Schwarz. This defines a tempered distribution with gradient in H^1. The completion therefore uses distributional parameters modulo constants, not just Schwartz or H^1 scalar parameters.

Exercise 12.4 (Weak symplectic does not mean every gradient exists). On the real space $\mathcal{E}_T = H^1_{\mathrm{trans}} \times L^2_{\mathrm{trans}}$, use the canonical form from Lecture 12. Let $\theta(r)$ be smooth, zero for $r \le 1$ and one for $r \ge 2$, and define the real transverse L^2 vector h by

$$\widehat{h}(k) = \theta(|k|)|k|^{-2} P_{\mathrm{T}}(k) e_1 .$$

(a) Prove that $h \in L^2$ but $h \notin H^1$. Show that $F(a,p) = \int h \cdot p \, \mathrm{d}^3x$ is continuous on $\mathcal{E}_T$.

(b) Show that no $X \in \mathcal{E}_T$ satisfies $\iota_X \Omega = DF$. Why does this not contradict the nondegeneracy proved in Lecture 12?

(c) Does finiteness of $\int (|\operatorname{curl} A|^2 + |\Pi|^2)$ imply membership in this inhomogeneous phase space? Test a nonzero constant potential with zero momentum.

Solution. For the first part, $P_{\mathrm{T}}(n)e_1$ has squared norm $1 - n_1^2$. Its angular integral is $8\pi/3 > 0$. For $r \ge 2$, the squared L^2 and H^1 norms therefore contain the radial factors

$$\int_2^\infty r^{-2} \mathrm{d}r < \infty, \qquad \int_2^\infty (1 + r^2) r^{-2} \mathrm{d}r = \infty .$$

The transition region is harmless. The profile is real and even, so h is real, and $P_{\mathrm{T}} h = h$ makes it transverse. Cauchy–Schwarz gives

$$|F(a,p)| \le \|h\|_2 \|p\|_2 ,$$

which proves continuity on the stated phase space.

For the second part, suppose $X = (X_A, X_\Pi)$ represents DF. Testing its contraction against $(0, q)$ gives

$$\int X_A \cdot q = \int h \cdot q \qquad (q \in L^2_{\mathrm{trans}}).$$

Choose $q = X_A - h$, an allowed transverse L^2 vector. Then $\|X_A - h\|_2^2 = 0$, so $X_A = h$, contradicting $X_A \in H^1$. Nondegeneracy only states that a vector pairing to zero with every tangent is zero. It does not assert that every continuous covector has a representing vector. This is the distinction expressed by *weakly symplectic*.

For the final part, take $A = e_1$ and $\Pi = 0$ on all of $\mathbb{R}^3$. Its curl and physical energy vanish, but $A \notin L^2$. It therefore lies outside $H^1_{\mathrm{trans}} \times L^2_{\mathrm{trans}}$. The inhomogeneous completion is a specified phase model; finite physical energy alone does not justify its canonical integral or its claimed domain.

Classical Field Theory — Exercise Sheet 13

Radiation estimates and boundary conditions

Solutions

Use Lecture 13's real Maxwell fields, retarded power theorem and amplitude $\mathcal{A}(u,n) = (4\pi)^{-1}\int \mathbf{J}(u+n\cdot y, y)\mathrm{d}^3y$. Here $n\in\mathbb{S}^2$, $u=t-r$, and $P_{\mathrm{T}}(n)=I-nn^{\mathsf{T}}$. The third problem also uses Lecture 6's time-reflected advanced kernel. The singular-source formula in the last problem is supplied separately.

Exercise 13.1 (A general slow-time source). Let $\rho_0(s,y)$ be scalar and $J_0(s,y)$ vector-valued, both real and smooth with compact spacetime support, satisfying $\partial_s\rho_0 + \mathrm{div}_y J_0 = 0$. For $0<\varepsilon\le 1$ put

$$\rho_\varepsilon(t,y)=\rho_0(\varepsilon t,y),\qquad \mathbf{J}_\varepsilon(t,y)=\varepsilon J_0(\varepsilon t,y),\qquad p_0(s)=\int y\rho_0(s,y)\mathrm{d}^3y.$$

Define finite constants

$$M_0=\int \sup_s|\partial_s J_0(s,y)|\mathrm{d}^3y,$$
$$M_1=\int |y|\sup_s|\partial_s^2 J_0(s,y)|\mathrm{d}^3y,$$
$$M_2=\frac12\int |y|^2\sup_s|\partial_s^3 J_0(s,y)|\mathrm{d}^3y.$$

(a) Verify conservation and prove $p_0'(s)=\int J_0(s,y)\mathrm{d}^3y$. Find the scaled dipole $p_\varepsilon(t)=\int y\rho_\varepsilon(t,y)\mathrm{d}^3y$ and its second derivative.

(b) Prove, uniformly for all $u\in\mathbb{R}$, the explicit absolute estimate

$$\left|P_\varepsilon(u)-\frac{|\ddot p_\varepsilon(u)|^2}{6\pi}\right|\le\frac{2M_0M_1+M_1^2}{4\pi}\varepsilon^5.$$

(c) Improve the integrated error to $O(\varepsilon^6)$ by retaining the first delayed-sampling correction. Give an explicit bound using M_0,M_1,M_2. Why does this improvement not assert that the directional power density has an $O(\varepsilon^6)$ error?

Solution. For the first part, both terms of the conservation law acquire the same factor ε. Compact spatial support permits coordinate integration by parts, giving

$$p_{0,i}'=-\int y_i\partial_{y_j}J_{0,j}\mathrm{d}^3y=\int J_{0,i}\mathrm{d}^3y,\qquad p_\varepsilon(t)=p_0(\varepsilon t),\qquad \ddot p_\varepsilon(t)=\varepsilon^2\int\partial_s J_0(\varepsilon t,y)\mathrm{d}^3y.$$

The scaled current is compact in spacetime for each positive ε, so Lecture 13 applies, although the temporal support grows as ε decreases.

For the second part, set $s=\varepsilon u$ and $a(s)=\int\partial_s J_0(s,y)\mathrm{d}^3y$. Differentiation of the amplitude and the mean-value theorem give

$$\mathcal{A}_\varepsilon'(u,n)=\frac{\varepsilon^2 a(s)}{4\pi}+R,\qquad |a(s)|\le M_0,\qquad |R|\le\frac{\varepsilon^3M_1}{4\pi}.$$

Since P_{T} is a contraction, the difference of the squared transverse norms is bounded by twice the product of these bounds plus $|R|^2$. Integrating over a sphere of area 4π yields

$$\left|P_\varepsilon-\frac{|\varepsilon^2 a|^2}{6\pi}\right|\le\frac{2\varepsilon^5M_0M_1+\varepsilon^6M_1^2}{4\pi}.$$

Here the leading integral uses $\int |n\times a|^2 \mathrm{d}\Omega = 8\pi|a|^2/3$. The dipole identity above and $\varepsilon \le 1$ prove the requested estimate.

For the final part, Taylor's formula with a bounded second derivative of $\partial_s J_0$ gives

$$4\pi\mathcal{A}'_\varepsilon = \varepsilon^2 a(s) + \varepsilon^3 m(s,n) + \varepsilon^4 b_\varepsilon(s,n), \quad m = \int (n\cdot y)\partial_s^2 J_0(s,y)\mathrm{d}^3y, \quad |m| \le M_1, \quad |b_\varepsilon| \le M_2.$$

The vector $m(s,-n) = -m(s,n)$, while $P_{\mathrm{T}}(-n) = P_{\mathrm{T}}(n)$. Thus $\int P_{\mathrm{T}}a \cdot P_{\mathrm{T}}m\mathrm{d}\Omega = 0$. The remaining cross terms and squares are bounded by

$$\left| P_\varepsilon - \frac{|\ddot{p}_\varepsilon|^2}{6\pi} \right| \le \frac{\varepsilon^6}{4\pi}\left(2M_0M_2 + M_1^2 + 2M_1M_2 + M_2^2\right).$$

All constants are independent of u and ε. At a fixed direction the first cross term generally remains nonzero; its cancellation is a property of the integrated power.

Exercise 13.2 (A nonradial source at zero acceleration). Choose a nonnegative $f \in C_c^\infty(B_1(0))$ with $\int f = 1$ and nonzero first moment $\mu = \int y f(y)\mathrm{d}^3y$. Let $a \in \mathbb{R}^3 \setminus \{0\}$ and choose a smooth compactly supported p_0 equal to as^3 on $|s| \le 1$. Use the conserved dipole source

$$p(t) = p_0(\varepsilon t), \qquad \rho = -p(t)\cdot\nabla f, \qquad \mathbf{J} = \dot{p}(t)f, \qquad 0 < \varepsilon \le 1.$$

Compute the exact retarded power $P(0)$. Show that it is positive although $\ddot{p}(0) = 0$, and evaluate it for $a = e_3$, $\mu = e_3/4$. Explain what this rules out about a relative error estimate. You may use the angular moments

$$\int n_i n_j \mathrm{d}\Omega = \frac{4\pi}{3}\delta_{ij}, \qquad \int n_i n_j n_k n_l \mathrm{d}\Omega = \frac{4\pi}{15}(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

Solution. The sampled acceleration is exactly $\ddot{p}(n\cdot y) = 6\varepsilon^3(n\cdot y)a$. The amplitude and power therefore are

$$\mathcal{A}'(0,n) = \frac{6\varepsilon^3}{4\pi}(n\cdot\mu)a, \qquad P(0) = \frac{36\varepsilon^6}{16\pi^2}\int (n\cdot\mu)^2\left(|a|^2 - (n\cdot a)^2\right)\mathrm{d}\Omega.$$

Contracting the supplied second and fourth moments gives

$$\int (n\cdot\mu)^2|a|^2\mathrm{d}\Omega = \frac{4\pi}{3}|a|^2|\mu|^2, \qquad \int (n\cdot\mu)^2(n\cdot a)^2\mathrm{d}\Omega = \frac{4\pi}{15}\left(|a|^2|\mu|^2 + 2(a\cdot\mu)^2\right).$$

Their difference yields

$$P(0) = \frac{6\varepsilon^6}{5\pi}\left(2|a|^2|\mu|^2 - (a\cdot\mu)^2\right).$$

Cauchy–Schwarz bounds its parenthesis below by $|a|^2|\mu|^2 > 0$. For $a = e_3$ and $\mu = e_3/4$, that parenthesis is $1/16$, giving $P(0) = 3\varepsilon^6/(40\pi)$. Such an f exists: translate a normalized radial smooth bump of radius less than $1/4$ to centre $e_3/4$. But the instantaneous leading dipole power is zero. An estimate of the error bounded by a finite constant times that leading value would force $P(0) = 0$, contradicting the exact result. The absolute estimates of the first problem allow it.

Exercise 13.3 (The half-retarded, half-advanced field). For a real smooth compact spacetime conserved source, let $(\phi_{\mathrm{em}}^{\mathrm{ret}}, \mathbf{A}_{\mathrm{ret}})$ be its retarded potentials. Define the advanced potentials by replacing $t - |x-y|$ by $t + |x-y|$ in the same $1/(4\pi|x-y|)$ integrals. Time reflection of Lecture 6's kernel gives these smooth sourced solutions. Let the symmetric potentials be half the sum of the two, and form their Maxwell fields.

(a) At future null infinity, $t = r + u$, show that the advanced field is exactly zero for all sufficiently large r (at each fixed u). Find the symmetric field's limiting outward power in terms of the retarded power.

(b) At past null infinity, $t = v - r$, define

$$\overline{\mathcal{A}}(v,n) = \frac{1}{4\pi}\int \mathbf{J}(v - n\cdot y, y)\mathrm{d}^3y, \qquad \overline{\mathcal{V}}(v,n) = \frac{1}{4\pi}\int \rho(v - n\cdot y, y)\mathrm{d}^3y.$$

Derive the leading advanced electric and magnetic coefficients and the signed outward power of the symmetric field. Primes now mean ∂_v. Does averaging the potentials cancel radiation at future null infinity?

Solution. For the first part, enclose the spatial support in B_R and the temporal support in $[-T,T]$. At $t = r + u$ the advanced source time is at least $2r + u - R$. Once this exceeds T, the source and all its derivatives vanish throughout the potential integrals. Hence the advanced potentials and fields are zero there. The symmetric field is exactly half the retarded field; the quadratic flux is one quarter of its retarded value.

For the second part, at $t = v - r$ the retarded source time is at most $v - 2r + R$, so the retarded fields vanish for large r. Expanding the advanced distance as in Lecture 13 gives

$$\mathbf{A}_{\mathrm{adv}} = \frac{\overline{\mathcal{A}}(v,n)}{r} + O(r^{-2}), \qquad \phi_{\mathrm{em}}^{\mathrm{adv}} = \frac{\overline{\mathcal{V}}(v,n)}{r} + O(r^{-2}).$$

Differentiating the exact source integrals at fixed t gives the leading Cartesian derivative $+n_i\overline{\mathcal{V}}'/r$ for a scalar potential; angular derivatives contribute only $O(r^{-2})$, by the same bounded-support estimates as in Lecture 13. The total spatial divergence of $\mathbf{J}(v - n\cdot y, y)$ integrates to zero and equals the integral of $\mathrm{div}_y\,\mathbf{J} - n\cdot\mathbf{J}_t$. Conservation therefore yields

$$\overline{\mathcal{V}}' = -n\cdot\overline{\mathcal{A}}', \qquad E_* = -(n\overline{\mathcal{V}}' + \overline{\mathcal{A}}') = -P_{\mathrm{T}}\overline{\mathcal{A}}', \qquad B_* = n\times\overline{\mathcal{A}}' = -n\times E_*.$$

Consequently $(E_*\times B_*)\cdot n = -|E_*|^2$. The symmetric field has the limiting signed outward power

$$-\frac{1}{4}\int_{\mathbb{S}^2} |n\times\overline{\mathcal{A}}'(v,n)|^2\mathrm{d}\Omega.$$

It transports energy inward at past infinity and outward at future infinity. These are different asymptotic regions, so the two fields do not cancel there. No pointwise equality between the past and future power functions has been assumed.

Exercise 13.4 (Liénard–Wiechert: a supplied singular-source formula). Let $z\colon \mathbb{R} \to \mathbb{R}^3$ be a smooth complete worldline with $|\dot{z}(s)| \le v_* < 1$. Fix an event (t, x) with $x \neq z(t)$. Let $Q \in \mathbb{R}$ be its charge. For this problem only, take as given the point-source potential prescription

$$\phi_{\mathrm{em}}(t,x) = \frac{Q}{4\pi}\int \frac{\delta(t-s-|x-z(s)|)}{|x-z(s)|}\mathrm{d}s, \qquad \mathbf{A}(t,x) = \frac{Q}{4\pi}\int \frac{\dot{z}(s)\delta(t-s-|x-z(s)|)}{|x-z(s)|}\mathrm{d}s.$$

The delta integrals are understood locally at their roots. You may use the one-variable identity $\delta(g(s)) = \sum_j \delta(s-s_j)/|g'(s_j)|$ for simple roots.

(a) Prove existence and uniqueness of the retarded root $t = s_{\mathrm{ret}} + |x - z(s_{\mathrm{ret}})|$. Evaluate the two potentials in terms of $R = |x - z(s_{\mathrm{ret}})|$, $n = (x - z(s_{\mathrm{ret}}))/R$ and $v = \dot{z}(s_{\mathrm{ret}})$.

(b) For $z(s) = (3s/5, 0, 0)$ at $(t, x) = (1, 0)$, compute the retarded time, Jacobian and scalar potential. Explain exactly which smooth-source hypotheses fail and why this calculation establishes no self-force law.

Solution. For the first part, put $h(s) = s + |x - z(s)|$. The speed bound gives, whenever $s_2 > s_1$,

$$h(s_2) - h(s_1) \ge (1 - v_*)(s_2 - s_1) > 0.$$

Also $h(s) \to +\infty$ as $s \to +\infty$, since $h(s) \ge s$. For negative s, $h(s) \le (1-v_*)s + |x - z(0)| \to -\infty$. Continuity and strict monotonicity give exactly one root $h(s) = t$. If its distance vanished, it would have $s = t$ and $x = z(t)$, excluded by the hypothesis. Thus $R > 0$ at the root and differentiation there is valid. For $g(s) = t - h(s)$ it gives

$$g'(s_{\mathrm{ret}}) = -(1 - n \cdot v), \qquad 1 - n \cdot v \ge 1 - v_* > 0.$$

The supplied delta identity now evaluates the prescriptions as

$$\phi_{\mathrm{em}}(t,x) = \frac{Q}{4\pi R(1 - n \cdot v)}, \qquad \mathbf{A}(t,x) = \frac{Qv}{4\pi R(1 - n \cdot v)}.$$

For the second part, $s = 5/8$ gives $R = 3/8$, $n = -e_1$ and $v = 3e_1/5$, hence $s + R = 1$ and $1 - n \cdot v = 8/5$. Substitution yields $\phi_{\mathrm{em}} = 5Q/(12\pi)$ and $\mathbf{A} = Qe_1/(4\pi)$. The associated current has a spatial point delta and persists for all time; it is neither a smooth spacetime source nor compactly supported in time. Thus this is an application of the explicitly supplied singular-source prescription, not an application of Lecture 13's smooth-source theorem. The fields are evaluated off the worldline. No finite value of the field on the charge, and no equation for its self-force, has been defined.

Classical Field Theory — Exercise Sheet 14

First-order systems and real structures

Solutions

All spinor entries commute. Use Lecture 14's Hermitian inner product, unitary Fourier convention and real form $\Omega(h,g) = 2\,\mathrm{Im}\int h^\dagger g$. Problems 1 and 3 supply new four-component matrices; Problem 4 returns to the actual two-component matrices of Lecture 14.

Exercise 14.1 (Four components in three space dimensions). Let I_2 be the two-by-two identity and put

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \Gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \Gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}.$$

Define $\beta_D = \Gamma^0$, $\alpha_D^i = \Gamma^0\Gamma^i$ and $H_D(k) = \alpha_D^i k_i + m\beta_D$, where $m \geq 0$.

(a) Verify every Clifford anticommutator for signature $(+,-,-,-)$. Prove that $H_D(k)$ is Hermitian and find its eigenvalues, multiplicities and orthogonal spectral projectors, including the exceptional $m = k = 0$ case.

(b) For $\mathrm{i}\psi_t = (-\mathrm{i}\alpha_D^i\partial_i + m\beta_D)\psi$, derive the conserved current $\rho = \psi^\dagger\psi$, $j^i = \psi^\dagger\alpha_D^i\psi$ and prove $|n\cdot j| \leq \rho$ for every unit vector n. Explain why verifying these matrices alone does not prove the spin representation or Lorentz-covariance theorem cited in Lecture 14.

Solution. For the first part, direct multiplication gives $\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij}I_2$. The block products therefore give

$$(\Gamma^0)^2 = I_4, \quad \Gamma^0\Gamma^i + \Gamma^i\Gamma^0 = 0, \quad \Gamma^i\Gamma^j + \Gamma^j\Gamma^i = -2\delta_{ij}I_4, \qquad \alpha_D^i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}.$$

Both α_D^i and β_D are Hermitian. Their anticommutators give

$$H_D^2 = (|k|^2 + m^2)I_4, \qquad \mathrm{tr}\, H_D = 0.$$

For $\nu > 0$, the Hermitian spectral theorem permits only eigenvalues $\pm\nu$; trace zero gives two of each. The projectors

$$P_\pm = \frac{1}{2}(I_4 \pm H_D/\nu)$$

are Hermitian, square to themselves, sum to the identity and have trace two. At $m = k = 0$, the zero matrix has a four-dimensional zero eigenspace; the displayed division by ν is not used there.

For the second part, substitute the evolution and its adjoint into the product derivative. The two mass terms cancel and give

$$\partial_t(\psi^\dagger\psi) = -\partial_i(\psi^\dagger\alpha_D^i\psi).$$

For $A_n = n_i\alpha_D^i$, Hermiticity and $A_n^2 = I_4$ imply $|\psi^\dagger A_n\psi| \leq |\psi|\,|A_n\psi| = |\psi|^2$. This proves the claimed bound. The calculation verifies a Clifford representation and a current. It does not construct the action of the spin group on Lorentz transformations, which is an additional theorem.

Exercise 14.2 (A constant symmetric-hyperbolic system). Let $A^1, \dots, A^d$ be constant Hermitian N-by-N matrices and B a constant skew-Hermitian matrix. For complex columns consider

$$w_t + A^i \partial_i w + Bw = 0, \qquad c_* = \sup_{|n|=1} \|n_i A^i\|_{\mathrm{op}}.$$

Here the operator norm is the Euclidean matrix norm.

(a) Construct the unique distributional solution in $C(\mathbb{R}; L^2)$ for each L^2 datum and prove norm preservation. Derive the smooth local current.

(b) Using Lecture 5's moving-ball argument, prove that data supported in a closed set K remain supported in $\{x : \mathrm{dist}(x, K) \le c_*|t|\}$. Give the limiting argument for rough data and include $c_* = 0$.

(c) For $d = N = 2$, take $A^1 = \sigma_1$, $A^2 = 2\sigma_3$ and $B = \mathrm{i}\sigma_2$. Compute c_* and exhibit a column and normal attaining the current bound. Would Hermitian $B = I_2$ preserve the same norm?

Solution. For the first part, put

$$\widehat{w}(t, k) = e^{-t(\mathrm{i}k_i A^i + B)} \widehat{w}(0, k).$$

The exponential is unitary at real t, k. It is pointwise continuous in t, so dominated convergence and Plancherel give a norm-preserving $C(\mathbb{R}; L^2)$ evolution. Smooth frequency cutoffs give bounded-frequency solutions; their uniform L^2 convergence on compact time intervals passes the differential equation to compact spacetime tests. For uniqueness, cut off a zero-data difference in frequency. Its bounded generator makes the distributional time equation a norm-primitive equation, as in Lecture 14. Its squared norm has zero derivative, so it vanishes. Remove the cutoff to obtain uniqueness. For a smooth solution, Hermiticity of the A^i and $B^\dagger = -B$ give

$$\partial_t \rho + \partial_i j^i = 0, \qquad |j \cdot n| = |w^\dagger (n_i A^i) w| \le c_* \rho.$$

The skew lower-order term cancels rather than adding damping.

For the second part, first take smooth compact initial data; the Fourier formula gives a smooth solution since every differentiated integrand is bounded by a polynomial times the initial Schwartz profile. For $c_* > 0$, let $E(t)$ be its charge on $B_{R - c_* t}(a)$, $0 \le t < R/c_*$. Polar-coordinate FTC and Lecture 5's ball divergence formula give

$$E'(t) = -\int_{\partial B_{R - c_* t}(a)} (j \cdot n + c_* \rho) \mathrm{d}S \le 0.$$

Smooth compact data are dense in L^2. Unitarity gives convergence of the data and evolved fields in L^2; hence their squared norms on these balls converge. The same inequality holds for arbitrary data. If $\mathrm{dist}(a, K) > c_* t$, choose $c_* t < R < \mathrm{dist}(a, K)$: the initial ball charge is zero, so the evolved field vanishes on a neighbourhood of a. A countable cover proves the almost-everywhere support statement. For $c_* = 0$, every A^i is zero (take $n = e_i$); evolution is the pointwise unitary matrix e^{-tB} and preserves support exactly. Reversing time proves the negative-time assertion.

For the last part, the Pauli anticommutators give

$$(n_1 A^1 + n_2 A^2)^2 = (n_1^2 + 4n_2^2) I_2.$$

On the unit circle its norm is at most two, with equality at $n = e_2$. For $w = e_1$ one has $\rho = 1$ and $j \cdot e_2 = 2$. Replacing B by I_2 instead gives $\partial_t \rho + \partial_i j^i = -2\rho$; its Fourier formula contains e^{-t} times a unitary matrix, so the squared norm is multiplied by e^{-2t}.

Exercise 14.3 (Weyl sectors and a mass coupling). Use the four-component matrices of Problem 1. Define $\Gamma^5 = \mathrm{i}\Gamma^0\Gamma^1\Gamma^2\Gamma^3$ and $P_{\mathrm{ch},\pm} = (I_4 \pm \Gamma^5)/2$. Compute Γ^5 and its two eigenspaces. Write the Dirac Hamiltonian in the coordinates

$$\psi = \frac{1}{\sqrt{2}}\begin{pmatrix} u+v \\ u-v \end{pmatrix}, \qquad u, v \in \mathbb{C}^2.$$

Determine when the two chiral subspaces are invariant under evolution, and derive the two Weyl equations in the massless case.

Solution. Multiplication of the four blocks gives

$$\Gamma^5 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}.$$

It is Hermitian and squares to I_4. Its positive and negative spaces consist of columns $(u,u)^T/\sqrt{2}$ and $(v,-v)^T/\sqrt{2}$, respectively. The associated projectors are $P_{\mathrm{ch},\pm}$ as defined in the question. For the unitary matrix

$$W = \frac{1}{\sqrt{2}}\begin{pmatrix} I_2 & I_2 \\ I_2 & -I_2 \end{pmatrix},$$

block multiplication yields

$$W^\dagger \Gamma^5 W = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \qquad W^\dagger H_D(k) W = \begin{pmatrix} \sigma\cdot k & mI_2 \\ mI_2 & -\sigma\cdot k \end{pmatrix}.$$

Thus in position space

$$\mathrm{i}u_t = -\mathrm{i}\sigma\cdot\nabla u + mv, \qquad \mathrm{i}v_t = +\mathrm{i}\sigma\cdot\nabla v + mu.$$

At zero mass these are independent two-component Weyl equations. At positive mass a nonzero column in either sector produces an immediate component in the other. No independent massive evolution on a single chiral sector follows by discarding that equation.

Exercise 14.4 (A commuting Majorana-type real subspace). Return to Lecture 14's two-component matrices $\alpha = -\sigma_3$, $\beta = \sigma_1$ and define the real-linear, antilinear-in-complex-scalars map

$$C\psi = \sigma_3\overline{\psi}.$$

Prove $C^2 = I$, characterize its fixed space in L^2, and show that Dirac evolution preserves it. Derive the evolution of its two real entries. Restrict the real form Ω to that fixed space. Is the result nondegenerate? Explain why this calculation does not give a classical commuting replacement for a quantum Majorana field or its statistics.

Solution. Conjugating twice gives $C^2 = \sigma_3\overline{\sigma_3} = I$. The fixed condition says that the first entry is real and the second is purely imaginary. Write it as $\psi = (a, \mathrm{i}b)^T$. In the position-space evolution

$$\psi_t = -\alpha\psi_x - \mathrm{i}m\beta\psi,$$

conjugation reverses the sign of i, while anticommutation reverses the mass matrix sign again. Commutation of σ_3 with α then shows that $C\psi$ solves the same equation. The L^2 uniqueness theorem proves invariance, also for rough data. Substitution gives the real distributional equations

$$a_t = a_x + mb, \qquad b_t = -b_x - ma.$$

For two fixed tangent columns $h = (a, \mathrm{i}b)^T$ and $g = (c, \mathrm{i}d)^T$, their inner product integrand is $ac + bd$, which is real. Hence

$$\Omega(h, g) = 2\,\mathrm{Im} \int (ac + bd)\mathrm{d}x = 0.$$

There are nonzero fixed columns, so the restriction is degenerate. Multiplication by i does not preserve this fixed space, and Lecture 14's test $\Omega(h, \mathrm{i}h) = 2\|h\|_2^2$ is not an internal test here. The present example concerns commuting classical entries and a real subspace of their evolution. Quantum fermionic variables and statistics are additional structures, not consequences of this restriction.

Classical Field Theory — Exercise Sheet 15

Beyond the one-dimensional cubic proof

Solutions

Use the mass-one Klein–Gordon group $U_d(t)$ on $\mathcal{E}_d = H^1(\mathbb{R}^d) \times L^2(\mathbb{R}^d)$, with Hilbert norm $\|(f,g)\|_{\mathrm{lin}}^2 = \|f\|_{H^1}^2 + \|g\|_2^2$. Lecture 15 proves the forced estimate in one dimension; Problem 2 checks its dimension-independent Fourier proof. All fields below are real.

Exercise 15.1 (A general potential and an affine background). Let $V \in C^3(\mathbb{R})$ and let b be a smooth bounded solution of $-b'' + V'(b) = 0$ on $\mathbb{R}$. Do not assume $b \in L^2$. For $\phi = b + v$ consider $\phi_{tt} - \phi_{xx} + V'(\phi) = 0$.

(a) Derive the mass-one perturbation force $N_b(v)$, and show that it maps $H^1(\mathbb{R})$ locally Lipschitzly to $L^2(\mathbb{R})$, vanishing at zero. Give a constant on a ball of radius R using the maximum of $|V''|$ on a specified compact interval.

(b) For every initial ball $\|(v_0, g_0)\|_{\mathrm{lin}} \le B$, $B \ge 1$, give a common contraction radius and positive time. Prove existence, uniqueness among weak energy solutions, continuous dependence, and the finite-lifespan alternative. Which derivatives of V did the estimates use?

Solution. For the first part, cancellation of $-b'' + V'(b)$ gives

$$v_{tt} - v_{xx} + v = N_b(v), \qquad N_b(v) = v - [V'(b+v) - V'(b)], \qquad N_b(0) = 0.$$

For $\|v\|_{H^1}, \|w\|_{H^1} \le R$, the one-dimensional embedding bounds both perturbations by $R/\sqrt{2}$. Their background-shifted values lie in $[-M, M]$, where $M = \|b\|_\infty + R/\sqrt{2}$. The scalar identity

$$V'(b+v) - V'(b+w) = (v-w) \int_0^1 V''(b + w + \theta(v-w))\mathrm{d}\theta$$

gives $\|N_b(v) - N_b(w)\|_2 \le L_R \|v - w\|_{H^1}$ with $L_R = 1 + \max_{[-M,M]} |V''|$. Taking $w = 0$ proves membership in L^2. This argument uses a difference of the two forces; it never assigns an L^2 norm to the background separately.

For the second part, let $X_T = C([0,T]; \mathcal{E}_1)$ and put

$$(\Phi z)(t) = U_1(t) z_0 + \int_0^t U_1(t-s)(0, N_b(v(s)))\mathrm{d}s, \quad R = 2B, \quad T = \frac{1}{2L_R}.$$

Isometry and the forced estimate give

$$\|\Phi z\|_{X_T} \le B + T L_R R = R, \qquad \|\Phi z - \Phi \widetilde{z}\|_{X_T} \le \tfrac{1}{2} \|z - \widetilde{z}\|_{X_T}.$$

The closed radius-R ball is complete, so Banach's theorem gives a unique fixed point. The forcing lemma identifies it as a weak energy solution. Conversely, the force of any such solution is continuous in L^2; subtract its Duhamel construction and use L4's homogeneous uniqueness. Thus every weak energy solution is mild. Continuity puts it inside a suitable contraction ball for a short interval, giving local uniqueness; restart at each common time to obtain uniqueness on any overlap. For data in the same initial ball, subtraction gives $\|z - \widetilde{z}\|_{X_T} \le 2\|z_0 - \widetilde{z}_0\|_{\mathrm{lin}}$. Restarting also gives continuous dependence on compact intervals in a common bounded ball.

Unique solutions glue to a maximal one. If a finite endpoint T_* had a sequence $t_j \uparrow T_*$ with bounded norms, the common positive time for that ball would extend a restart beyond T_*, a contradiction. Therefore $\|z(t)\|_{\mathrm{lin}} \to \infty$ at a finite endpoint. The estimates used continuity of V'' on bounded intervals, not V'''; the assumed C^3 regularity is more than needed here.

Exercise 15.2 (The cubic survives in three dimensions). Take as given the Sobolev estimate $\|u\|_{L^6(\mathbb{R}^3)} \le C_S \|u\|_{H^1(\mathbb{R}^3)}$ for every real $u \in H^1$, where C_S is a fixed positive constant. Consider

$$u_{tt} - \Delta u + u^3 = 0, \qquad (u(0), u_t(0)) \in \mathcal{E}_3.$$

(a) Justify the three-dimensional version of the forced mass-one energy estimate, including the weak equation. Prove local existence, energy-class uniqueness and continuous dependence by a Duhamel contraction, with a common lifespan for a bounded initial ball.

(b) Would the same $H^1 \to L^2$ force argument handle u^5? For a nonzero real $\eta \in C_c^\infty(\mathbb{R}^3)$, compute the scaling of $u_\Lambda(x) = \Lambda^{1/2}\eta(\Lambda x)$, $\Lambda \ge 1$, in H^1, and of its cubic and quintic forces in L^2. Limit the conclusion to this contraction argument.

Solution. For the first part, the mass-one Fourier rotation on $(\nu\widehat{u}, \widehat{p})$ is orthogonal pointwise in $k \in \mathbb{R}^3$. Consequently U_3 is an isometry and its norm integral obeys

$$\left\| \int_0^t U_3(t-s)(0, F(s)) \mathrm{d}s \right\|_{\mathrm{lin}} \le \int_0^t \|F(s)\|_2 \mathrm{d}s.$$

The integral is a continuous energy pair. Smooth compact frequency cutoffs of $F \in C([0,T]; L^2)$ converge uniformly, by the compact-image finite-net argument of Lecture 15. On each cutoff, differentiating the sine integral gives $u_t = p$ and $p_t = \Delta u - u + F$ in L^2. The displayed bound passes pairs uniformly in energy norm to the limit; compact test functions pass the equation to the limit as well. The zero initial traces converge. L4's uniqueness identifies every forced energy solution with this construction plus its homogeneous evolution. This proves the needed dimension change without a nonlinear existence input.

Rewrite the equation as $u_{tt} - \Delta u + u = N(u) = u - u^3$. Hölder and the given embedding give, on the radius-R H^1 ball,

$$\|u^3 - v^3\|_2 \le \|u - v\|_6 (\|u\|_6^2 + \|u\|_6\|v\|_6 + \|v\|_6^2) \le 3C_S^3 R^2 \|u - v\|_{H^1}.$$

Thus $N : H^1 \to L^2$ is locally Lipschitz with $N(0) = 0$ and $L_R = 1 + 3C_S^3 R^2$. With $R = 2B$, $T = 1/(2L_R)$ the two ball estimates are $B + TL_R R = R$ and $TL_R = 1/2$. Banach's theorem constructs the mild solution. The forced identification, short-time uniqueness and overlap restart from Problem 1 give weak energy-class uniqueness, continuous dependence and the same continuation alternative.

For the second part, changing variable $y = \Lambda x$ gives

$$\|u_\Lambda\|_{H^1}^2 = \Lambda^{-2}\|\eta\|_2^2 + \|\nabla\eta\|_2^2, \quad \|u_\Lambda^3\|_2 = \|\eta^3\|_2, \quad \|u_\Lambda^5\|_2 = \Lambda\|\eta^5\|_2.$$

A locally Lipschitz map on H^1 is bounded on a sufficiently small ball about zero. Multiplying this family by any fixed small positive constant still makes the quintic norms unbounded while its H^1 norms fit that ball. Thus the quintic is not locally Lipschitz from H^1 to L^2 there. This refutes that force estimate, not local theory in other spaces or by other methods.

Exercise 15.3 (Sine-Gordon on a weighted path space). Let b be a smooth bounded stationary sine-Gordon background, $-b'' + \sin b = 0$, and write $\phi = b + v$. For any fixed $T > 0$, equip $X_T = C([0,T]; \mathcal{E}_1)$ with the equivalent norm

$$\|z\|_a = \sup_{0 \le t \le T} e^{-at} \|z(t)\|_{\mathrm{lin}}, \qquad a > 0.$$

Derive the reference mass-one Duhamel force. Show that the Duhamel map is a contraction on the entire space X_T for $a = 4$, with no radius restriction. Prove existence and uniqueness on every finite interval and give a continuous-dependence bound. Why is an amplitude-dependent local constant unnecessary? Do not assume that b or $\sin b$ belongs to L^2.

Solution. Subtracting the stationary equation gives the force

$$N_b(v) = v - \sin(b+v) + \sin b, \qquad N_b(0) = 0, \qquad \partial_v N_b(v) = 1 - \cos(b+v) \in [0,2].$$

Hence $\|N_b(v) - N_b(w)\|_2 \le 2\|v - w\|_2$, independently of the background and amplitudes. In particular the force of every continuous energy path is continuous in L^2; no separate integrability of $\sin b$ is used. The Duhamel map of Problem 1 therefore maps the entire Banach space X_T to itself. For two paths its difference obeys

$$e^{-at}\|\Phi z(t) - \Phi \widetilde{z}(t)\|_{\mathrm{lin}} \le 2\|z - \widetilde{z}\|_a \int_0^t e^{-a(t-s)}\,\mathrm{d}s \le \frac{2}{a}\|z - \widetilde{z}\|_a.$$

For $a = 4$ it is a strict contraction without restricting to a ball. Banach's theorem gives a unique fixed point on each prescribed finite interval. Forced identification again gives exactly the weak energy solution. Restrictions of solutions constructed for different T agree by uniqueness, so they give a global solution; the negative direction follows by time reversal. For two data, the homogeneous contribution has weighted norm at most the initial difference. Absorbing the contraction term gives weighted difference at most twice that value. Removing the weight gives

$$\sup_{[0,T]} \|z(t) - \widetilde{z}(t)\|_{\mathrm{lin}} \le 2e^{4T}\|z_0 - \widetilde{z}_0\|_{\mathrm{lin}}.$$

The global bound on cosine, rather than a supremum over an expanding field range, is what removes amplitude dependence.

Exercise 15.4 (Defocusing continuation with the analytic input stated). For the three-dimensional cubic solution of Problem 2, assume additionally that the following energy identity holds throughout its maximal energy-class lifespan:

$$E_0 = \int_{\mathbb{R}^3} \left[\frac{1}{2}u_t^2 + \frac{1}{2}|\nabla u|^2 + \frac{1}{4}u^4\right] \mathrm{d}^3x.$$

This is an additional hypothesis at the weak energy regularity, not a consequence of L3's classical balance. Prove that the initial integral is finite and that the solution is global. Give a bound for its full $\mathcal{E}_3$ norm on every finite time interval, explicitly recovering the L^2 displacement term.

Solution. Hölder gives $\|u_0\|_4^4 \le \|u_0\|_2 \|u_0\|_6^3 < \infty$, so the prescribed initial energy is finite. Every term in the assumed conserved energy is nonnegative. It follows that

$$\|u_t(t)\|_2 \le \sqrt{2E_0}, \qquad \|\nabla u(t)\|_2 \le \sqrt{2E_0}.$$

The energy solution has the L^2 primitive $u(t) = u_0 + \int_0^t u_t(s)\mathrm{d}s$, hence

$$\|u(t)\|_2 \le \|u_0\|_2 + |t|\sqrt{2E_0}, \qquad \|(u(t), u_t(t))\|_{\mathrm{lin}}^2 \le \left(\|u_0\|_2 + |t|\sqrt{2E_0}\right)^2 + 4E_0.$$

This stays finite on every finite time interval, contradicting the continuation alternative at any finite maximal endpoint. Global existence here follows from Problem 2 plus the explicitly assumed weak conservation identity. The argument has not proved that analytic input from the classical balance at insufficient regularity.

Classical Field Theory — Exercise Sheet 16

Energy, plateaux and local blow-up

Solutions

Use the general one-dimensional local theorem proved on Sheet 15: for $V \in C^3$, $V'(0) = 0$, data in $H^1(\mathbb{R}) \times L^2(\mathbb{R})$ have a unique weak energy solution, and a finite maximal endpoint forces its full energy-space norm to tend to infinity. Its reference linear mass is one. The conservation argument below must be made at that weak regularity.

Exercise 16.1 (A nonnegative potential beyond the quartic). Let $V \in C^3(\mathbb{R})$ satisfy $V \geq 0$ and $V(0) = V'(0) = 0$. For the energy solution of $u_{tt} - u_{xx} + V'(u) = 0$ with data (u_0, g_0), set

$$E(t) = \int_{\mathbb{R}} \left[\frac{1}{2} u_t^2 + \frac{1}{2} u_x^2 + V(u) \right] \mathrm{d}x.$$

(a) Prove finiteness, continuity and conservation of E by smoothing the actual solution in space. Justify the potential chain rule and the endpoint limits. Do not assume a density theorem for nonlinear solutions.

(b) Prove global existence and give a finite-time bound on the full $H^1 \times L^2$ norm. For $V(s) = s^2/2 + s^6/6$, obtain a stronger bound uniform in time. Identify which part of the proof needs $V \geq 0$.

Solution. For the first part, fix a compact time interval K inside the lifespan. The one-dimensional embedding bounds $|u|$ by some M on $K \times \mathbb{R}$. Since $V(0) = V'(0) = 0$, boundedness of V'' on $[-M, M]$ gives

$$|V(a)| \leq C_M a^2, \quad |V'(a)| \leq C_M |a|, \quad |V(a) - V(b)| \leq C_M (|a| + |b|)|a - b|.$$

It follows that $V(u) \in C(K; L^1)$ and $V'(u) \in C(K; L^2)$. Thus the initial integral and the time-dependent energies are finite and continuous.

Choose a nonnegative compact smooth unit-mass mollifier and denote spatial convolution by a subscript ε. Compactness of the time images and the approximate-identity theorem give

$$u_\varepsilon \to u \text{ in } C(K; H^1), \qquad p_\varepsilon \to u_t \text{ in } C(K; L^2), \qquad |u_\varepsilon| \leq M.$$

The smoothed equation is

$$\partial_t p_\varepsilon = \partial_x^2 u_\varepsilon - (V'(u))_\varepsilon, \qquad (u_\varepsilon)_t = p_\varepsilon, \qquad R_\varepsilon = V'(u_\varepsilon) - (V'(u))_\varepsilon.$$

For fixed ε, the right side of the first equation is continuous in L^2. The norm-primitive argument makes p_ε continuously norm differentiable there. Also $\partial_t \partial_x u_\varepsilon = \partial_x p_\varepsilon \in C(K; L^2)$.

To justify the potential derivative, Taylor's theorem on the bounded range gives

$$\left| \int [V(u_\varepsilon(t+h)) - V(u_\varepsilon(t)) - V'(u_\varepsilon(t))(u_\varepsilon(t+h) - u_\varepsilon(t))] \mathrm{d}x \right| \leq C_M \|u_\varepsilon(t+h) - u_\varepsilon(t)\|_2^2.$$

The right side is $O(h^2)$ because u_ε is C^1 in L^2. Divide by h and use $V'(u_\varepsilon(t)) \in L^2$ to obtain $\frac{\mathrm{d}}{\mathrm{d}t} \int V(u_\varepsilon) = \int V'(u_\varepsilon) p_\varepsilon$. The derivative is continuous by the same Lipschitz bound on V'. The kinetic and spatial squared norms are also differentiable. Integration by parts with large spatial cutoffs leaves an error at most $C\|p_\varepsilon\|_2 \|\partial_x u_\varepsilon\|_2 / R$, tending to zero. Consequently

$$\frac{\mathrm{d}}{\mathrm{d}t} E[u_\varepsilon, p_\varepsilon] = \int p_\varepsilon R_\varepsilon \mathrm{d}x.$$

The residual tends to zero uniformly in $C(K;L^2)$: split it into $V'(u_\varepsilon)-V'(u)$ and $V'(u)-(V'(u))_\varepsilon$. The first is bounded by $C_M\|u_\varepsilon - u\|_2$ and the second converges by the compact-image approximate-identity argument. The velocities are uniformly bounded. Integrate between any two times in K and pass to the limit. The potential difference estimate above gives uniform convergence of the endpoint integrals, while the two quadratic terms converge by the norm limits. Hence $E(t) = E(s)$, including $s = 0$ whenever it is one of the endpoints. No approximation by exact smooth nonlinear solutions was used.

For the second part, positivity and conservation give $\|u_t\|_2, \|u_x\|_2 \le \sqrt{2E_0}$. The L^2 primitive gives

$$\|u(t)\|_2 \le \|u_0\|_2 + |t|\sqrt{2E_0}, \qquad \|(u(t),u_t(t))\|_{\mathrm{lin}}^2 \le \big(\|u_0\|_2 + |t|\sqrt{2E_0}\big)^2 + 4E_0.$$

Every finite-time endpoint is excluded by the continuation alternative. For $V(s) = s^2/2+s^6/6$, the energy is at least half the entire squared linear norm, so that squared norm is bounded by $2E_0$ for all time. The conservation proof used bounded derivatives on the solution range and the two vanishing conditions at zero. Nonnegativity is needed to extract the bounds that imply continuation, not to establish conservation itself.

Exercise 16.2 (Sine-Gordon energy does not bound every phase norm). Take $V(u) = 1 - \cos u$ and background zero.

(a) Use Problem 1 to prove energy conservation and global energy-class evolution. Verify explicitly the initial potential integrability and continuity needed for that argument.

(b) Choose a fixed nondecreasing $\eta \in C^\infty(\mathbb{R})$, zero for $s \le 0$ and one for $s \ge 1$. For $R \ge 1$ take initial displacement

$$u_R(x) = 2\pi\eta(x + R + 1)\eta(R + 1 - x)$$

and zero velocity. Compute its energy in terms of two integrals on $[0, 1]$ and show that it is independent of R, while $\|u_R\|_2 \to \infty$. Does this contradict global existence or the finite-time bound of Problem 1?

Solution. For the first part, the potential is nonnegative and smooth with $V(0) = V'(0) = 0$. Its integral is finite since $0 \le 1 - \cos u \le u^2/2$. Moreover, the fundamental theorem of calculus and $|\sin s| \le |s|$ give

$$|V(a) - V(b)| \le (|a| + |b|)|a - b|.$$

Cauchy–Schwarz proves continuity in L^1 along an L^2-continuous energy path. All hypotheses of Problem 1 apply. Its conservation proof and continuation bounds give global evolution, in agreement with Sheet 15's independent weighted-contraction construction.

For the second part, $u_R = 2\pi$ on $[-R, R]$ and is zero outside $[-R - 1, R + 1]$. On the left ramp it is $2\pi\eta(s)$ with $s = x + R + 1 \in [0, 1]$; the right ramp has the reversed coordinate and the same energy. The plateau and exterior contribute no gradient or potential. Thus

$$E[u_R, 0] = 4\pi^2 \int_0^1 \eta'(s)^2 \mathrm{d}s + 2\int_0^1 [1 - \cos(2\pi\eta(s))]\mathrm{d}s.$$

This is a fixed finite positive number. However, integrating the squared plateau gives

$$\|u_R\|_2^2 \ge \int_{-R}^{R} 4\pi^2 \mathrm{d}x = 8\pi^2 R \longrightarrow \infty.$$

Each datum is compact and smooth and has its own finite displacement norm. Problem 1's bound retains that initial norm, so it does not claim a common bound depending only on energy. The family contradicts no continuation result.

Exercise 16.3 (Focusing blow-up from compact data on the line). Change the potential to $V(u) = -u^4/4$ on $\mathbb{R}$, so $u_{tt} - u_{xx} = u^3$. Fix $0 < T < R$ and choose a real $\chi \in C_c^\infty(\mathbb{R})$ with $0 \le \chi \le 1$ and $\chi = 1$ on $[-R, R]$. Take

$$u_0 = \frac{\sqrt{2}}{T}\chi, \qquad g_0 = \frac{\sqrt{2}}{T^2}\chi.$$

For this conditional exercise, assume additionally that the unique energy solution for the displayed data has a C^2 representative throughout its maximal energy-class lifespan $[0, T_*)$. This is a regularity hypothesis, and assumes no global existence.

(a) Compute its finite initial focusing energy. Compare on the cone $|x| \le R - t$ with $q(t) = \sqrt{2}/(T - t)$ for every common time $0 \le t < \min\{T, T_*\}$. Prove the needed local uniqueness rather than assuming a global finite-energy theorem for q.

(b) Deduce $T_* \le T$. State carefully what was proved if the maximal solution terminates earlier than T. Why do these data improve on simply placing the spatially constant ODE on all of $\mathbb{R}$?

Solution. For the first part, direct substitution of the data gives

$$E_0 = \int \left[\frac{1}{2}g_0^2 + \frac{1}{2}(u_0')^2 - \frac{1}{4}u_0^4\right] \mathrm{d}x = T^{-2}\int \chi'^2 \mathrm{d}x + T^{-4}\int (\chi^2 - \chi^4)\mathrm{d}x.$$

Both integrals are finite and nonnegative. The cutoff is nonconstant, so the derivative integral is positive. Thus the data have positive finite energy and belong to $H^1 \times L^2$. The local theorem applies despite the focusing sign. The ODE field satisfies $q_{tt} = q^3$ with the same data on $[-R, R]$.

Fix any $t_1 < \min\{T, T_*\}$. The assumed regularity makes the difference $w = u - q$ classical on the compact cone up to t_1. It satisfies

$$w_{tt} - w_{xx} = a(t, x)w, \qquad a = u^2 + uq + q^2.$$

The coefficient is bounded on that compact cone. Use the positive local energy

$$\mathcal{E}(t) = \frac{1}{2}\int_{-R+t}^{R-t} (w_t^2 + w_x^2 + w^2)\mathrm{d}x.$$

The moving-interval FTC gives

$$\begin{aligned}\mathcal{E}'(t) = &-\frac{1}{2}[(w_t - w_x)^2 + w^2](t, R-t) - \frac{1}{2}[(w_t + w_x)^2 + w^2](t, -R+t)\\ &+ \int_{-R+t}^{R-t} (a+1)ww_t \mathrm{d}x \le C\mathcal{E}(t), \qquad C = \sup_{\text{cone}} |a+1|.\end{aligned}$$

Here $2|ww_t| \le w^2 + w_t^2$ controls the last term. The initial energy is zero. Differentiating $e^{-Ct}\mathcal{E}(t)$ shows it is nonincreasing and nonnegative, hence zero. Thus $u = q$ on the cone up to every such t_1, in particular at its centre. Only a local positive comparison energy was used; q is not an L^2 field on the line.

For the second part, if $T_* > T$, continuity into H^1 would bound $\|u(t)\|_\infty$ on the compact interval $[0, T]$. But $u(t, 0) = \sqrt{2}/(T - t) \to \infty$ there, a contradiction. Hence $T_* \le T$ and

the local continuation alternative forces the full energy norm to diverge at its actual finite endpoint. If $T_* < T$, the comparison asserted equality only up to that common lifespan; it did not assert existence until T or locate the earlier blow-up. The cutoff produces genuine finite-energy line data, unlike a nonzero spatially constant ODE datum. The conservation part of Problem 1 also applies to this signed potential, but its positivity step fails: a conserved focusing energy cannot bound the full evolution norm.

Classical Field Theory — Exercise Sheet 17

Forced and accidental massless directions

Solutions

Use the real scalar kinetic coefficient $1/2$ and the orbit–Hessian lemma of Lecture 17. Counts of massless modes below refer to the linearised field. A vacuum background need not belong to L^2; perturbation data may be Schwartz or in the earlier energy class.

Exercise 17.1 (The full orthogonal orbit). For $N \geq 3$, $\lambda, v > 0$, consider

$$V(u) = \frac{\lambda}{8}(|u|^2 - v^2)^2, \qquad u \in \mathbb{R}^N, \qquad u_0 = vn, \quad |n| = 1,$$

with the usual $O(N)$ action and canonical kinetic term. Find the vacuum set, the stabiliser at u_0, its dimension, and the orbit tangent. Derive the Hessian and diagonalise the linearised field using the radial and tangential projectors, without first rotating n to a coordinate axis. Identify which masses are symmetry-forced. Finally discuss $N = 1$ separately: does its broken discrete symmetry force a massless mode?

Solution. Nonnegativity shows that the minima are exactly $|u| = v$. The stabiliser fixes n and restricts to an arbitrary orthogonal transformation of $n^\perp$, so

$$\dim O(N) = \frac{N(N-1)}{2}, \qquad \dim H = \frac{(N-1)(N-2)}{2}, \qquad \dim O(N)/H = N - 1.$$

For every skew matrix T, $n \cdot Tn = 0$. Conversely, for $w \perp n$, $T = (wn^{\mathsf{T}} - nw^{\mathsf{T}})/v$ satisfies $Tu_0 = w$. Thus the orbit tangent is exactly $n^\perp$. Direct differentiation gives

$$\nabla V(u) = \frac{\lambda}{2}(|u|^2 - v^2)u, \qquad D^2V(u) = \frac{\lambda}{2}\big[(|u|^2 - v^2)I + 2uu^{\mathsf{T}}\big].$$

At u_0, this is $\lambda v^2 nn^{\mathsf{T}}$. For $P_\parallel = nn^{\mathsf{T}}$ and $P_\perp = I - P_\parallel$, the perturbation $h = nh_\parallel + h_\perp$ therefore obeys

$$(\Box + \lambda v^2)h_\parallel = 0, \qquad \Box h_\perp = 0.$$

The orthogonal decomposition preserves the canonical kinetic coefficient. All $N - 1$ zero modes are orbit directions, and no additional kernel remains. For $N = 1$, the vacuum set is the two points $\{-v, v\}$ and the stabiliser of either point in $O(1)$ is trivial. Both groups have dimension zero; there is no continuous orbit tangent. The Hessian is the positive scalar λv^2, so breaking this discrete symmetry gives no Goldstone mode.

Exercise 17.2 (A flat singlet still has a nonlinear restoring force). Add a real scalar z with its own canonical kinetic term and potential

$$\widetilde{V}(u, z) = \frac{\lambda}{8}(|u|^2 - v^2)^2 + z^4.$$

Let $O(N)$ act on u and trivially on z. Find the vacuum set, the Hessian kernel at $(vn, 0)$, and the subspace forced by the symmetry. Can the extra singlet direction be tangent to a curve of vacua? On the spatial torus, set $u = vn$ and take a homogeneous singlet $z(0) = A > 0$, $z_t(0) = 0$. Derive its nonlinear period as a definite integral and its dependence on A. Explain how a zero linearised mass is compatible with this nonlinear oscillation. These homogeneous data are not claimed to be L^2 data on $\mathbb{R}^d$.

Solution. The minima require $|u| = v$ and $z = 0$. At $(vn, 0)$ the Hessian and its kernel are

$$D^2\widetilde{V} = \begin{pmatrix} \lambda v^2 n n^{\mathsf{T}} & 0 \\ 0 & 0 \end{pmatrix}, \qquad \ker D^2\widetilde{V} = (n^{\perp} \times \{0\}) \oplus (\{0\} \times \mathbb{R}).$$

The orbit tangent is just $n^{\perp} \times \{0\}$, of dimension $N - 1$. Every curve of vacua has its singlet coordinate identically zero, so its tangent cannot be the extra singlet direction. That zero is accidental.

For the torus solution the singlet equation is $z_{tt} + 4z^3 = 0$. Multiplication by z_t gives

$$\frac{1}{2} z_t^2 + z^4 = A^4, \qquad |z_t| = \sqrt{2(A^4 - z^4)}.$$

The orbit goes between $-A$ and A; uniqueness for this smooth mechanical ODE and its time-reversal symmetry give four equal quarter-periods. Changing variable $z = As$ in one quarter gives

$$T_A = 4 \int_0^A \frac{\mathrm{d}z}{\sqrt{2(A^4 - z^4)}} = \frac{2\sqrt{2}}{A} \int_0^1 (1 - s^4)^{-1/2} \mathrm{d}s.$$

The integral is finite because its endpoint singularity is proportional to $(1 - s)^{-1/2}$. Thus the nonlinear restoring force gives a finite period at every nonzero amplitude, while $T_A \to \infty$ as $A \downarrow 0$. The zero Hessian captures the infinitesimal limit, not absence of the cubic restoring force or the presence of a vacuum curve.

Exercise 17.3 (An ineffective group factor belongs to the stabiliser). Take $G = SO(3) \times U(1)$ acting on a real triplet by $(R, e^{\mathrm{i}\theta})u = Ru$. Use the radial potential of Problem 1 with $N = 3$ and vacuum $u_0 = v(2, -1, 2)/3$. Determine the full stabiliser and the differential of the orbit map $(c, \theta) \mapsto c \times u_0$ on $\mathbb{R}^3 \oplus \mathbb{R}$. Find a basis of its kernel, the number of broken directions and the Hessian nullity. Why would counting $\dim G$ or only counting rotations that fix u_0 give a wrong answer?

Solution. The rotation stabiliser consists of rotations around the line through $(2, -1, 2)$, hence is $SO(2)$. Every phase factor fixes every triplet. Therefore

$$H = SO(2)_{u_0} \times U(1), \qquad \dim G = 4, \quad \dim H = 2, \quad \dim G/H = 2.$$

The infinitesimal action is $c \times u_0$, independent of θ. Its kernel requires c parallel to $(2, -1, 2)$ and leaves θ arbitrary. A basis of the kernel in the four real parameters is

$$(2, -1, 2, 0), \qquad (0, 0, 0, 1).$$

Its rank is two. The Hessian is $\lambda u_0 u_0^{\mathsf{T}}$ of rank one, so its nullity is also two and matches the orbit tangent. Counting all four generators would include both unbroken parameters. Subtracting only the one-dimensional rotation stabiliser would still incorrectly count the ineffective phase as broken. The full stabiliser is what appears in the orbit–Hessian lemma.

Exercise 17.4 (Kinetic normalisation before diagonalising masses). For a real three-component perturbation h, consider the quadratic density

$$L = \frac{1}{2}\partial_\mu h^{\mathsf{T}} K \partial^\mu h - \frac{1}{2} h^{\mathsf{T}} M h, \qquad K = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 9 & 9 \\ 0 & 9 & 9 \end{pmatrix}.$$

The real phase data may be taken Schwartz. Verify the factorisation $K = S^{\mathsf{T}} S$ for

$$S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Derive the canonical momentum, and find the corresponding change of coordinates and momenta that preserves $\int \delta h_a \wedge \delta p_a$. Compute the physical squared masses and compare them with the raw eigenvalues of M. State the general eigenproblem and explain what happens to the kernel dimension under this kinetic normalisation.

Solution. Multiplying the given matrices gives $S^{\mathsf{T}} S = K$ and $\det S = -1$, so K is positive definite. The velocity derivative gives $p = K h_t$. For $Q = Sh$ and $P = S^{-\mathsf{T}} p$, the canonical potential is preserved:

$$p^{\mathsf{T}} \delta h = P^{\mathsf{T}} \delta Q.$$

Taking its negative exterior derivative proves preservation of the canonical form, including the momentum transformation's inverse transpose. Also $P = S h_t = Q_t$, so the kinetic term in Q is canonical. The transformed density is

$$L = \frac{1}{2}\partial_\mu Q^{\mathsf{T}} \partial^\mu Q - \frac{1}{2} Q^{\mathsf{T}} D Q, \qquad D = S^{-\mathsf{T}} M S^{-1} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{pmatrix}.$$

Thus its squared masses are $0, 4, 9$, whereas the nonzero two-by-two block of M itself has eigenvalues $0, 18$ in addition to its separate eigenvalue4. In general the field equation is $K \Box h + M h = 0$, giving

$$M w = m^2 K w$$

for a mode of squared mass m^2. An invertible congruence preserves the kernel dimension, even though it does not preserve the nonzero raw eigenvalues. Here the zero direction $(0, 1, -1)^T$ becomes $Q = e_1$. A symmetry-forced kernel therefore remains massless under normalisation, but physical nonzero masses must use the kinetic pairing as well as the potential Hessian.

Classical Field Theory — Exercise Sheet 18

Spectrum and moving kinks

Solutions

Retain $b(x) = \tanh x$, $V(s) = \frac{1}{2}(1-s^2)^2$ and $H_K = -\partial_x^2 + 4 - 6\operatorname{sech}^2 x$ with domain $H^2 \subset L^2$. Lecture 18 proves $H_K = A^\dagger A$ for $A = \partial_x + 2\tanh x$ and $\ker H_K = \operatorname{span}\{\operatorname{sech}^2 x\}$. Use its H^1 form $\mathfrak{q}[h] = \|Ah\|_2^2$ and density/integration-by-parts results.

Exercise 18.1 (The full spectrum through two adjoint pairs). The following analytic spectral theorem is given for this problem: if W is real, bounded and smooth, with $W(x) \to 0$ at both ends, then $-\partial_x^2 + 4 + W$ on $H^2(\mathbb{R})$ is self-adjoint, its essential spectrum is $[4, \infty)$, and every spectral point outside that interval is an isolated finite-multiplicity eigenvalue. This follows from Teschl, *Mathematical Methods in Quantum Mechanics* (2009), Lemmas 9.33 and 9.35, pp. 215–216, after shifting by four; the discrete/essential spectrum convention is in §6.4, p. 145. This input supplies more than the formal factorisation in Lecture 18. Set $B = \partial_x + \tanh x$ with domain H^1.

(a) Find $B^\dagger$ and the product domains, and prove $AA^\dagger = B^\dagger B + 3$ and $BB^\dagger + 3 = -\partial_x^2 + 4$. Use these identities to classify all L^2 eigenvalues of H_K, including possible embedded or threshold eigenvalues. Check the domains when intertwining.

(b) Normalise its two bound states. Exhibit a bounded solution at spectral value four and explain why it is not another L^2 eigenstate. Using the given theorem, state the full spectrum.

Solution. For the first part, compact integration by parts gives $B^\dagger = -\partial_x + \tanh x$ on H^1. Conversely a vector in the adjoint domain has distributional derivative in L^2, because the coefficient is bounded, so that is the full domain. Multiplication by $\tanh x$ preserves H^1. As in Lecture 18, $u \in H^1$ and $Bu \in H^1$ are equivalent to $u \in H^2$; the same argument applies to $B^\dagger$ and $A^\dagger$. All product domains in the following identities are therefore H^2:

$$AA^\dagger = -\partial_x^2 + 4 - 2\operatorname{sech}^2 x = B^\dagger B + 3, \qquad BB^\dagger + 3 = -\partial_x^2 + 4 =: H_0.$$

The identities first follow by product differentiation on compact tests, then extend by H^2 density.

Suppose $H_K u = \lambda u$ with $u \in H^2$ nonzero. Self-adjointness makes λ real, and factorisation makes it nonnegative. The zero case is the known one-dimensional kernel. The equation $u'' = (4 - 6\operatorname{sech}^2 x - \lambda)u$ and the bounded derivatives of its coefficient bootstrap u into every integer Sobolev space: Leibniz' rule makes multiplication by this coefficient bounded on each integer H^s; if $u \in H^s$, its right side is in H^s, so Fourier transformation gives $u \in H^{s+2}$. Products and first derivatives in the following intertwinings thus belong to the actual operator domains. For $\lambda > 0$, $v = Au$ is nonzero and

$$(B^\dagger B + 3)v = \lambda v.$$

If $Bv \neq 0$, then $z = Bv \in H^2 \cap L^2$ obeys $H_0 z = \lambda z$. Fourier transformation gives $(k^2 + 4 - \lambda)\widehat{z} = 0$. The zero set of this multiplier is empty or finite, hence has measure zero; no nonzero L^2 Fourier profile can be supported there. Thus $Bv = 0$. Solving the first-order equation gives $v = C\operatorname{sech} x$, and then necessarily $\lambda = 3$. The map A is injective on a positive eigenspace, so

this candidate has multiplicity at most one. This excludes all other eigenvalues, including those in $[4,\infty)$.

For the second part, $A^\dagger \operatorname{sech} x = 3\operatorname{sech} x \tanh x$ constructs a nonzero eigenfunction at three. The substitution $z = \tanh x$ gives

$$\int \operatorname{sech}^4 x\,\mathrm{d}x = \frac{4}{3}, \qquad \int \operatorname{sech}^2 x \tanh^2 x\,\mathrm{d}x = \frac{2}{3}.$$

Thus the positive normalisations are

$$\psi_0 = \frac{\sqrt{3}}{2}\operatorname{sech}^2 x, \qquad \psi_3 = \sqrt{\frac{3}{2}}\operatorname{sech} x \tanh x.$$

They are orthogonal by parity. Applying the formal differential expressions of $A^\dagger B^\dagger$ to the free threshold solution one gives $3\tanh^2 x - 1$; this step does not apply the L^2 operators to the non-L^2 constant. Direct substitution verifies that the resulting smooth profile solves the differential equation $H_K u = 4u$. Half this function tends to one at both ends, so it is bounded but not L^2. Finally the given theorem supplies essential spectrum $[4,\infty)$ and allows only isolated eigenvalues elsewhere. The classification above proves

$$\sigma(H_K) = \{0,3\} \cup [4,\infty).$$

No claim about nonlinear stability follows from this spectral calculation.

Exercise 18.2 (A moving kink and the sign of momentum). For $|v_{\mathrm{tr}}| < 1$, set $\gamma = (1-v_{\mathrm{tr}}^2)^{-1/2}$ and

$$\phi(t,x) = \tanh\bigl(\gamma(x - v_{\mathrm{tr}}t - x_0)\bigr).$$

Verify the nonlinear field equation directly. At fixed time compute the charge, full energy and the lower-index translation integral $P_1 = \int \phi_t \phi_x \mathrm{d}x$. Find the physical raised momentum P^1 and the relation between its square and the energy square. Check the result at $v_{\mathrm{tr}} = 3/5$. Justify membership in the affine class $\phi - b \in H^1$, $\phi_t \in L^2$.

Solution. Write $\xi = \gamma(x - v_{\mathrm{tr}}t - x_0)$. The field equation becomes

$$\phi_{tt} - \phi_{xx} + V'(\phi) = \bigl[\gamma^2(v_{\mathrm{tr}}^2 - 1) + 1\bigr] b''(\xi) = 0.$$

At fixed time the spatial endpoint limits remain -1 and 1, so the charge is one. The derivatives and potential are

$$\phi_x = \gamma \operatorname{sech}^2 \xi, \qquad \phi_t = -\gamma v_{\mathrm{tr}} \operatorname{sech}^2 \xi, \qquad V(\phi) = \frac{1}{2}\operatorname{sech}^4 \xi.$$

Since $\mathrm{d}x = \mathrm{d}\xi/\gamma$ and $\int \operatorname{sech}^4 \xi \mathrm{d}\xi = 4/3$,

$$E = \frac{\gamma^2(1+v_{\mathrm{tr}}^2)+1}{2\gamma}\frac{4}{3} = \frac{4}{3}\gamma, \qquad P_1 = -\gamma v_{\mathrm{tr}}\frac{4}{3}.$$

Lecture 3 gives $P^1 = -P_1$, whence $E^2 - (P^1)^2 = 16\gamma^2(1-v_{\mathrm{tr}}^2)/9 = 16/9$. For $v_{\mathrm{tr}} = 3/5$, $\gamma = 5/4$ and the values are $E = 5/3$, $P_1 = -1$, $P^1 = 1$. Both the boosted profile and b approach the same vacua exponentially at each fixed time. Their difference and its derivative therefore decay exponentially, as does ϕ_t. This proves the stated affine Sobolev membership.

Exercise 18.3 (Removing the linear translation drift). Use the spectral theorem for the self-adjoint operator in Problem 1 as an additional analytic input: on the orthogonal complement of its zero mode, the form-domain inequality is $\mathfrak{q}[h] \geq 3\|h\|_2^2$ for $h \in H^1$ with $(\psi_0, h) = 0$. Prove $\|h\|_{H^1}^2 \leq 2\mathfrak{q}[h]$ on this subspace. For any real solution of $h_{tt} + H_K h = 0$ with

$$h \in C(H^2) \cap C^1(H^1) \cap C^2(L^2),$$

prove that $(\psi_0, h(0)) = (\psi_0, h_t(0)) = 0$ preserves this orthogonality, and deduce a uniform bound on $\|h(t)\|_{H^1}^2 + \|h_t(t)\|_2^2$ from its initial quadratic energy. No existence theorem for this new linear equation is required: the assertion is conditional on a solution in the displayed class.

Solution. The form identity and the coefficient bound give

$$\|h_x\|_2^2 \leq \mathfrak{q}[h] + 2\|h\|_2^2, \qquad \|h\|_{H^1}^2 \leq \mathfrak{q}[h] + 3\|h\|_2^2 \leq 2\mathfrak{q}[h].$$

For the stated solution regularity, $c(t) = (\psi_0, h(t))$ is twice continuously differentiable and self-adjointness gives $c'' = -(\psi_0, H_K h) = -(H_K \psi_0, h) = 0$. Both initial values vanish, hence $c(t) = 0$ throughout the interval. The actual H^2 domain and the H^1 time derivative justify integration by parts when differentiating the quadratic form. It gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}_{\mathrm{lin}} = (h_t, h_{tt} + H_K h) = 0.$$

Applying the form bound at each time yields

$$\|h(t)\|_{H^1}^2 + \|h_t(t)\|_2^2 \leq 2\mathfrak{q}[h(t)] + \|h_t(t)\|_2^2 \leq 4\mathcal{E}_{\mathrm{lin}}(0).$$

Nonlinear evolution does not satisfy this projected linear equation. A nonlinear moving centre and its orthogonality condition would require additional arguments; the estimate itself establishes only the stated linear control.

Exercise 18.4 (The nonlinear energy remainder on a fixed slice). For real $h \in H^1(\mathbb{R})$, compute the exact remainder in $E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] - \mathfrak{q}[h]/2$. Prove a bound using only $\|h\|_{H^1}$. If $(\psi_0, h) = 0$, show that for $\|h\|_{H^1} \leq 1/(16\sqrt{2})$,

$$E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] \geq \frac{1}{8}\|h\|_{H^1}^2.$$

Why is this static estimate insufficient by itself to prove nonlinear orbital stability of the kink?

Solution. The polynomial expansion gives

$$V(b+h) = V(b) + V'(b)h + \tfrac{1}{2}V''(b)h^2 + 2bh^3 + \tfrac{1}{2}h^4.$$

The kinetic cross term and the linear potential term integrate to zero by $-b'' + V'(b) = 0$. This holds first for compact smooth perturbations, then for H^1 perturbations by density, since $b', b'', V'(b) \in L^2$. All cubic and quartic terms are integrable by the one-dimensional embedding. Thus the exact remainder is

$$\mathcal{R}[h] = \int (2bh^3 + \tfrac{1}{2}h^4)\mathrm{d}x.$$

Using $|b| \le 1$ and $\|h\|_\infty \le \|h\|_{H^1}/\sqrt{2}$ gives

$$|\mathcal{R}[h]| \le 2\|h\|_\infty \|h\|_2^2 + \tfrac{1}{2}\|h\|_\infty^2 \|h\|_2^2 \le \sqrt{2}\|h\|_{H^1}^3 + \tfrac{1}{4}\|h\|_{H^1}^4.$$

Under the orthogonality condition, $\mathfrak{q}[h]/2 \ge \|h\|_{H^1}^2/4$. Writing $\delta = 1/(16\sqrt{2})$, the remainder coefficient satisfies $\sqrt{2}\delta + \delta^2/4 = 1/16 + 1/2048 < 1/8$. Subtracting it proves the requested lower bound. This controls a fixed slice satisfying a fixed orthogonality condition. It neither selects a moving translation parameter nor proves that the nonlinear solution remains on such slices. Those are among the missing steps in an orbital-stability proof.

Classical Field Theory — Exercise Sheet 19

Adjoint identities, matter and surface currents

Solutions

All fields and coefficients on this sheet are smooth. Variations and gauge parameters in action identities have compact support inside the coordinate region. Use the Euler sign $\delta S = \int E_a \delta\phi^a$ and the connection conventions of Lecture 19. No equations of motion are imposed unless explicitly stated.

Exercise 19.1 (A second-order adjoint and the elementary converse). On an open coordinate patch $U \subset \mathbb{R}^{1+1}$ with coordinates (t, x), let a scalar parameter $\varepsilon \in C_c^\infty(U)$ produce two variations

$$R(\varepsilon)^1 = (1+tx)\varepsilon_{tt} + 2(t+x^2)\varepsilon_{tx} + e^t\varepsilon_x, \qquad R(\varepsilon)^2 = (1+t^2)\varepsilon_{xx} + x\varepsilon.$$

Compute $R^*(E_1, E_2)$, first in divergence form and then with every coefficient derivative expanded. Prove the compact pairing identity. Suppose an action has the compact first variation $\int E_a[\phi]h^a$ and $R^*_\phi E[\phi] = 0$ on U for every smooth field. Prove that $R_\phi(\varepsilon)$ is an infinitesimal gauge symmetry. Explain why the same converse holds for any finite-order operator $R_\phi(\varepsilon)^a = \sum_I r^{aI}_\alpha[\phi]\partial_I\varepsilon^\alpha$, without interpreting it as a converse for arbitrary conserved currents.

Solution. Repeated coordinate integration by parts gives

$$R^*E = \partial_t^2[(1+tx)E_1] + 2\partial_t\partial_x[(t+x^2)E_1] - \partial_x(e^tE_1) + \partial_x^2[(1+t^2)E_2] + xE_2.$$

For example the mixed term expands as $2(t+x^2)E_{1,tx} + 2E_{1,x} + 4xE_{1,t}$. The first term adds $(1+tx)E_{1,tt} + 2xE_{1,t}$. The other terms therefore yield the full expression

$$R^*E = (1+tx)E_{1,tt} + 2(t+x^2)E_{1,tx} + 6xE_{1,t} + (2-e^t)E_{1,x} + (1+t^2)E_{2,xx} + xE_2.$$

Each discarded derivative contains a derivative of the compact parameter. Extend by zero and use Fubini/FTC to remove it. This proves $\int E_a R(\varepsilon)^a = \int (R^*E)\varepsilon$ without a boundary theorem.

For the converse, apply the supplied first variation to the compact variation $h = R_\phi(\varepsilon)$. At the fixed field its coefficients are smooth, so this variation is still compactly supported in U. The pairing identity and the assumed off-shell identity give

$$\delta S[\phi]R_\phi(\varepsilon) = \int (R^*_\phi E[\phi])\varepsilon = 0.$$

The same finite integrations by parts give $(R^*_\phi E)_\alpha = \sum_I (-1)^{|I|}\partial_I(r^{aI}_\alpha E_a)$ for arbitrary finite order. Hence the same conclusion holds on U. Holding the field fixed does not freeze the evaluated coefficients under coordinate differentiation. No assertion that an arbitrary conserved current comes from a symmetry was used or proved.

Exercise 19.2 (The Euler term carried by charged matter). Write $A_{\mathrm{em}} = A_\mu \mathrm{d}x^\mu$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and let $u = (u_1, u_2)^{\mathsf{T}}$ be two real scalar fields. Given $q, \lambda, v > 0$, set

$$T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D_\mu u = \partial_\mu u - q A_\mu T u, \qquad L = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} D_\mu u \cdot D^\mu u - \frac{\lambda}{8}(|u|^2 - v^2)^2.$$

The gauge variations are $\delta A_\mu = \partial_\mu \varepsilon$ and $\delta u = q\varepsilon T u$.

(a) Verify covariance of $D_\mu u$, derive both Euler arrays by compact variation, and derive the off-shell Noether identity containing them.

(b) On a coordinate patch take $A_\mu = 0$ and $u = (t, t^2)^{\mathsf{T}}$. Compute the ordinary divergence of the total gauge Euler array and its balancing scalar Euler term. Does the pure Maxwell identity remain an identity for that total array by itself?

(c) For a real orthogonal representation with skew generators t_a, use instead the supplied variations $\delta A_\mu = D_\mu \varepsilon$, $\delta u = -\varepsilon^a t_a u$. Assume these variations are a gauge symmetry for every compact parameter. If the compact first variation is $\int [\langle E_A^\mu, \delta A_\mu \rangle + E_u \cdot \delta u]$, derive its covariant component identity. Here A is a connection as in Lecture 19, and the Lie algebra basis is orthonormal in its invariant pairing.

Solution. For the first part, direct differentiation cancels $q(\partial_\mu \varepsilon) T u$ against the potential variation and leaves $\delta(D_\mu u) = q\varepsilon T(D_\mu u)$. Skewness of T and radiality of the potential show $\delta L = 0$. The covariant product rule, with $T^{\mathsf{T}} = -T$, permits compact integration by parts. It gives

$$E_u = -D_\mu D^\mu u - \frac{\lambda}{2}(|u|^2 - v^2)u, \qquad E_A^\mu = \partial_\nu F^{\nu\mu} - q(Tu) \cdot D^\mu u.$$

In particular the matter contribution to E_A^μ has a minus sign, because $\delta_A D_\mu u = -q\delta A_\mu T u$. Substituting the gauge variation into the first variation yields

$$0 = \int \left[E_A^\mu \partial_\mu \varepsilon + q E_u \cdot Tu\, \varepsilon \right] = \int \left[-\partial_\mu E_A^\mu + q E_u \cdot Tu \right] \varepsilon.$$

The compact smooth-test fundamental lemma proves the local off-shell identity.

For the second part, $Tu = (-t^2, t)^{\mathsf{T}}$ and $(Tu) \cdot u_t = t^2$. All spatial derivatives and the curvature vanish. Thus

$$E_A^0 = -qt^2, \quad E_A^i = 0, \quad \partial_\mu E_A^\mu = -2qt.$$

The radial potential term is orthogonal to Tu, while $-u_{tt} \cdot Tu = -(0, 2) \cdot (-t^2, t) = -2t$. Hence $qE_u \cdot Tu = -2qt$, exactly balancing the divergence. Except at $t = 0$, that divergence is nonzero. The identity $\partial_\mu \partial_\nu F^{\nu\mu} = 0$ still holds for the pure curvature part, but it does not annihilate the total matter-coupled Euler array.

For the final part, invariance and compact integration by parts give

$$0 = \int \left[-(D_\mu E_A^\mu)^a - E_u \cdot t_a u \right] \varepsilon^a.$$

The fundamental lemma proves $(D_\mu E_A^\mu)^a + E_u \cdot t_a u = 0$. This sign uses the explicitly supplied connection variation and matter variation; it must not be guessed by replacing an ordinary derivative in the abelian formula without also translating its coupling convention.

Exercise 19.3 (Noncommuting constant connections). Let $T_a = -\mathrm{i}\sigma_a/2$, so $[T_a, T_b] = \epsilon_{abc} T_c$. In pure Yang–Mills theory with coupling $g > 0$, take first $A_1 = T_1$, $A_2 = T_2$, $A_0 = A_3 = 0$, all constant. Compute every nonzero curvature and Euler component $E^\nu = g^{-2} D_\mu F^{\mu\nu}$, and check the covariant identity. Repeat after changing A_3 to $T_1 + T_2$. Exhibit the separate nonzero commutator contributions to $D_\nu E^\nu$ in that second case. What can these constant examples establish about an ordinary-divergence identity for arbitrary nonconstant connections?

Solution. For the first connection the only independent curvature is $F_{12} = [T_1, T_2] = T_3$, with $F_{21} = -T_3$. All temporal curvature and E^0 vanish. Spatial raising gives $F^{ij} = F_{ij}$, so

$$E^1 = g^{-2}[T_2, -T_3] = -g^{-2}T_1, \qquad E^2 = g^{-2}[T_1, T_3] = -g^{-2}T_2, \qquad E^3 = 0.$$

Each $[A_i, E^i]$ vanishes in this first case. For the second connection the independent curvatures are

$$F_{12} = T_3, \qquad F_{13} = T_3, \qquad F_{23} = -T_3,$$

with the reversed-index negatives and all temporal entries zero. Summing the two relevant commutators for each spatial Euler component gives

$$E^1 = g^{-2}(-2T_1 + T_2), \quad E^2 = g^{-2}(T_1 - 2T_2), \quad E^3 = -g^{-2}(T_1 + T_2), \quad E^0 = 0.$$

Their ordinary derivatives vanish because the fields are constant, but

$$[A_1, E^1] = g^{-2}T_3, \qquad [A_2, E^2] = -g^{-2}T_3, \qquad [A_3, E^3] = 0.$$

The nonzero terms cancel, verifying the covariant identity without erasing its commutators separately. Ordinary divergence happens to vanish for these constant fields. That observation cannot prove an off-shell ordinary-divergence identity for general connections; Lecture 19's variable-connection guard exhibits why the covariant derivative is essential.

Exercise 19.4 (A local superpotential theorem and a Maxwell verification). For this exercise, assume the following local decomposition statement. It is an additional hypothesis here, not a theorem to infer from the fundamental lemma alone. For a finite-order local Lagrangian of unconstrained $\mathbb{R}^N$-valued fields on a star-shaped coordinate patch of spacetime dimension at least two, a finite-order infinitesimal gauge symmetry has a local Noether current of the form

$$j^\mu_\varepsilon = W^\mu_\varepsilon + \partial_\nu U^{\nu\mu}_\varepsilon, \qquad U^{\nu\mu}_\varepsilon = -U^{\mu\nu}_\varepsilon.$$

The current and these expressions depend linearly on the parameter and finitely many of its derivatives, with smooth finite-order local field coefficients. The remainder W vanishes when the Euler expressions and their derivatives vanish. This is a local statement on the full jet space; no global topological superpotential theorem is supplied.

Verify it directly for

$$L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + A_\mu j^\mu_{\text{ext}}, \qquad \partial_\mu j^\mu_{\text{ext}} = 0, \qquad \delta A_\mu = \partial_\mu \varepsilon,$$

where the external current is fixed and smooth. Derive the gauge Noether current, its Euler remainder and an explicit U. On solutions, compare its spatial integral for compact spatial parameters with a parameter equal to a constant ε_∞ outside a ball. For the latter comparison assume a smooth Maxwell field whose external current has compact spatial support on the slice and total charge $Q = \int j^0_{\text{ext}} \mathrm{d}^3x$. Use $F_{i0} = E_i$ and an outward sphere orientation to find the surface sign. No Schwartz or L^2 condition on the potential is imposed in this boundary comparison.

Solution. The source conservation law gives $\delta L = j^\mu_{\text{ext}}\partial_\mu \varepsilon = \partial_\mu(\varepsilon j^\mu_{\text{ext}})$. The momentum derivative with respect to $\partial_\mu A_\nu$ is $-F^{\mu\nu}$, so Lecture 2's formula gives

$$j^\mu_\varepsilon = -F^{\mu\nu}\partial_\nu \varepsilon - \varepsilon j^\mu_{\text{ext}}, \qquad E^\mu_A = \partial_\nu F^{\nu\mu} + j^\mu_{\text{ext}}.$$

Antisymmetry of F and the ordinary product rule give the exact off-shell decomposition

$$j^{\mu}_{\varepsilon} = -\varepsilon E^{\mu}_{A} + \partial_{\nu} U^{\nu\mu}_{\varepsilon}, \qquad U^{\nu\mu}_{\varepsilon} = \varepsilon F^{\nu\mu}.$$

This verifies the assumed decomposition here with an Euler remainder, by direct calculation. The background source need not satisfy a field equation; its prescribed conservation is what ensured gauge invariance up to a divergence.

On a solution the time-slice density is a spatial divergence, since $U^{00} = 0$. For a compact spatial parameter, U^{i0} vanishes outside a compact set, so coordinate cutoff integration gives $\int j^{0}_{\varepsilon} = 0$. For a parameter constant outside a ball containing the source support, its current density is still compactly supported: the parameter derivatives and the source term vanish outside that ball. Using Lecture 5's ball formula, for every sufficiently large sphere,

$$\int j^{0}_{\varepsilon} \mathrm{d}^3x = \int_{S^2_r} \varepsilon_{\infty} F^{i0} n_i \mathrm{d}S = -\varepsilon_{\infty} \int_{S^2_r} \mathbf{E} \cdot n \mathrm{d}S = -\varepsilon_{\infty} Q.$$

The minus sign comes from $F^{i0} = -E_i$. The electric Gauss charge itself is $Q = +\int \mathbf{E} \cdot n$; its sign was not changed. Compact-parameter identities cannot discard the surface term of a parameter with a nonzero limit. The explicit Maxwell calculation establishes this boundary comparison without claiming that the assumed local statement solves a global gauge or topological problem.

Classical Field Theory — Exercise Sheet 20

Nonabelian constraints and local reduction

Solutions

Use $T_a = -\mathrm{i}\sigma_a/2$ for $\mathfrak{su}(2)$, so $[T_a, T_b] = \epsilon_{abc}T_c$ and $\langle T_a, T_b\rangle = \delta_{ab}$. All spatial fields are smooth on $\mathbb{T}^3$. Write $\mathcal{V}_{\mathrm{per}} = (2\pi)^3$ for its volume. The canonical bracket has the order fixed in Lectures 7 and 20.

Exercise 20.1 (A homogeneous constraint manifold). Restrict the spatial fields to constants $A_i = a_i^a T_a$ and $\Pi_i = p_i^a T_a$. A parameter $\varepsilon = \varepsilon^a T_a$ is also constant.

(a) Pull the canonical symplectic form back to these constants and derive their finite-dimensional bracket. Compute the generator $C[\varepsilon] = \mathcal{V}_{\mathrm{per}} \sum_i \langle \Pi_i, [A_i, \varepsilon]\rangle$ and its Hamiltonian vector field. Prove its constraint bracket, including the volume factor and sign.

(b) At $A_i = T_i$ for $i = 1, 2, 3$, write $\Pi_i = \sum_j M_{ij}T_j$ for a real matrix M. Classify all M satisfying $\sum_i [A_i, \Pi_i] = 0$. Give one nonscalar matrix in the constraint set and one outside it. Explain why checking a single homogeneous state cannot prove the constraint algebra on the full periodic field space.

Solution. The canonical form on the homogeneous subspace is

$$\Omega_{\mathrm{hom}} = \mathcal{V}_{\mathrm{per}} \sum_{i,a} \mathrm{d}a_i^a \wedge \mathrm{d}p_i^a.$$

Solving $\iota_{X_F}\Omega_{\mathrm{hom}} = \mathrm{d}F$ gives $X_F = (\mathcal{V}_{\mathrm{per}}^{-1}F_p, -\mathcal{V}_{\mathrm{per}}^{-1}F_a)$ and therefore

$$\{F, G\}_{\mathrm{hom}} = \frac{1}{\mathcal{V}_{\mathrm{per}}} \sum_{i,a} (F_{a_i^a} G_{p_i^a} - F_{p_i^a} G_{a_i^a}).$$

The reciprocal volume is essential.

Vary the displayed generator, holding its parameter fixed. Invariance of the Lie-algebra pairing gives

$$(C[\varepsilon])_{\Pi_i} = \mathcal{V}_{\mathrm{per}}[A_i, \varepsilon], \qquad (C[\varepsilon])_{A_i} = \mathcal{V}_{\mathrm{per}}[\varepsilon, \Pi_i].$$

The reciprocal volume in the bracket cancels these factors and gives the variations

$$\delta A_i = [A_i, \varepsilon], \qquad \delta\Pi_i = [\Pi_i, \varepsilon].$$

Substitution of both gradients into that bracket yields

$$\{C[\varepsilon], C[\zeta]\} = \mathcal{V}_{\mathrm{per}} \sum_i \big(\langle [\varepsilon, \Pi_i], [A_i, \zeta]\rangle - \langle [A_i, \varepsilon], [\zeta, \Pi_i]\rangle\big).$$

Move the outer parameter commutators with pairing invariance. The result is

$$-\mathcal{V}_{\mathrm{per}} \sum_i \langle \Pi_i, [\varepsilon, [A_i, \zeta]] + [[A_i, \varepsilon], \zeta]\rangle = -\mathcal{V}_{\mathrm{per}} \sum_i \langle \Pi_i, [A_i, [\varepsilon, \zeta]]\rangle.$$

The matrix Jacobi identity gives the last equality. This is $-C[[\varepsilon, \zeta]]$.

At the chosen configuration the Gauss array is

$$\sum_i [T_i, M_{ij}T_j] = (M_{23} - M_{32})T_1 + (M_{31} - M_{13})T_2 + (M_{12} - M_{21})T_3.$$

Its three coefficients vanish exactly when M is symmetric. For example,

$$M = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ 0 & 3 & -1 \end{pmatrix}$$

satisfies the constraint and is not scalar. The matrix with $M_{12} = 1$ and every other entry zero instead gives T_3. These examples test a subset of phase space. The full field-space algebra also involves spatial derivatives of arbitrary smooth parameters and must be proved with their integration by parts.

Exercise 20.2 (Integrating a residual gauge generator). For a fixed smooth periodic $\varepsilon(x)$, let $h_s(x) = \exp(s\varepsilon(x))$ and define

$$A_i(s) = h_s^{-1}A_i h_s + h_s^{-1}\partial_i h_s, \qquad \Pi_i(s) = h_s^{-1}\Pi_i h_s.$$

Here s is the transformation parameter, not physical time.

(a) Prove that these arrays solve $\partial_s A_i(s) = D_i^{(s)}\varepsilon$ and $\partial_s \Pi_i(s) = [\Pi_i(s), \varepsilon]$. Show that the transformation preserves the canonical form, the temporal-gauge Hamiltonian and the Gauss constraint surface.

(b) Take $\varepsilon = \sin x^1\, T_3$, $A_1 = T_1$, $A_2 = T_3$, $A_3 = 0$, $\Pi_1 = T_2$ and $\Pi_2 = \Pi_3 = 0$. Compute the transformed fields and their Gauss array. Is this example on the constraint surface? What changes if a finite gauge transformation depends on physical time?

Solution. The identities $\partial_s h_s = \varepsilon h_s$ and $\partial_s h_s^{-1} = -h_s^{-1}\varepsilon$ give

$$\partial_s A_i(s) = h_s^{-1}[A_i, \varepsilon]h_s + h_s^{-1}(\partial_i\varepsilon)h_s, \qquad \partial_s \Pi_i(s) = [\Pi_i(s), \varepsilon].$$

Differentiating $h_s^{-1}\varepsilon h_s = \varepsilon$ in x^i gives

$$h_s^{-1}(\partial_i\varepsilon)h_s = \partial_i\varepsilon + [h_s^{-1}\partial_i h_s, \varepsilon].$$

Combining these two equations proves the asserted equation for $A_i(s)$.

For variations of the initial phase point the matrix h_s is fixed, so $\delta A_i(s) = h_s^{-1}\delta A_i h_s$ and $\delta\Pi_i(s) = h_s^{-1}\delta\Pi_i h_s$. Trace invariance preserves the canonical two-form. The product rule, with $\partial_i h_s^{-1} = -h_s^{-1}(\partial_i h_s)h_s^{-1}$, gives

$$\partial_i(h_s^{-1}Xh_s) = h_s^{-1}(\partial_i X)h_s - [h_s^{-1}\partial_i h_s, h_s^{-1}Xh_s].$$

Adding the connection commutator proves the covariant-derivative identity. For the curvature, put $b_i = h_s^{-1}\partial_i h_s$ and $\widehat{A}_i = h_s^{-1}A_i h_s$. The same inverse derivative gives

$$\partial_i\widehat{A}_j = h_s^{-1}(\partial_i A_j)h_s - [b_i, \widehat{A}_j], \qquad \partial_i b_j - \partial_j b_i = -[b_i, b_j].$$

The second equality follows by differentiating both b terms; their mixed second derivatives cancel. Since $A_i(s) = \widehat{A}_i + b_i$, substitution gives

$$\begin{aligned} F_{ij}(s) = h_s^{-1}(\partial_i A_j - \partial_j A_i)h_s - [b_i, \widehat{A}_j] + [b_j, \widehat{A}_i] - [b_i, b_j] \\ + [\widehat{A}_i + b_i, \widehat{A}_j + b_j] = h_s^{-1}F_{ij}h_s. \end{aligned}$$

Thus both the kinetic and curvature squares in the Hamiltonian are unchanged. The covariant-derivative identity gives $\mathcal{D}(s) = h_s^{-1}\mathcal{D}h_s$, preserving its zero set.

For the specified parameter, conjugation by $\exp(-\alpha T_3)$ rotates T_1 to $\cos\alpha T_1 - \sin\alpha T_2$ and T_2 to $\sin\alpha T_1 + \cos\alpha T_2$. The derivative of h_s contributes $s\cos x^1 T_3$ to $A_1(s)$. Thus the full nonzero arrays are

$$A_1(s) = \cos\alpha T_1 - \sin\alpha T_2 + s\cos x^1 T_3, \qquad A_2(s) = T_3,$$
$$\Pi_1(s) = \sin\alpha T_1 + \cos\alpha T_2.$$

The components $A_3(s)$, $\Pi_2(s)$ and $\Pi_3(s)$ are zero. Writing $\alpha' = s\cos x^1$, their derivative term is

$$\partial_1\Pi_1(s) = \alpha'(\cos\alpha\, T_1 - \sin\alpha\, T_2).$$

Its commutator with the derivative contribution to $A_1(s)$ is the negative of this expression. The remaining commutator of the two rotated T_1, T_2 directions is T_3. Hence $\mathcal{D}(s) = T_3$, also obtained by conjugating the original $\mathcal{D} = T_3$. Gauge invariance does not turn nonconstraint data into constraint data.

In temporal gauge $A_0 = 0$, a physical-time-dependent h gives $A_0' = h^{-1}\partial_t h$. It is zero exactly when $\partial_t h = 0$. The parameter flow above has no physical-time dependence.

Exercise 20.3 (A linearized reduced bracket and its limit). Fix a real constant η and the background $A_1^0 = \eta T_3$, $A_2^0 = A_3^0 = 0$, $\Pi^0 = 0$. Work only with smooth variations a_i, p_i and parameters having zero spatial mean. This excludes the entire zero Fourier mode from this problem. Put $D_i^0 = \partial_i + [A_i^0, \cdot\,]$.

(a) Derive the linearized Gauss equation $D_i^0 p_i = 0$ and the gauge variation $a_i \mapsto a_i + D_i^0\varepsilon$. For the Coulomb condition $\partial_i a_i = 0$, compute the eigenvalues of $M = -\partial_i D_i^0$ on each nonzero Fourier mode, using $T_+ = T_1 + \mathrm{i}T_2$, $T_- = T_1 - \mathrm{i}T_2$ and T_3.

(b) For $|\eta| < 1$, prove that M^{-1} maps smooth zero-mean arrays to themselves. Define

$$P = 1 + D^0 M^{-1}\,\mathrm{div}, \qquad P^* = 1 + \nabla M^{-1} D^0 \cdot\,.$$

Prove that Pa is divergence-free and gauge invariant, while P^*p satisfies linearized Gauss. Prove $P^2 = P$. For real smooth zero-mean vector smearings f, g, compute the canonical bracket of $\mathcal{A}[f] = \int\langle f, Pa\rangle$ and $\mathcal{P}[g] = \int\langle g, P^*p\rangle$.

(c) At $\eta = 2$, find a real zero-mean parameter with $M\varepsilon = 0$ but $D^0\varepsilon \neq 0$. Explain why this prevents using the preceding inverse there. Does the calculation establish a global nonlinear reduced bracket?

Solution. At $\Pi^0 = 0$, variation of $D_i\Pi_i$ contains no term $[a_i, \Pi_i^0]$, leaving $D_i^0 p_i$. The linearized residual gauge action is $a \mapsto a + D^0\varepsilon$ and leaves p unchanged. On the mode $e^{\mathrm{i}n\cdot x}T_+$ the first covariant derivative has multiplier $\mathrm{i}(n_1 - \eta)$; the other two have $\mathrm{i}n_2, \mathrm{i}n_3$. Contracting with $-\partial_i$ gives $|n|^2 - \eta n_1$. The T_- and neutral directions give respectively $|n|^2 + \eta n_1$ and $|n|^2$. Reality pairs conjugate modes, so this diagonalization respects real arrays.

For nonzero integer n, $|n_1| \le |n|^2$. Thus all three eigenvalues are at least $(1 - |\eta|)|n|^2 > 0$ when $|\eta| < 1$. Dividing rapidly decreasing Fourier coefficients by them preserves rapid decrease. It therefore defines M^{-1} on smooth zero-mean functions. The real operator M is self-adjoint: the constant D^0 commutes with the spatial derivatives, and their periodic adjoints are their negatives.

Since $\mathrm{div}\, D^0 = -M$, the projection satisfies

$$\mathrm{div}\, P = 0, \qquad PD^0 = 0, \qquad P^2 = P + D^0 M^{-1}\,\mathrm{div}\, P = P.$$

The same commuting constant coefficients give $D^0 \cdot \nabla = -M$. Hence $D^0 \cdot P^* = 0$. Periodic integration by parts identifies the stated P^* as the real pairing adjoint of P. The gradients of the two observables are respectively $P^* f$ in the a variable and Pg in the p variable. Consequently

$$\{\mathcal{A}[f], \mathcal{P}[g]\} = \int \langle P^* f, Pg \rangle = \int \langle f, P^2 g \rangle = \int \langle f, Pg \rangle.$$

The brackets within either family vanish. The Coulomb representative in a gauge orbit is unique: solving $\operatorname{div}(a + D^0 \varepsilon) = 0$ requires $\varepsilon = M^{-1} \operatorname{div} a$ and gives Pa. On constrained momenta $P^* p = p$. The remaining pairing between $\operatorname{im} P$ and $\operatorname{im} P^*$ is nondegenerate. Indeed, for $a = Pa \neq 0$, choosing $p = P^* a$ gives $(a, p) = (Pa, a) = \|a\|_2^2 > 0$; the adjoint argument handles a nonzero p. Thus the bracket just computed is the projected canonical bracket of this linear constraint quotient, rather than an assumption about nonlinear reduction.

Set $\zeta = x^1 + x^2$ and $\varepsilon = T_1 \cos \zeta - T_2 \sin \zeta$. At $\eta = 2$ it gives

$$D_1^0 \varepsilon = T_1 \sin \zeta + T_2 \cos \zeta, \qquad D_2^0 \varepsilon = -T_1 \sin \zeta - T_2 \cos \zeta, \qquad D_3^0 \varepsilon = 0.$$

These derivatives are nonzero, but their ordinary divergence is zero. Thus $M\varepsilon = 0$ and the inverse construction fails. Even in its invertible range, the calculation treats only a linearized problem and excludes zero modes. It proves neither a nonlinear slice nor a global quotient theorem. It supplies an explicit local reduced-bracket calculation within those limits.

Classical Field Theory — Exercise Sheet 21

Vector mass matrices and longitudinal energy

Solutions

Use the signature $(+,-,-,-)$ and canonically normalised real scalar kinetic coefficient $1/2$. The questions concern quadratic theories near a fixed vacuum. They make no assertion about a global unitary gauge.

Exercise 21.1 (An orbit determines a vector mass matrix). Let a compact Lie group act orthogonally on $\mathbb{R}^N$, and choose a basis of its Lie algebra, represented by skew real generators T_a, $1 \le a \le r$. Let u_0 be a vacuum of an invariant C^2 potential. Fix $g > 0$, and give the scalar the kinetic term

$$\frac{1}{2}(D_\mu\phi)\cdot(D^\mu\phi), \qquad D_\mu\phi = \partial_\mu\phi + gA^a_\mu T_a\phi.$$

The vector kinetic terms are canonical at the quadratic level.

(a) Derive the vector squared-mass matrix at $\phi = u_0$. Prove that it is nonnegative and identify its kernel and rank. Include generators which act trivially in the representation.

(b) For the three-dimensional rotation representation, take

$$T_a w = e_a \times w, \qquad u_0 = (v/\sqrt{2}, v/\sqrt{2}, 0), \qquad v > 0.$$

Compute the full matrix, its orthonormal eigenvectors and the vector masses. Which gauge-field combination remains massless?

Solution. For $\phi = u_0 + h$, the part of the scalar kinetic term quadratic in the vector alone is

$$\frac{1}{2}g^2A^a_\mu A^{b\mu}(T_a u_0)\cdot(T_b u_0) = \frac{1}{2}A^a_\mu \mathsf{M}_{ab}A^{b\mu}.$$

The remaining quadratic terms include the scalar kinetic and scalar–vector mixing; they do not alter this coefficient matrix. For a real array z its quadratic form is

$$z^a\mathsf{M}_{ab}z^b = g^2\left|\sum_a z^aT_a u_0\right|^2 \ge 0.$$

It vanishes exactly when the displayed orbit direction vanishes. Thus the kernel is the Lie algebra of the stabiliser, including any ineffective generators. Lecture 17 identifies the rank of the orbit map as $\dim G - \dim H$, giving the asserted rank.

For the rotation representation the cross-product identity gives

$$\mathsf{M} = g^2(v^2 I - u_0 u_0^{\mathsf{T}}).$$

Substitution gives

$$\mathsf{M} = g^2v^2\begin{pmatrix} 1/2 & -1/2 & 0 \\ -1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

An orthonormal eigenbasis in generator space is

$$z_0 = \frac{1}{\sqrt{2}}(1,1,0), \qquad z_1 = \frac{1}{\sqrt{2}}(1,-1,0), \qquad z_2 = (0,0,1).$$

Their squared masses are respectively 0, g^2v^2 and g^2v^2. Canonical vector kinetic terms make their positive mass values gv. The unbroken generator rotates about u_0; its gauge component is $z_0 \cdot A$. The two orthogonal generator directions move the vacuum.

Exercise 21.2 (The charged and neutral electroweak combinations). The following bosonic representation data are given. A complex doublet $\Phi \in \mathbb{C}^2$ has kinetic term $\frac{1}{2}(D_\mu\Phi)^\dagger D^\mu\Phi$, potential $\lambda(\Phi^\dagger\Phi - v^2)^2/8$, and vacuum $\Phi_0 = (0,v)^\mathsf{T}$. Here $g, g', v, \lambda > 0$ and

$$D_\mu = \partial_\mu - \mathrm{i}\left(gW_\mu^a\frac{\sigma_a}{2} + g'B_\mu\frac{I_2}{2}\right).$$

All four real vector fields have canonical quadratic Maxwell kinetic terms. The Pauli matrices and the hypercharge generator $Y = I_2/2$ specify the representation; no fermion sector is included.

(a) Find the full real squared-mass matrix in the order (W^1, W^2, W^3, B). Diagonalise its neutral block and define Z and the massless photon A_γ with unit kinetic normalisation. Define $W^\pm = (W^1 \mp \mathrm{i}W^2)/\sqrt{2}$ and find their common mass.

(b) Find the generator annihilating Φ_0 and the coefficient of its photon coupling in D_μ. Find the radial scalar mass and compare the quadratic number of physical modes in the uncoupled limit $g, g' \to 0$ with the broken quadratic description at positive couplings. Explain why W^+ and W^- do not count as two independent complex vectors.

This is the bosonic mass calculation of the electroweak model. For the original construction see Weinberg, *Physical Review Letters* **19** (1967), 1264–1266, equations (4), (7)–(12). The signs and scalar normalisation to use are those explicitly given above.

Solution. At the vacuum the vector part of the derivative is

$$D_\mu\Phi_0 = -\frac{\mathrm{i}v}{2}\begin{pmatrix} g(W_\mu^1 - \mathrm{i}W_\mu^2) \\ -gW_\mu^3 + g'B_\mu \end{pmatrix}.$$

Its squared norm, multiplied by the kinetic coefficient $1/2$, gives

$$L_{\text{mass}} = \frac{v^2}{8}\left[g^2(W_\mu^1W^{1\mu} + W_\mu^2W^{2\mu}) + (-gW_\mu^3 + g'B_\mu)(-gW^{3\mu} + g'B^\mu)\right].$$

Thus the real squared-mass matrix is

$$\mathsf{M} = \frac{v^2}{4}\begin{pmatrix} g^2 & 0 & 0 & 0 \\ 0 & g^2 & 0 & 0 \\ 0 & 0 & g^2 & -gg' \\ 0 & 0 & -gg' & g'^2 \end{pmatrix}.$$

The orthonormal neutral eigenvectors are $(g,-g')/g_Z$ and $(g',g)/g_Z$, with eigenvalues $g_Z^2v^2/4$ and zero. Their orthogonal change of variables preserves the quadratic kinetic terms and defines

$$Z = \frac{gW^3 - g'B}{g_Z}, \qquad A_\gamma = \frac{g'W^3 + gB}{g_Z}.$$

Their masses are $g_Z v/2$ and zero. The two charged real directions have eigenvalue $g^2v^2/4$. Combining them into $W^\pm = (W^1 \mp \mathrm{i}W^2)/\sqrt{2}$ gives $m_W = gv/2$.

The generator $Q_{\mathrm{em}} = \sigma_3/2 + Y$ annihilates $(0, v)^{\mathsf{T}}$. The inverse neutral rotation is

$$W^3 = (gZ + g'A_\gamma)/g_Z, \qquad B = (-g'Z + gA_\gamma)/g_Z.$$

Consequently

$$gW^3\frac{\sigma_3}{2} + g'BY = \frac{g^2\sigma_3/2 - g'^2Y}{g_Z}Z + \frac{gg'}{g_Z}Q_{\mathrm{em}}A_\gamma.$$

Thus its photon coupling is $e = gg'/g_Z$.

Along the radial line $\Phi = (0, v+h)^{\mathsf{T}}$, the kinetic term is $\frac{1}{2}(\partial h)^2$ and the quadratic potential is $\frac{1}{2}\lambda v^2 h^2$. Hence $m_h = \sqrt{\lambda}\, v$. The vacuum orbit of the supplied action has three independent tangents: the mass matrix has rank three, and those generator directions move Φ_0. The remaining scalar direction is radial. In the uncoupled limit $g, g' \to 0$, there are four massless real vectors with two polarisations each and four real scalar components, giving $8 + 4 = 12$. The scalar potential has one radial curvature λv^2 and three orbit zeros, so the scalar count includes one massive and three massless components. After the split there are three massive real vectors with three polarisations each, the photon with two, and the radial scalar with one, giving $9 + 2 + 1 = 12$. Since W^1 and W^2 are real, $W^- = \overline{W^+}$. They encode two real massive vectors, not two freely specifiable complex vectors. This count concerns the local quadratic theory only.

Exercise 21.3 (The positive reduced Proca Hamiltonian). Use Lecture 21's variables $B = -\beta\,\mathrm{d}t + \mathbf{b}\cdot\mathrm{d}x$, mass $m > 0$ and spatial momentum $\mathbf{\Pi}$. Take real Schwartz data on $\mathbb{R}^3$. Before elimination the Hamiltonian is

$$H_B = \int_{\mathbb{R}^3}\left[\frac{1}{2}|\mathbf{\Pi}|^2 + \frac{1}{2}|\operatorname{curl}\mathbf{b}|^2 + \frac{1}{2}m^2|\mathbf{b}|^2 + \beta\operatorname{div}\mathbf{\Pi} - \frac{1}{2}m^2\beta^2\right]\mathrm{d}^3x.$$

Its scalar constraints are $\pi_\beta = 0$ and $\operatorname{div}\mathbf{\Pi} - m^2\beta = 0$.

(a) Eliminate the scalar pair and derive the reduced Hamiltonian and both spatial Hamilton equations with the remaining canonical form. Prove positivity and show that a zero-energy real datum is zero.

(b) At a nonzero Fourier vector k, split the spatial amplitudes into longitudinal and transverse parts. Find the frequencies and a canonical normalisation of each real longitudinal Fourier quadrature. Treat $k = 0$ separately. Why does the energy formula not have a regular mass-zero limit for fixed arbitrary longitudinal momentum?

Individual Fourier modes here analyse the multiplier; they are not claimed to be finite-energy plane-wave fields on $\mathbb{R}^3$.

Solution. The terms depending on β can be written

$$\beta\operatorname{div}\mathbf{\Pi} - \frac{1}{2}m^2\beta^2 = -\frac{1}{2}m^2\left(\beta - \frac{\operatorname{div}\mathbf{\Pi}}{m^2}\right)^2 + \frac{|\operatorname{div}\mathbf{\Pi}|^2}{2m^2}.$$

The secondary constraint sets the square to zero. Since $\pi_\beta = 0$, the pullback of the scalar canonical term vanishes, leaving the spatial canonical form. The reduced Hamiltonian is therefore

$$H_{\mathrm{red}} = \frac{1}{2}\int\left[|\mathbf{\Pi}|^2 + |\operatorname{curl}\mathbf{b}|^2 + m^2|\mathbf{b}|^2 + m^{-2}|\operatorname{div}\mathbf{\Pi}|^2\right]\mathrm{d}^3x.$$

Its remaining bracket is the spatial canonical bracket.

Schwartz integration by parts in its variation gives

$$(H_{\mathrm{red}})_{\mathbf{\Pi}} = \mathbf{\Pi} - m^{-2}\nabla \operatorname{div} \mathbf{\Pi}, \qquad (H_{\mathrm{red}})_{\mathbf{b}} = \operatorname{curl}\operatorname{curl} \mathbf{b} + m^2 \mathbf{b}.$$

Hamilton's equations therefore give

$$\mathbf{b}_t = \mathbf{\Pi} - m^{-2}\nabla \operatorname{div} \mathbf{\Pi}, \qquad \mathbf{\Pi}_t = -\operatorname{curl}\operatorname{curl} \mathbf{b} - m^2 \mathbf{b}.$$

Every energy term is nonnegative. If the energy is zero, the momentum norm and m^2 times the spatial-field norm both vanish, so the datum is zero.

For $k \neq 0$, the transverse symbols obey

$$\partial_t \widehat{\mathbf{b}}_T = \widehat{\mathbf{\Pi}}_T, \qquad \partial_t \widehat{\mathbf{\Pi}}_T = -(|k|^2 + m^2)\widehat{\mathbf{b}}_T.$$

The longitudinal symbols instead obey

$$\partial_t b_L = \left(1 + \frac{|k|^2}{m^2}\right) \Pi_L, \qquad \partial_t \Pi_L = -m^2 b_L.$$

A second derivative gives frequency squared $|k|^2 + m^2$ in both cases. For one real quadrature the longitudinal Hamiltonian is $\frac{1}{2}[(\nu^2/m^2)\Pi_L^2 + m^2 b_L^2]$. Put $Q_L = (m/\nu)b_L$ and $P_L = (\nu/m)\Pi_L$. This reciprocal rescaling preserves $\mathrm{d}b_L \wedge \mathrm{d}\Pi_L$ and turns it into $\frac{1}{2}(P_L^2 + \nu^2 Q_L^2)$. At $k = 0$ no direction is selected as longitudinal; all three spatial axes obey frequency m.

For a fixed nonzero longitudinal momentum at $k \neq 0$, the term $|k|^2|\Pi_L|^2/(2m^2)$ diverges as $m \downarrow 0$. The scalar constraint elimination itself divides by m^2. It therefore cannot describe Maxwell's mass-zero gauge reduction by simply substituting $m = 0$. This agrees with the singular phase-variable replacement in Lecture 21.

Classical Field Theory — Exercise Sheet 22

Vortex orientation, scalar reduction and moduli

Solutions

Throughout this sheet $q = v = \lambda = 1$, $D_j = \partial_j - \mathrm{i}A_j$, $B = \partial_1 A_2 - \partial_2 A_1$, and the plane has orientation $\mathrm{d}x_1 \wedge \mathrm{d}x_2$. Use Lecture 22's energy and current $K_j = \frac{1}{2}\operatorname{Im}(\overline{\Phi} D_j \Phi)$. Every sector statement retains its finite-energy, uniform vacuum-modulus and absolute magnetic-integrability assumptions.

Exercise 22.1 (Reflection and higher-winding antivortices). Let $R(x_1, x_2) = (x_1, -x_2)$. For smooth fields set $\widetilde{\Phi} = \Phi \circ R$ and $\widetilde{A} = R^* A$.

(a) Track B, K, the energy and counterclockwise winding under this reflection. Derive the square completion for winding $-N$, $N \geq 1$, and prove the bound and equality equations in the reflected sector.

(b) For real radial profiles use $\Phi = f(r)e^{-\mathrm{i}N\theta}$ and $A = \alpha(r)\mathrm{d}\theta$. Derive both equality ODEs, the flux and the boundary conditions on the positive vacuum branch. Give sufficient smooth Cartesian core forms and the first corrections to $f = s_N r^N + \cdots$ and $\alpha = -r^2/4 + \cdots$. Here $s_N > 0$ is a local slope coefficient, not a proved global shooting value.

Solution. The pulled-back components are $\widetilde{A}_1 = A_1 \circ R$ and $\widetilde{A}_2 = -A_2 \circ R$. Thus $\widetilde{B} = -B \circ R$. The covariant derivatives obey $D_1\widetilde{\Phi} = (D_1\Phi)\circ R$ and $D_2\widetilde{\Phi} = -(D_2\Phi)\circ R$. Their squared norms, the magnetic square and scalar potential therefore give the same energy after the change of variables. The current components transform as a one-form, giving $\operatorname{curl}\widetilde{K} = -(\operatorname{curl} K)\circ R$. A counterclockwise circle becomes clockwise, so winding changes sign.

Apply the positive completion of Lecture 22 to the unreflected fields and express it through the reflected ones. Dropping tildes gives

$$e_1 = \frac{1}{2}|(D_1 - \mathrm{i}D_2)\Phi|^2 + \frac{1}{2}\left[B + \frac{1 - |\Phi|^2}{2}\right]^2 - \frac{B}{2} - \operatorname{curl} K.$$

The reflected sector has flux $-2\pi N$. Reflection preserves all absolute-integrability and modulus assumptions. It also transports the lecture's sequence of circles with vanishing current integral. Integration of this identity along that sequence yields

$$E_1 = \pi N + \frac{1}{2}\int_{\mathbb{R}^2}\left[|(D_1 - \mathrm{i}D_2)\Phi|^2 + \left(B + \frac{1 - |\Phi|^2}{2}\right)^2\right]\mathrm{d}^2x.$$

The squares prove the bound and vanish exactly for $(D_1 - \mathrm{i}D_2)\Phi = 0$, $B = -(1 - |\Phi|^2)/2$.

The polar derivative formula, including the angular potential, gives

$$(D_1 - \mathrm{i}D_2)\Phi = e^{-\mathrm{i}(N+1)\theta}\left[f' - \frac{N + \alpha}{r}f\right], \qquad B = \frac{\alpha'}{r}.$$

Hence the radial equations are

$$f' = \frac{N + \alpha}{r}f, \qquad \frac{\alpha'}{r} = -\frac{1}{2}(1 - f^2).$$

Sufficient smooth core forms are $f = r^N F(r^2)$ and $\alpha = r^2 G(r^2)$, with smooth real F, G. Indeed, $\Phi = (x_1 - \mathrm{i}x_2)^N F(r^2)$ and $A = G(r^2)(-x_2 \mathrm{d}x_1 + x_1 \mathrm{d}x_2)$ are smooth. They imply $f(0) = \alpha(0) = 0$. Choose $f \to 1$ at infinity. Radial flux is $2\pi[\alpha(\infty) - \alpha(0)] = -2\pi N$, so $\alpha \to -N$.

The second equation first gives $\alpha = -r^2/4 + O(r^{2N+2})$. The first then gives $(\log(f/r^N))' = \alpha/r = -r/4 + O(r^{2N+1})$. Thus $f = s_N r^N(1 - r^2/8 + O(r^4))$. Substituting its square in $\alpha' = -r/2 + rf^2/2$ gives

$$\alpha = -\frac{r^2}{4} + \frac{s_N^2}{4N+4} r^{2N+2} + O(r^{2N+4}).$$

This is a local necessary expansion of a smooth radial equality field. It does not establish a global solution for every s_N.

Exercise 22.2 (The scalar equation with its point sources). Let a positive critical vortex in the stated sector have finitely many zeros $z_j \in \mathbb{R}^2$ of multiplicities $m_j \geq 1$. Away from the zeros put $w = \log|\Phi|^2$. Assume as local zero data that $w = 2m_j \log|x - z_j| + w_j$ near z_j, where w_j is smooth. This local regular remainder is a hypothesis, not a claim about arbitrary smooth fields with zeros. Use the distributional identity $\Delta \log|x - z_j| = 2\pi\delta_{z_j}$ as a given.

(a) On a phase patch derive A_1, A_2 in terms of w and the real phase. Eliminate the potential from the two Bogomolny equations.

(b) Prove the resulting equation on the whole plane as a distribution, including its point-source coefficient. Determine $\int_{\mathbb{R}^2}(1 - e^w)\mathrm{d}^2x$ from the sector flux when $\sum_j m_j = N$. Does this reduction prove existence for prescribed zeros?

Solution. On a phase patch the real and imaginary parts of the first equality equation give

$$A_1 = \partial_1\theta + \frac{1}{2}\partial_2 w, \qquad A_2 = \partial_2\theta - \frac{1}{2}\partial_1 w.$$

Mixed phase derivatives cancel, leaving $B = -\Delta w/2$ away from the zeros. The second equality equation is $B = (1 - e^w)/2$, so $\Delta w + 1 - e^w = 0$ there.

Near z_j, the given logarithmic part contributes $4\pi m_j \delta_{z_j}$ to the distributional Laplacian. The function $e^w = |x - z_j|^{2m_j} e^{w_j}$ is smooth across that point. The regular function $\Delta w_j + 1 - e^w$ vanishes on the punctured neighbourhood, so by continuity it vanishes throughout it. Combining these local statements with the equation off the zero set gives

$$\Delta w + 1 - e^w = 4\pi \sum_j m_j \delta_{z_j}$$

as distributions on the plane. The coefficient is $4\pi m_j$, not $2\pi m_j$, because the logarithm is that of the squared modulus.

Since $1 - e^w = 2B$ extends smoothly and B is absolutely integrable, the integral exists and the positive sector flux gives

$$\int_{\mathbb{R}^2}(1 - e^w)\mathrm{d}^2x = 2\int_{\mathbb{R}^2} B\,\mathrm{d}^2x = 4\pi N.$$

No uncontrolled integral of a Laplacian at infinity was needed. The calculation starts with an existing equality field and prescribed local zero behaviour. Solving the singular nonlinear equation, with its asymptotics and reconstructing a smooth potential, requires the separately cited existence theory.

Exercise 22.3 (Parameters at a two-vortex collision). A *moduli space* here is the set of critical positive N-vortex solutions with the vacuum asymptotics, modulo smooth gauge transformations. Use the following stronger geometric result as a *given*: this space is a nonsingular complex manifold, and the coefficients of

$$P(z) = \prod_{j=1}^{N} (z - z_j)$$

are smooth complex coordinates, including at coincident zeros. This input is not inferred from existence and uniqueness as a set alone. For its zero-coordinate description see Alonso Izquierdo, Garcia Fuertes, Manton and Mateos Guilarte, *JHEP* **01** (2024) 020, §5, pp. 10–11, equation (5.1) and discussion. Taubes's existence/uniqueness result is the one cited in Lecture 22.

(a) Count the real moduli and the moduli after fixing the mean zero position. Explain why ordering the zeros overcounts configurations.

(b) For $N = 2$, write $P(z) = z^2 - sz + p$. Replace (s, p) by the centre $Z = s/2$ and discriminant $\Delta = s^2 - 4p$. Locate the collision set and determine its real codimension. Explain why (z_1, z_2) and their separation are poor smooth coordinates there.

Solution. The leading coefficient of P is fixed to one, leaving N independent complex coefficients. The given coordinate theorem makes their space real $2N$-dimensional. Permuting roots does not change the polynomial or the corresponding gauge class, so labelled roots are redundant. The coefficient of z^{N-1} is $-\sum_j z_j$. Fixing the mean position to zero sets this one complex coefficient to zero and leaves $2N - 2$ real parameters.

For $N = 2$, put $Z = s/2$ and $\Delta = s^2 - 4p$. The inverse transformation is $s = 2Z$, $p = Z^2 - \Delta/4$, so it is a smooth invertible change of complex coordinates. It gives

$$P(z) = (z - Z)^2 - \frac{\Delta}{4}, \qquad z_{1,2} = Z \pm \frac{\sqrt{\Delta}}{2}.$$

The collision occurs exactly when $\Delta = 0$. The coordinate Z remains arbitrary, giving a complex one-dimensional, real two-dimensional subset of the real four-dimensional space. The ambient coefficient chart is nonsingular there.

The root-to-coefficient Jacobian has determinant $z_1 - z_2$ up to sign, so it degenerates at a collision. The inverse requires a square root and a choice of root labels. For centred roots at $\pm d$, one has $\Delta = 4d^2$. Thus d, or the nonnegative distance $2|d|$, is not a smooth coordinate at the collision even though Δ is. These are consequences of the supplied smooth-coordinate input and polynomial algebra, not a proof of that analytic moduli-space input.

Exercise 22.4 (Designing a radial computation). For the positive unit-vortex ODEs $f' = (1 - \alpha)f/r$ and $\alpha'/r = (1 - f^2)/2$, design a computation on $[r_0, R]$, where $0 < r_0 \ll 1 < R$.

(a) Derive the coefficients through r^5 in f and through r^6 in α, using the free coefficient $s_0 = f'(0)$.

(b) Treat s_0 as an unknown parameter. Give three boundary residuals for a two-equation boundary-value solver using the core series and $f(R) = 1$. Explain why adding $\alpha(R) = 1$ independently would overdetermine this particular setup.

(c) Specify separate tests of origin truncation, outer-boundary truncation and solver accuracy. What further information is needed before a numerical residual could prove existence of a full-plane vortex? No numerical run is required in this problem.

Solution. Write $f = s_0 r + c_3 r^3 + c_5 r^5 + \cdots$ and $\alpha = a_2 r^2 + a_4 r^4 + a_6 r^6 + \cdots$. The magnetic

equation gives

$$2a_2 = \frac{1}{2}, \qquad 4a_4 = -\frac{s_0^2}{2}, \qquad 6a_6 = -s_0 c_3.$$

The first equation gives $3c_3 = c_3 - a_2 s_0$ and $5c_5 = c_5 - a_2 c_3 - a_4 s_0$. Therefore

$$a_2 = \frac{1}{4}, \quad c_3 = -\frac{s_0}{8}, \quad a_4 = -\frac{s_0^2}{8}, \quad a_6 = \frac{s_0^2}{48}, \quad c_5 = \frac{s_0}{128} + \frac{s_0^3}{32}.$$

Let $f_{\mathrm{core}}(r; s_0)$ and $\alpha_{\mathrm{core}}(r; s_0)$ denote these truncated polynomials. The three endpoint residuals are

$$f(r_0) - f_{\mathrm{core}}(r_0; s_0), \qquad \alpha(r_0) - \alpha_{\mathrm{core}}(r_0; s_0), \qquad f(R) - 1.$$

The two first-order equations contribute two integration constants and the unknown s_0 contributes one parameter. Three scalar residuals match that count. Adding an independent fourth one generally overdetermines it. The computed value of $\alpha(R) - 1$ is instead a useful diagnostic.

To test origin truncation, decrease r_0 while retaining small solver error and fixed R, or increase the core-series order. To test outer truncation, increase R and compare the fitted s_0, the profiles on a common interior interval and the tail diagnostics. To test the solver, refine its mesh/tolerance at fixed endpoints and measure both ODE residuals and endpoint residuals independently. A small collocation residual is not the same as a proved uniform solution error, and none of these finite tests controls the entire infinite tail. A rigorous numerical existence proof would additionally require validated error bounds and control of the singular-origin and infinite-boundary limits. The cited Taubes theorem supplies existence by a separate argument.

Classical Field Theory — Exercise Sheet 23

A second KdV bracket and further conserved quantities

Solutions

Use Lecture 23's field space $\mathcal{S} = \mathcal{S}(\mathbb{R}, \mathbb{R})$, gradient space $\mathcal{B} = \mathbb{R} \oplus \mathcal{S}$ and admissible smooth functionals $\mathcal{A}_{\mathrm{KdV}}$. Write $D_x = \partial_x$ for spatial differentiation. Every time-dependent conclusion is conditional on a C^1 path in $\mathcal{S}$ satisfying KdV.

Exercise 23.1 (A variable Hamiltonian operator). For $u \in \mathcal{S}$ define the linear operator $J_2[u] : \mathcal{B} \to \mathcal{S}$ by

$$J_2[u]b = -b''' + 4ub' + 2u'b.$$

The prime means D_x, including when applied to a gradient. Define $\{F, G\}_2 = \int_{\mathbb{R}} F_u J_2[u] G_u \, \mathrm{d}x$.

(a) Prove skewness by integration by parts with both gradients in $\mathcal{B}$, keeping the endpoint primitive. Derive the gradient of $\{F, G\}_2$ and prove closure of the stated class.

(b) Prove Jacobi, also for $J_2 + \alpha D_x$ with every constant $\alpha \in \mathbb{R}$. Use the Hessian symmetry from Lecture 23 and account for the extra terms from differentiating the operator itself. A useful notation is $W(a, b) = ab' - a'b$.

(c) For $P = \frac{1}{2}\int u^2$ and $H = \int (u^3 + u_x^2/2)$, prove $J_2 P_u = D_x H_u$. Is the Gardner mass Casimir also a Casimir of this bracket?

The second KdV operator is discussed in Olver, *Applications of Lie Groups to Differential Equations*, second edition, Examples 7.6 and 7.9, pp. 438–439 and 444. The operator above states the course's scaling and signs explicitly; its proof is part of this problem.

Solution. Take $a, b \in \mathcal{B}$ for the skewness calculation. For the functional calculation below set $a = F_u$, $b = G_u$ and $H_F = D(F_u)(u)$, with $H_G = D(G_u)(u)$ defined in the same way. All terms in $J_2 b$ are Schwartz: differentiation removes the constant part of b, while the coefficients u, u' are Schwartz. Direct differentiation gives the skew primitive

$$aJ_2 b + bJ_2 a = D_x\big[-ab'' + a'b' - a''b + 4uab\big].$$

At each endpoint every displayed term tends to zero. The first three contain a derivative of a gradient, and the last contains u. Integrating proves skewness on $\mathcal{B}$.

Differentiate the bracket in a Schwartz direction h. The derivative of its operator, with b fixed, is $4hb' + 2h'b$. Hessian symmetry and skewness move the other two derivatives, giving

$$\begin{aligned} D\{F, G\}_2(u)h &= \int \big[(H_F h) J_2 b + a J_2(H_G h) + a(4hb' + 2h'b)\big]\mathrm{d}x \\ &= \int h\big[H_F J_2 b - H_G J_2 a + 2(ab' - a'b)\big]\mathrm{d}x. \end{aligned}$$

The last integration uses $4ab'h + 2abh' = 2(ab' - a'b)h + D_x(2abh)$, whose endpoint term vanishes. Thus the gradient is $H_F J_2 b - H_G J_2 a + 2W(a, b)$. It belongs to $\mathcal{B}$ because both Hessian directions $J_2 b, J_2 a$ belong to $\mathcal{S}$, and $W(a, b)$ is Schwartz. The displayed formula is smooth by the same composition and product calculus as Lecture 23. It proves bracket closure.

For Jacobi write also $c = K_u$. Substitution of this gradient in the cyclic sum $\{\{F, G\}_2, K\}_2 + \{\{G, K\}_2, F\}_2 + \{\{K, F\}_2, G\}_2$ cancels the six Hessian terms in pairs by symmetry. Every

Hessian direction is J_2a, J_2b or J_2c in $\mathcal{S}$. The remaining sum is

$$2\int \big[W(a,b)J_2c + W(b,c)J_2a + W(c,a)J_2b\big]\mathrm{d}x.$$

The coefficients multiplying u and u' vanish pointwise:

$$W(a,b)c' + W(b,c)a' + W(c,a)b' = 0, \qquad W(a,b)c + W(b,c)a + W(c,a)b = 0.$$

Expanding the six products proves each equality. For the third-derivative part put $T = W(a,b)c'' + W(b,c)a'' + W(c,a)b''$. Its derivative is

$$T' = W(a,b)c''' + W(b,c)a''' + W(c,a)b'''.$$

Indeed, the extra terms are the cyclic sum $(ab'' - a''b)c''$, which cancels in pairs. Each term of T contains derivatives of gradients and tends to zero at both ends. The remaining Jacobi sum is therefore $-2\int T' = 0$. Bilinearity, skewness and the product rule follow from the definition and the usual product gradient. This proves the Poisson assertion. Adding αD_x preserves skewness and closure. In the remaining Jacobi expression its extra contribution is $2\alpha\int[W(a,b)c' + W(b,c)a' + W(c,a)b'] = 0$. Thus every $J_2 + \alpha D_x$ is Poisson: the two brackets are compatible.

Finally, $P_u = u$ and $H_u = 3u^2 - u''$ give

$$J_2P_u = -u''' + 4uu' + 2u'u = 6uu' - u''' = D_xH_u.$$

Thus $u_t = J_2P_u$ is the same KdV equation as $u_t = D_xH_u$. The mass gradient is one, and $J_21 = 2u'$ is generally nonzero. Fix a nonconstant profile u_* and the smearing $g = u_*'$. At that field the linear observable $u(g)$ detects it: $\{u(g), M\}_2[u_*] = 2\int (u_*')^2 > 0$. The Gardner and second-operator degeneracies differ.

Exercise 23.2 (Finding a higher polynomial invariant). For real constants A, B, C put

$$I_{A,B,C}[u] = \int_{\mathbb{R}} \big(Au^4 + Bu(u')^2 + C(u'')^2\big)\mathrm{d}x.$$

Determine all triples such that $DI_{A,B,C}(u)(6uu' - u''') = 0$ for every $u \in \mathcal{S}$. Conclude conservation along every stated Schwartz KdV path. Show necessity as well as sufficiency; do not infer independence of two integrals merely by looking at different integrands.

Solution. Integration by parts gives

$$(I_{A,B,C})_u = 4Au^3 - B(u')^2 - 2Buu'' + 2Cu''''.$$

Pair this gradient with $6uu' - u'''$. Terms which are single total derivatives have zero integral. For the others use

$$\int u^2u'u'' = -\int u(u')^3, \qquad \int (u')^2u''' = -2\int u'(u'')^2, \qquad \int uu''u''' = -\frac{1}{2}\int u'(u'')^2.$$

These are derivatives of $u^2(u')^2$, $(u')^2u''$ and $u(u'')^2$. They also imply $\int u^3u''' = 3\int u(u')^3$ and $\int uu'u'''' = \frac{5}{2}\int u'(u'')^2$. Consequently the complete directional derivative is

$$DI_{A,B,C}(u)(6uu' - u''') = (6B - 12A)\int u(u')^3 + (30C - 3B)\int u'(u'')^2.$$

This proves sufficiency if $B = 2A$ and $B = 10C$.

For necessity let $u_* = (1 + x)e^{-x^2}$. Gaussian integration gives

$$\int u_*(u_*')^3 = \frac{\sqrt{\pi}}{8}, \qquad \int u_*'(u_*'')^2 = \frac{32\sqrt{3\pi}}{81}.$$

Both are nonzero. Replace u_* by au_* for arbitrary real a. The first integral scales as a^4 and the second as a^3. A polynomial in a vanishes identically only when both coefficients vanish. Thus $B = 2A$ and $B = 10C$, equivalently $(A, B, C) = C(5, 10, 1)$, including the zero triple. The chain rule along a $C^1(\mathcal{S})$ solution now proves conservation. No existence theorem for arbitrary initial data was used.

Exercise 23.3 (Classifying Gardner Casimirs). A Casimir of Lecture 23's Gardner bracket is an admissible functional commuting with every admissible functional. Prove that these are exactly

$$F[u] = \psi(M[u]), \qquad M[u] = \int_{\mathbb{R}} u \, \mathrm{d}x,$$

for smooth $\psi : \mathbb{R} \to \mathbb{R}$. Constant scalar functionals are included. Explain why this does not say that every local polynomial density is a nonlinear function of the density u. Compare the local-functional discussion in Olver, Example 7.13, p. 446: the present admissible class also allows nonlocal composites $\psi(M)$.

Solution. Let $a = F_u[u] \in \mathcal{B}$ at a fixed field. For every Schwartz g, the Casimir property gives

$$0 = \{F, u(g)\} = \int ag' = -\int a'g.$$

Here a' is Schwartz. The compact-test fundamental lemma makes $a' = 0$, so a is spatially constant. Its value may depend on the entire field.

If u and v have the same mass, the straight segment $u_s = u + s(v - u)$ stays in $\mathcal{S}$ and has that mass. The chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}s}F[u_s] = \int F_u[u_s](v - u) = 0,$$

because the gradient is spatially constant and $\int (v - u) = 0$. Thus F is constant on every affine mass fibre. Choose $\eta \in \mathcal{S}$ with integral one and set $\psi(s) = F[s\eta]$. It is smooth by composition, and u and $M[u]\eta$ lie in the same fibre. Hence $F[u] = \psi(M[u])$.

Conversely, this composite has gradient $\psi'(M[u])$, a constant element of $\mathcal{B}$ depending smoothly on u. It is admissible, and for any admissible G,

$$\{F, G\} = \psi'(M[u]) \int D_x G_u[u] \, \mathrm{d}x = 0.$$

The integral vanishes because the gradient's derivative is Schwartz and its constant part has the same limit at both ends. This includes constant ψ. For example $M[u]^2$ is such a functional, whereas $\int u(x)^2 \mathrm{d}x$ is the momentum functional and has nonzero gradient derivative for a nonconstant field. Nonlinear composition with the integrated mass is different from integrating a nonlinear density.

Classical Field Theory — Exercise Sheet 24

A higher Lax flow and operator domains

Solutions

Use $D_x = \partial_x$ and the ordered commutator $[B, L] = BL - LB$. All operator calculations in the first problem are on $C_c^\infty(\mathbb{R}, \mathbb{C})$, with smooth real coefficient u. A displayed differential expression does not specify its Hilbert-space domain.

Exercise 24.1 (Determining a fifth-order Lax flow). Let $L = -D_x^2 + u$. For real constants $r_1, \dots, r_6$ consider

$$B_5 = 16D_x^5 + r_1 u D_x^3 + r_2 u' D_x^2 + (r_3 u'' + r_4 u^2) D_x + r_5 u''' + r_6 u u'.$$

Products with u and its derivatives here mean multiplication before the indicated derivative of the test.

(a) Determine the constants so that $[B_5, L]$ is multiplication for every smooth u. Find the resulting scalar coefficient and write the corresponding evolution equation $L_t = [B_5, L]$.

(b) Prove that the resulting B_5 is formally skew-symmetric on compact smooth tests. Distinguish this from a statement about a closed skew-adjoint operator on L^2.

Solution. For a test ψ, write $[B_5, L]\psi = \sum_{j=0}^{4} C_j \psi^{(j)}$. Leibniz's rule gives the coefficients of differentiated tests:

$$\begin{aligned}
C_4 &= (80 + 2r_1)u', \\
C_3 &= (160 + r_1 + 2r_2)u'', \\
C_2 &= (160 + r_2 + 2r_3)u''' + (3r_1 + 4r_4)uu', \\
C_1 &= (80 + r_3 + 2r_5)u'''' \\
&\quad + (2r_2 + 2r_4 + 2r_6)(u')^2 + (3r_1 + 2r_4 + 2r_6)uu''.
\end{aligned}$$

For example, $[16D_x^5, u]$ contributes $80u'\psi^{(4)} + 160u''\psi''' + 160u'''\psi'' + 80u''''\psi' + 16u^{(5)}\psi$. Each other term is obtained by the same product rule and commutation with $-D_x^2$ or multiplication by u.

To force $C_j = 0$ for every smooth u, its finite jets at any point may be prescribed independently by a polynomial times a cutoff. The first two equations give $r_1 = -40$ and $r_2 = -60$. The two terms of C_2 then give $r_3 = -50$ and $r_4 = 30$. The u'''' and $(u')^2$ terms give $r_5 = -15$ and $r_6 = 30$. The last uu'' equation is then also zero. The undifferentiated coefficient before substitution is

$$C_0 = (16 + r_5)u^{(5)} + (r_1 + r_6)uu''' + (r_2 + r_3 + 3r_6)u'u'' + r_4 u^2 u'.$$

Thus the unique operator in this ansatz is

$$B_5 = 16D_x^5 - 40uD_x^3 - 60u'D_x^2 + (30u^2 - 50u'')D_x + 30uu' - 15u''',$$

and its commutator is multiplication by

$$K_5[u] = u^{(5)} - 10uu''' - 20u'u'' + 30u^2 u'.$$

The equation $L_t = [B_5, L]$ is $u_t = K_5[u]$. No existence result for that evolution is asserted by this calculation.

For the formal adjoint, compact integration by parts gives $(aD_x^j)^* = (-D_x)^j \circ a$ for real a. Let $c_1 = 30u^2 - 50u''$ and $c_0 = 30uu' - 15u'''$. Expansion gives

$$\begin{aligned} B_5^* &= -16D_x^5 + 40D_x^3 \circ u - 60D_x^2 \circ u' - D_x \circ c_1 + c_0 \\ &= -16D_x^5 + 40uD_x^3 + 60u'D_x^2 - c_1 D_x \\ &\quad + 40u''' - 60u''' - c_1' + c_0. \end{aligned}$$

Since $c_1' = 60uu' - 50u'''$, the last coefficient equals $15u''' - 30uu' = -c_0$. Hence $B_5^* = -B_5$ on tests. This means $(f, B_5 g) = -(B_5 f, g)$ for compact smooth f, g. It does not identify a closed operator or its Hilbert adjoint domain.

Exercise 24.2 (Constancy for a sufficiently regular eigenbranch). Let $u \in C^1(I; \mathcal{S}(\mathbb{R}, \mathbb{R}))$ solve $u_t = K_5[u]$ on an open interval I. Suppose $\lambda \in C^1(I; \mathbb{R})$ and $g \in C^1(I; H^7(\mathbb{R}, \mathbb{C}))$, with $g(t) \neq 0$ and $L(t)g(t) = \lambda(t)g(t)$ at every time. The Sobolev space H^7 uses the Fourier norm $\|\langle k \rangle^7 \widehat{g}\|_2$.

Use the following operator fact as a *given*: a real bounded multiplication perturbation of $-D_x^2$ is self-adjoint on the common domain $H^2(\mathbb{R})$. This is the relative-bound-zero case of Kato, *Perturbation Theory for Linear Operators*, V, Theorem 4.3, p. 287. Prove that λ is constant. Specify the domain needed for moving L across an inner product. Does this calculation require simplicity or isolation, and does it construct a branch or prove a theorem about the whole spectrum?

Solution. Multiplication by u and all its spatial derivatives preserves the Sobolev spaces used here, since those coefficient derivatives are bounded. The eigenvalue equation $g'' = (u - \lambda)g$ successively gives $g \in H^m$ for every integer m, so it has a smooth representative. Only the assumed H^7 regularity is needed for $B_5 g \in H^2$. The equation $Lg = \lambda g$ also implies $B_5(Lg) = \lambda B_5 g$. Both $LB_5 g$ and $B_5 Lg$ belong to L^2.

The assumed time regularity permits differentiation into L^2:

$$u_t g + Lg_t = \lambda' g + \lambda g_t.$$

The real eigenvalue and self-adjointness give $(g, Lg_t) = (Lg, g_t) = \lambda(g, g_t)$. Subtracting these terms yields

$$\lambda' \|g\|_2^2 = (g, u_t g) = (g, [B_5, L]g) = \lambda(g, B_5 g) - (Lg, B_5 g) = 0.$$

The commutator equality is the finite differential formula proved in the first problem, now with both compositions justified on this function. Moving L uses $g \in H^2$ and $B_5 g \in H^2$. Since g is nonzero at every time, $\lambda' = 0$ throughout I.

The proof did not need an eigenspace of dimension one or an isolated eigenvalue. Such properties are important in the separate problem of obtaining differentiable eigenbranches. Here the branch and its strong regularity were hypotheses. Neither their existence nor an evolution operator on all of L^2 has been constructed.

Exercise 24.3 (The same interior expression with moving endpoints). Fix $\ell > 0$ and a smooth positive nonconstant function $\kappa(t)$. On $L^2(0,\ell)$ consider the expression $L = -\partial_x^2$ with domain

$$\mathcal{D}_t = \left\{ f \in H^2(0,\ell) : f(0) = 0, \quad f'(\ell) = \beta(t) f(\ell) \right\}, \qquad \beta(t) = \kappa(t)\coth(\kappa(t)\ell).$$

The trace values in this formula have their usual continuous Sobolev representatives on a finite interval.

(a) Show that this realization is symmetric by its endpoint Green formula. Find a nonzero smooth eigenfunction with a nonconstant eigenvalue. Self-adjointness is not required for this task.

(b) With $B = 0$, verify $L_t\psi = [B, L]\psi = 0$ on every compactly supported interior test. Explain why this does not contradict the preceding conditional eigenbranch result or establish constancy of the varying eigenvalue.

Solution. For $f, g \in \mathcal{D}_t$, integration by parts gives

$$(f, Lg) - (Lf, g) = [\overline{f'}g - \overline{f}g']_0^\ell.$$

At zero both undifferentiated values vanish. At ℓ, both derivatives equal the same real coefficient $\beta(t)$ times their values, so the two products cancel. This proves symmetry on the stated domain.

The function $g(t,x) = \sinh(\kappa(t)x)$ obeys $g(t,0) = 0$ and

$$g_x(t,\ell) = \kappa\cosh(\kappa\ell) = \kappa\coth(\kappa\ell)\sinh(\kappa\ell) = \beta(t)g(t,\ell).$$

It is nonzero and $-g_{xx} = -\kappa(t)^2 g$. Its eigenvalue is therefore nonconstant when κ is nonconstant.

The differential coefficient of L is independent of time, so on every compact interior test $L_t\psi = 0 = [0, L]\psi$. This is only a common interior-test identity. Differentiation of the endpoint condition instead gives

$$(g_t)'(\ell) - \beta(t)g_t(\ell) = \beta'(t)g(\ell).$$

Whenever $\kappa'(t) \neq 0$, its right side is nonzero. Indeed, with $z = \kappa\ell > 0$, $\mathrm{d}(\kappa\coth(\kappa\ell))/\mathrm{d}\kappa = (\sinh z\cosh z - z)/\sinh^2 z > 0$. The numerator is zero at zero and has derivative $2\sinh^2 z > 0$. Thus g_t need not belong to $\mathcal{D}_t$. The cancellation using a fixed self-adjoint domain in the preceding problem is unavailable. A zero identity for the interior differential expression alone does not control time-dependent boundary domains.

Classical Field Theory — Exercise Sheet 25

Analytic extension and direct scattering

Solutions

Use $L = -\partial_x^2 + u$, the Wronskian $W(f,g) = fg' - f'g$, and $f_- = af_+(\cdot,-k) + bf_+(\cdot,k)$. For real $k \neq 0$, the right reflection amplitude is $r = b/a$. All potentials in this sheet are real Schwartz functions.

Exercise 25.1 (An upper-half-plane Volterra construction). For $\operatorname{Im} k \geq 0$ define

$$H(s,k) = \int_0^s e^{2ik\sigma} d\sigma, \qquad s \geq 0.$$

Thus $H(s,k) = (e^{2iks} - 1)/(2ik)$ for $k \neq 0$ and $H(s,0) = s$. Consider

$$m(x,k) = 1 + \int_x^\infty H(y-x,k)u(y)m(y,k)dy.$$

(a) For every finite R, construct the unique bounded solution on $x \geq R$ by iteration, uniformly for k in the closed upper half-plane. Prove that $m(x,k)$ is holomorphic in $\operatorname{Im} k > 0$ for each fixed x, and continuous up to its real boundary, including $k = 0$.

(b) Prove $m \to 1$ and $m_x \to 0$ as $x \to +\infty$. Show that $f_+(x,k) = e^{ikx}m(x,k)$ solves $Lf_+ = k^2 f_+$. State the two right asymptotic conditions at zero energy. Does this prove a nonzero Wronskian basis at $k = 0$?

The complex-analysis inputs may be used as givens: dominated parameter integrals of holomorphic functions are holomorphic, and locally uniform limits of holomorphic functions are holomorphic. These are the parameter-integral theorem and the uniform-limit result proved in *Analysis*, Lecture 10.

Solution. For $s \geq 0$, the defining integral gives $|H(s,k)| \leq s$ throughout the closed upper half-plane. Put $m_0 = 1$ and let m_j be the jth Volterra iterate. For $x \geq R$, each kernel factor at an ordered integration point y is at most $y - R$. Hence ordered integration yields

$$|m_j(x,k)| \leq \frac{W_R^j}{j!}, \qquad W_R = \int_R^\infty (y-R)|u(y)|dy < \infty.$$

The series $\sum_{j\geq 0} m_j$ converges uniformly in $x \geq R$ and in all k in that closed half-plane. Dominated convergence gives the integral equation. If a bounded difference satisfies its homogeneous equation, j iterations bound it by its supremum times $W_R^j/j!$. Letting $j \to \infty$ proves uniqueness.

For each finite list of ordered points the product of the kernels is holomorphic in k. Its integrable majorant is the product of the weights $(y-R)|u(y)|$, independent of k on the closed upper half-plane. The given parameter-integral theorem therefore makes every iterate holomorphic in the open upper half-plane. Their locally uniform sum is holomorphic. The same majorant and dominated convergence give continuity of every iterate on the closure, including $k = 0$. Uniform convergence of the series preserves that continuity. Solutions constructed with different R agree on overlaps by uniqueness, so they define m at every finite x.

The refined bound at a chosen x is

$$|m(x,k)-1| \le e^{W(x)}-1, \qquad W(x)=\int_x^\infty (y-x)|u(y)|\mathrm{d}y.$$

It tends to zero as $x \to +\infty$. Differentiating the integral with its lower endpoint gives

$$m_x = -\int_x^\infty e^{2\mathrm{i}k(y-x)}u(y)m(y,k)\mathrm{d}y, \qquad m_{xx} = um - 2\mathrm{i}km_x.$$

The first formula is justified by an integrable majorant on each right half-line; its modulus is at most $e^{W_R}\int_x^\infty |u(y)|\mathrm{d}y$, which tends to zero. The second follows from FTC and one more differentiation. Thus

$$f_+'' = e^{\mathrm{i}kx}(-k^2m + 2\mathrm{i}km_x + m_{xx}) = (u-k^2)f_+.$$

It has the relative tails $e^{-\mathrm{i}kx}f_+ \to 1$ and $e^{-\mathrm{i}kx}f_+' \to \mathrm{i}k$. At zero these say $f_+(x,0) \to 1$ and $f_+'(x,0) \to 0$. The construction includes this boundary value, but no pair with Wronskian $2\mathrm{i}k \neq 0$ exists by that formula at $k=0$. It proves neither a zero-energy scattering basis nor absence of a resonance.

Exercise 25.2 (Reflectionless scattering by intertwining). Fix $\kappa > 0$ and set

$$L_\kappa = -\partial_x^2 - 2\kappa^2\operatorname{sech}^2(\kappa x), \qquad A = \partial_x + \kappa\tanh(\kappa x), \quad A^\dagger = -\partial_x + \kappa\tanh(\kappa x).$$

(a) Factor $L_\kappa + \kappa^2$ and the free operator plus κ^2. Use the resulting intertwining identity to construct both real-energy Jost solutions and determine a, b, r with the fixed Wronskian convention.

(b) Find the right-normalised negative-energy state and its inverse squared norm. Show that there are no other negative-energy L^2 states. Then give a bounded zero-energy solution which is not in L^2.

Use the Sobolev integration-by-parts and domain arguments of Lecture 18. A bounded non-L^2 zero-energy solution is called a *zero-energy resonance* in this problem.

Solution. The product rule gives

$$A^\dagger A = L_\kappa + \kappa^2, \qquad AA^\dagger = L_0 + \kappa^2, \qquad L_0 = -\partial_x^2.$$

Consequently $L_\kappa A^\dagger = A^\dagger L_0$ as differential expressions on smooth functions. In this calculation a free exponential is not asserted to belong to the L^2 operator domain. Applying this to $e^{\mathrm{i}kx}$ and dividing by $\kappa - \mathrm{i}k$ gives

$$f_+(x,k) = e^{\mathrm{i}kx}\frac{k+\mathrm{i}\kappa\tanh(\kappa x)}{k+\mathrm{i}\kappa}.$$

Its right coefficient and derivative have the required limits. Reflection in x constructs the left solution

$$f_-(x,k) = e^{-\mathrm{i}kx}\frac{k-\mathrm{i}\kappa\tanh(\kappa x)}{k+\mathrm{i}\kappa}.$$

Since $f_+(x,-k) = e^{-\mathrm{i}kx}(k-\mathrm{i}\kappa\tanh(\kappa x))/(k-\mathrm{i}\kappa)$, comparison gives

$$f_- = \frac{k-\mathrm{i}\kappa}{k+\mathrm{i}\kappa}f_+(x,-k).$$

Thus $a = (k-\mathrm{i}\kappa)/(k+\mathrm{i}\kappa)$ and $b = r = 0$. This also gives $W(f_-,f_+) = 2\mathrm{i}ka$ with the stated order. For real k, $|a| = 1$, in agreement with the flux identity.

At $k = \mathrm{i}\kappa$ the right solution is

$$f_+(x, \mathrm{i}\kappa) = \frac{1}{2} e^{-\kappa x}(1 + \tanh(\kappa x)) = \frac{1}{2}\operatorname{sech}(\kappa x).$$

It decays at both ends, has right coefficient one and squared norm $\frac{1}{4}\int \operatorname{sech}^2(\kappa x)\mathrm{d}x = 1/(2\kappa)$. Hence its inverse squared norm is 2κ. It is in the kernel of A, so its energy is $-\kappa^2$.

If $L_\kappa g = -\mu^2 g$ with $\mu > 0$ and $g \in H^2$, its equation and the bounded smooth potential bootstrap g to every H^m. The intertwining $AL_\kappa = L_0 A$ gives $L_0(Ag) = -\mu^2 Ag$ with $Ag \in H^2$. Free integration by parts implies $\|(Ag)'\|_2^2 = -\mu^2\|Ag\|_2^2$, so $Ag = 0$. Solving this first-order equation gives $g = C\operatorname{sech}(\kappa x)$, and its energy is necessarily $-\kappa^2$. This proves uniqueness of the negative level and its one-dimensional eigenspace.

Finally, letting $k \to 0$ in the right solution gives $\tanh(\kappa x)$. Direct differentiation also verifies $L_\kappa \tanh(\kappa x) = 0$. Its two endpoint limits are -1 and 1, so it is bounded but not in L^2. It is a resonance under the stated definition, not an additional bound state.

Exercise 25.3 (A weak asymmetric potential retains a phase). Let $u_\epsilon = \epsilon V$, with $\epsilon \in \mathbb{R}$, and

$$V(x) = e^{-(x-1)^2} + 2e^{-(x+2)^2}.$$

For each fixed real $k \neq 0$, derive the first-order expansions of a, b, r as $\epsilon \to 0$ and prove an $O(\epsilon^2)$ remainder bound. Evaluate the Gaussian coefficient and the leading reflection probability. Show that no negative-energy bound states occur when $\epsilon \geq 0$. The expansion is a fixed-k assertion; no uniform limit at $k = 0$ is requested.

Solution. The left Volterra equation is

$$f_-(x, k) = e^{-\mathrm{i}kx} + \int_{-\infty}^{x} \frac{\sin k(x-y)}{k}\, \epsilon V(y) f_-(y, k)\mathrm{d}y.$$

The same ordered iteration as Lecture 25 gives, on the whole line,

$$|f_-| \leq e^{\delta}, \qquad |f_- - e^{-\mathrm{i}kx}| \leq e^{\delta} - 1, \qquad \delta = \frac{|\epsilon|\|V\|_1}{|k|}.$$

Separating the sine and taking the two endpoint coefficients at the right gives the exact integral expressions

$$a = 1 - \frac{\epsilon}{2\mathrm{i}k}\int_{\mathbb{R}} e^{\mathrm{i}ky} V(y) f_-(y, k)\mathrm{d}y, \qquad b = \frac{\epsilon}{2\mathrm{i}k}\int_{\mathbb{R}} e^{-\mathrm{i}ky} V(y) f_-(y, k)\mathrm{d}y.$$

The derivative asymptotics follow from the differentiated Volterra equation, so these are the coefficients in the actual right Jost basis. Replacing f_- by its free term makes an error in each coefficient at most

$$\frac{|\epsilon|\|V\|_1}{2|k|}(e^{\delta} - 1) = O(\epsilon^2).$$

It follows that

$$a = 1 - \frac{\epsilon}{2\mathrm{i}k}\int V + O(\epsilon^2), \qquad b = \frac{\epsilon}{2\mathrm{i}k}\int e^{-2\mathrm{i}ky} V(y)\mathrm{d}y + O(\epsilon^2).$$

Since $a = 1 + O(\epsilon)$ and $a \neq 0$, division gives the same first-order coefficient for $r = b/a$.

Translation of the Gaussian integral yields

$$\int V = 3\sqrt{\pi}, \qquad \int e^{-2\mathrm{i}kx} V(x)\mathrm{d}x = \sqrt{\pi} e^{-k^2}(e^{-2\mathrm{i}k} + 2e^{4\mathrm{i}k}).$$

Thus

$$a = 1 - \frac{3\epsilon\sqrt{\pi}}{2\mathrm{i}k} + O(\epsilon^2),$$

$$r = \frac{\epsilon\sqrt{\pi}e^{-k^2}}{2\mathrm{i}k}(e^{-2\mathrm{i}k} + 2e^{4\mathrm{i}k}) + O(\epsilon^2).$$

and b has the same first-order term. The absolute square of the parenthesis is $5 + 4\cos 6k$. Squaring the amplitude expansion gives

$$|r|^2 = \frac{\epsilon^2\pi e^{-2k^2}}{4k^2}(5 + 4\cos 6k) + O(|\epsilon|^3).$$

For instance at $k = \pi/4$ the parenthesis is $-2 - \mathrm{i}$, so its coefficient after division by i has both real and imaginary parts. Taking the probability discards that phase.

For $\epsilon \geq 0$ the potential is nonnegative. A putative bound state $Lf = -\mu^2 f$ would give

$$\int (|f'|^2 + \epsilon V|f|^2)\mathrm{d}x = -\mu^2\|f\|_2^2,$$

which is impossible for $\mu > 0$ and nonzero f. This proves the stated absence without asserting that weak negative potentials have the same property.

Classical Field Theory — Exercise Sheet 26

Scattering evolution and soliton shifts

Solutions

Use the standing real KdV solution $u \in C^\infty(\mathbb{R}; \mathcal{S})$ of Lecture 24 and the Jost/scattering conventions of Lecture 25. The first problem fills the analytic step left sketched in Lecture 26. The later finite-rank construction prescribes its norming coefficients; it makes no additional theorem about birth of eigenvalues.

Exercise 26.1 (Time differentiation at real energy). Fix real $k \neq 0$. For the right Jost solution put $f_+(t,x,k) = e^{ikx} m(t,x,k)$, with

$$m = 1 + \int_x^\infty \frac{e^{2ik(y-x)} - 1}{2ik}\, u(t,y) m(t,y,k)\, \mathrm{d}y.$$

(a) Prove that this Volterra construction is continuously differentiable in t. On every compact time interval prove $m_t \to 0$ as $x \to +\infty$, together with the spatial derivative limits needed to compare both the function and derivative tails of $(f_+)_t - B f_+$. Here $B = -4\partial_x^3 + 6u\partial_x + 3u_x$.

(b) Prove the right and left temporal equations, justify differentiability of a, b from finite-point Wronskians, and prove $r(k,t) = r(k,t_0) e^{8ik^3(t-t_0)}$. Do not pair the oscillatory Jost solutions in L^2.

Solution. Fix a compact time interval J and a finite R. The real-energy kernel is bounded by $|k|^{-1}$. Define the tails

$$A_J(x) = |k|^{-1} \int_x^\infty \sup_{t\in J} |u(t,y)|\, \mathrm{d}y,$$
$$B_J(x) = |k|^{-1} \int_x^\infty \sup_{t\in J} |u_t(t,y)|\, \mathrm{d}y.$$

They are finite and tend to zero at the right end. The nth ordered iterate has bound $A_J(R)^n/n!$ on $x \geq R$. Differentiating one of its n coefficient factors and summing over all positions gives the bound $B_J(R) A_J(R)^{n-1}/(n-1)!$. Indeed, the time-differentiated point can occupy any position among the ordered remaining $n-1$ points. These scalar series converge uniformly. Termwise time differentiation is therefore justified, gives continuous m_t, and the refined tail bound is

$$|m_t(t,x,k)| \leq B_J(x) e^{A_J(x)} \longrightarrow 0 \quad \text{uniformly for } t \in J.$$

The same series gives $|m| \leq e^{A_J(x)}$ and $m \to 1$ uniformly there.

Differentiating in space, including the lower endpoint, gives

$$m_x = -\int_x^\infty e^{2ik(y-x)} u m\, \mathrm{d}y, \qquad m_{xx} = um - 2ikm_x.$$

The first expression tends uniformly to zero. For every $j \geq 0$, further differentiation obeys $m^{(j+2)} = D_x^j(um) - 2ikm^{(j+1)}$. Induction gives bounded spatial derivatives and their zero limits for all positive derivative orders, uniformly on J. Time differentiation of the integral for m_x gives

$$(m_t)_x = -\int_x^\infty e^{2ik(y-x)} (u_t m + u m_t)\, \mathrm{d}y.$$

The already proved uniform bounds and integrable tails justify it and make it tend to zero. Differentiating m_{xx} in time also gives $(m_t)_{xx} = u_t m + u m_t - 2ik(m_t)_x$, with zero right limit. These identities establish the local regularity needed below.

Thus, through every required spatial order, $e^{-ikx}\partial_x^j f_+ \to (ik)^j$. Also $e^{-ikx}(f_+)_t \to 0$ and $e^{-ikx}\partial_x(f_+)_t \to 0$. The limits through order four show that Bf_+ and its derivative have the same tails as $4ik^3 e^{ikx}$ and its derivative. Order five and the mixed second derivative justify applying L to $(f_+)_t - Bf_+$. Differentiating $Lf_+ = k^2 f_+$ and using the pointwise Lax formula gives

$$(L - k^2)\big((f_+)_t - Bf_+\big) = 0.$$

Hence $z = (f_+)_t - Bf_+ + 4ik^3 f_+$ solves that equation and has both zero right Jost tails. The solution $f_+ + z$ has exactly the same two normalisations as f_+, so Lecture 25's uniqueness forces $z = 0$. We have proved $(f_+)_t = Bf_+ - 4ik^3 f_+$.

The reflected Volterra estimates give the corresponding regularity at the left end. Apply the original Lax equation to that left solution; reflection is used for estimates, not to assert that $u(t,-x)$ solves the same KdV equation. Since $Be^{-ikx} \sim -4ik^3 e^{-ikx}$ there, uniqueness gives $(f_-)_t = Bf_- + 4ik^3 f_-$.

At a fixed finite spatial point the Wronskian formulas are

$$a = \frac{W(f_-, f_+)}{2ik}, \qquad b = \frac{W(f_+(\cdot,-k), f_-)}{2ik}.$$

The just-proved C^1 time dependence of functions and first derivatives makes both coefficients C^1. Differentiate $f_- = af_+(-k) + bf_+(k)$. Use the two temporal equations, including the reversed sign at $-k$, and cancel the B terms. The result is

$$a_t f_+(-k) + (b_t - 8ik^3 b) f_+(k) = 0.$$

Their nonzero Wronskian $2ik$ gives $a_t = 0$ and $b_t = 8ik^3 b$. Since $a \neq 0$, division and integration give $r(k,t) = r(k,t_0)e^{8ik^3(t-t_0)}$. No integral of a squared Jost function was used.

Exercise 26.2 (A positive finite-rank inverse equation). Take distinct $\kappa_j > 0$ and $c_j > 0$, $1 \le j \le N$, with $F(s) = \sum_j c_j e^{-\kappa_j s}$. Consider the right Marchenko equation

$$K(x,y) + F(x+y) + \int_x^\infty K(x,z)F(z+y)\,\mathrm{d}z = 0, \qquad y \ge x.$$

The finite positive reflectionless case of the inverse-scattering theorem is a *given*: these data are admissible and reconstruct a real rapidly decaying potential by $u = -2\,\mathrm{d}K(x,x)/\mathrm{d}x$. It is the zero-background specialization of the cited theorem in Lectures 25–26: Egorova–Grunert–Teschl, *Nonlinearity* **22** (2009), §4, Theorems 4.3–4.4, equations (4.26)–(4.30). Their $(\gamma_j^+)^2$ is c_j.

(a) Reduce the equation to a real symmetric $N \times N$ linear system. Prove that its matrix is positive definite and invertible for every real x.

(b) With $v_j(x) = \sqrt{c_j}e^{-\kappa_j x}$, define $\mathsf{C}_{jl} = v_j v_l/(\kappa_j + \kappa_l)$ and $\tau = \det(I + \mathsf{C})$. Prove $K(x,x) = (\log\tau)'$ and $u = -2(\log\tau)''$. Explain why no real pole arises from these positive data.

Solution. The integral equation itself makes $K(x,y)$ a linear combination of the N exponentials in y. Write $K = -\sum_j z_j e^{-\kappa_j y}$. Since the exponents are distinct, their coefficients give

$$z_j + c_j \sum_l \frac{e^{-(\kappa_j+\kappa_l)x}}{\kappa_j + \kappa_l} z_l = c_j e^{-\kappa_j x}.$$

After setting $q_j = z_j/\sqrt{c_j}$ and $v_j = \sqrt{c_j}e^{-\kappa_j x}$ this is

$$(I + \mathsf{C})q = v, \qquad \mathsf{C}_{jl} = \frac{v_j v_l}{\kappa_j + \kappa_l}.$$

For any real column a,

$$a^{\mathsf{T}}\mathsf{C}a = \int_x^\infty \left(\sum_j a_j\sqrt{c_j}e^{-\kappa_j z}\right)^2 \mathrm{d}z \geq 0.$$

Thus $a^{\mathsf{T}}(I + \mathsf{C})a \geq |a|^2$, proving invertibility and a strictly positive determinant at every finite x.

On the diagonal the kernel is $K(x,x) = -v^{\mathsf{T}}q = -v^{\mathsf{T}}(I + \mathsf{C})^{-1}v$. Differentiating each entry gives $\mathsf{C}' = -vv^{\mathsf{T}}$. Jacobi's finite determinant formula therefore yields

$$(\log\tau)' = \operatorname{tr}[(I + \mathsf{C})^{-1}\mathsf{C}'] = -v^{\mathsf{T}}(I + \mathsf{C})^{-1}v = K(x,x).$$

The supplied reconstruction formula gives $u = -2(\log\tau)''$. The entries are smooth and $\tau > 0$, so this expression and the kernel have no real poles. These finite-dimensional calculations solve the integral equation; the interpretation as complete scattering data remains the stated inverse-theorem input.

Exercise 26.3 (Two pulses and their shifts). Take $\kappa_1 > \kappa_2 > 0$ and prescribe $c_j(t) = c_j(0)e^{8\kappa_j^3 t}$, with $c_j(0) > 0$. Put

$$x_{j,0} = \frac{1}{2\kappa_j}\log\frac{c_j(0)}{2\kappa_j}, \qquad \eta_j = -2\kappa_j(x - 4\kappa_j^2 t - x_{j,0}).$$

(a) Evaluate the two-state determinant and find its interaction coefficient d_{12} in $\tau = 1 + e^{\eta_1} + e^{\eta_2} + d_{12}e^{\eta_1+\eta_2}$.

(b) Verify directly that $u = -2\partial_x^2\log\tau$ solves KdV. One may use, after deriving it by differentiating $\log\tau$,

$$\tau\tau_{xt} - \tau_x\tau_t + \tau\tau_{xxxx} - 4\tau_x\tau_{xxx} + 3\tau_{xx}^2 = 0$$

as a sufficient condition.

(c) Find the incoming and outgoing centre lines of each separated pulse and its net shift. Justify the two spatial derivatives in each moving-frame limit, rather than compare only pointwise values of τ.

The parameters $x_{j,0}$ describe the norming coefficients. They need not be actual minima positions during the interaction.

Solution. The two diagonal entries of C are e^{η_j}, while $\mathsf{C}_{12}^2 = 4\kappa_1\kappa_2 e^{\eta_1+\eta_2}/(\kappa_1+\kappa_2)^2$. Thus

$$\tau = 1 + e^{\eta_1} + e^{\eta_2} + \left(1 - \frac{4\kappa_1\kappa_2}{(\kappa_1+\kappa_2)^2}\right)e^{\eta_1+\eta_2}.$$

The coefficient is $d_{12} = (\kappa_1 - \kappa_2)^2/(\kappa_1+\kappa_2)^2$, strictly between zero and one.

For the PDE check set $q = \log\tau$. Dividing the proposed bilinear expression by τ^2 and differentiating the logarithm gives $q_{xt} + q_{xxxx} + 6q_{xx}^2$. Its x derivative, multiplied by -2, is $u_t - 6uu_x + u_{xxx}$ for $u = -2q_{xx}$.

To verify that expression, let $\zeta_j = -2\kappa_j$. Then $\partial_x \eta_j = \zeta_j$ and $\partial_t \eta_j = -\zeta_j^3$. In the symmetric exponential-pair expansion, constant/self pairs vanish. Pairs differing by only one η_j vanish by this dispersion relation. The remaining coefficient of $e^{\eta_1+\eta_2}$ is

$$3\zeta_1\zeta_2\left[d_{12}(\zeta_1+\zeta_2)^2 - (\zeta_1-\zeta_2)^2\right] = 0.$$

The chosen d_{12} proves its vanishing, hence the KdV equation.

In the faster frame $x = 4\kappa_1^2 t + X$, $\eta_2 \to +\infty$ as $t \to -\infty$ and $\eta_2 \to -\infty$ as $t \to +\infty$. The relevant determinants are consequently $e^{\eta_2}(1 + d_{12}e^{\eta_1})$ incoming and $1 + e^{\eta_1}$ outgoing. The incoming centre solves $d_{12}e^{\eta_1} = 1$, giving

$$x_1^{\text{in}} = 4\kappa_1^2 t + x_{1,0} + \frac{\log d_{12}}{2\kappa_1}, \qquad x_1^{\text{out}} = 4\kappa_1^2 t + x_{1,0}.$$

In the slower frame the roles are reversed: $\eta_1 \to -\infty$ incoming and $\eta_1 \to +\infty$ outgoing. Thus

$$x_2^{\text{in}} = 4\kappa_2^2 t + x_{2,0}, \qquad x_2^{\text{out}} = 4\kappa_2^2 t + x_{2,0} + \frac{\log d_{12}}{2\kappa_2}.$$

Subtracting the two intercepts gives $\Delta x_1 = -\log d_{12}/(2\kappa_1) > 0$ and $\Delta x_2 = \log d_{12}/(2\kappa_2) < 0$.

On each compact set of X, the discarded exponential has an exponentially small factor in $|t|$. Its first two X derivatives have the same factor, since each exponent is affine in X. The retained factors $1+e^{\eta_j}$ and $1+d_{12}e^{\eta_j}$ are bounded below by one. The same estimates therefore hold after two derivatives of their logarithms. They give separated profiles $-2\kappa_j^2 \operatorname{sech}^2$ with the centre lines just found, not merely a pointwise determinant comparison.

Classical Field Theory — Exercise Sheet 27

Matrix compatibility, formal charges and Bäcklund kinks

Solutions

The first two problems assume a smooth real KdV solution $u : I \times \mathbb{R} \to \mathbb{R}$, with I an open time interval, $u_t = 6uu_x - u_{xxx}$ and spatial period $L_{\mathrm{per}} > 0$. This is a conditional hypothesis, not an existence assertion on the circle. The last problem instead uses a real field on the whole line.

Exercise 27.1 (Converting the scalar KdV pair to matrices). For fixed $k \in \mathbb{C} \setminus \{0\}$ consider

$$-\psi_{xx} + u\psi = k^2\psi, \qquad \psi_t = -4\psi_{xxx} + 6u\psi_x + 3u_x\psi.$$

Set $\Psi = (\psi, \psi_x)^{\mathsf{T}}$.

(a) Eliminate higher spatial derivatives to derive $\Psi_x = U\Psi$, $\Psi_t = V\Psi$. Compute every entry of $U_t - V_x + [U, V]$ for an arbitrary smooth u.

(b) For the assumed periodic KdV field, state exactly what Lecture 27's monodromy theorem proves for these matrices. Does it establish the Poisson commutativity or independence of the resulting traces?

Solution. The spectral equation gives the spatial rows immediately: $\psi_x = \Psi_2$ and $(\Psi_2)_x = (u - k^2)\Psi_1$. Its derivative is $\psi_{xxx} = u_x\psi + (u - k^2)\psi_x$. Substitution into the temporal equation gives

$$\psi_t = -u_x\psi + (2u + 4k^2)\psi_x.$$

Differentiating it in x gives

$$(\psi_x)_t = [-u_{xx} + (2u + 4k^2)(u - k^2)]\psi + u_x\psi_x.$$

Thus the matrices are

$$U = \begin{pmatrix} 0 & 1 \\ u - k^2 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} -u_x & 2u + 4k^2 \\ -u_{xx} + (2u + 4k^2)(u - k^2) & u_x \end{pmatrix}.$$

For the curvature, matrix multiplication gives

$$[U, V] = \begin{pmatrix} -u_{xx} & 2u_x \\ -2(u - k^2)u_x & u_{xx} \end{pmatrix}.$$

Also

$$U_t - V_x = \begin{pmatrix} u_{xx} & -2u_x \\ u_t + u_{xxx} - (4u + 2k^2)u_x & -u_{xx} \end{pmatrix}.$$

Adding proves

$$U_t - V_x + [U, V] = \begin{pmatrix} 0 & 0 \\ u_t - 6uu_x + u_{xxx} & 0 \end{pmatrix}.$$

For the assumed KdV field it is zero. The matrices are smooth and periodic, so the period transport satisfies $T_t = [V(t, 0; k), T]$ and every $\operatorname{tr} T(t; k)^r$ is constant, for each fixed nonzero k and positive integer r. This uses the finite-matrix transport theorem with the stated basepoint. It does not prove that traces with different labels are independent or have zero mutual field-theory Poisson brackets.

Exercise 27.2 (Local charges from a formal Riccati series). On a patch with $\psi \neq 0$, the scalar equation suggests $\rho = \psi_x/\psi$ and $\rho_x + \rho^2 = u - k^2$. In this problem work purely with formal Laurent series: put $\zeta = 2ik$ and seek

$$\rho = \frac{\zeta}{2} + \sum_{j\geq 1} a_j[u]\zeta^{-j}.$$

Equality means equality of each coefficient; convergence and a globally nonvanishing actual ψ are not assumed.

(a) Find the recursion and the first five coefficients. Integrate them over one spatial period and identify the first three nonzero integrals.

(b) Prove the formal identity $\rho_t = \partial_x[-u_x+(2u+4k^2)\rho]$ under KdV. Conclude coefficientwise conservation. Explain which analytic conclusion about an actual spectral expansion has not been proved.

Solution. Put $w = \rho - \zeta/2$. Since $k^2 = -\zeta^2/4$, the Riccati equation becomes $w_x + \zeta w + w^2 = u$. Its constant coefficient gives $a_1 = u$. The coefficient of ζ^{-n} gives

$$a_{n+1} = -a_n' - \sum_{\substack{i+j=n\\ i,j\geq 1}} a_i a_j.$$

In increasing order this gives

$$\begin{aligned} &a_1 = u, \qquad a_2 = -u', \qquad a_3 = u'' - u^2,\\ &a_4 = -u''' + 4uu',\\ &a_5 = u'''' - 5(u')^2 - 6uu'' + 2u^3. \end{aligned}$$

Periodicity removes integrals of derivatives, including a_2 and a_4. It also gives $\int uu'' = -\int (u')^2$. Consequently

$$\int_0^{L_{\mathrm{per}}} a_1 = \int u, \qquad \int a_3 = -\int u^2, \qquad \int a_5 = \int [(u')^2 + 2u^3].$$

The integrals on the right are over the same period.

For the time identity put $S = -u_x + (2u+4k^2)\rho$. Differentiating the formal Riccati equation gives $(\partial_x + 2\rho)\rho_t = u_t$. A direct calculation, using $\rho_x + \rho^2 = u - k^2$, gives

$$\begin{aligned} (\partial_x + 2\rho)S_x &= -u_{xxx} + 4u_x(\rho_x + \rho^2) + (2u + 4k^2)(\rho_{xx} + 2\rho\rho_x)\\ &= -u_{xxx} + 4u_x(u - k^2) + (2u + 4k^2)u_x\\ &= 6uu_x - u_{xxx} = u_t. \end{aligned}$$

Thus $R = \rho_t - S_x$ obeys $(\partial_x + 2\rho)R = 0$. The series R is bounded above in powers of ζ. If its highest nonzero term were $r_m\zeta^m$, multiplication by the leading ζ in 2ρ would give $r_m\zeta^{m+1}$. Neither its x derivative nor the negative-power coefficients can cancel that highest term. Hence no such term exists and $R = 0$ formally.

The coefficient of each power therefore gives $\partial_t a_j = \partial_x b_j$ for a periodic differential polynomial b_j. Integrating that finite coefficient identity proves conservation. The reasoning never interchanges an infinite series with an integral. It proves neither convergence of the Laurent series nor an asymptotic expansion of an actual monodromy eigenvalue.

Exercise 27.3 (Bäcklund transformation and a sine-Gordon kink). On $\mathbb{R}^{1,1}$ use $\xi = (x+t)/2$, $\eta = (x-t)/2$. For two real C^2 fields $\phi, \widetilde{\phi}$, put $\alpha_B = (\phi + \widetilde{\phi})/2$ and $\beta_B = (\phi - \widetilde{\phi})/2$. For a fixed $a_B > 0$, the proposed *Bäcklund system* is

$$\partial_\xi \alpha_B = a_B \sin \beta_B, \qquad \partial_\eta \beta_B = a_B^{-1} \sin \alpha_B.$$

(a) Prove that any such pair solves $\phi_{\xi\eta} = \sin\phi$ and $\widetilde{\phi}_{\xi\eta} = \sin\widetilde{\phi}$.

(b) Put $\widetilde{\phi} = 0$ and solve for the branch $0 < \phi < 2\pi$ at finite points. Convert it to physical coordinates and find its velocity.

(c) For density $L = \frac{1}{2}(\phi_t^2 - \phi_x^2) - (1 - \cos\phi)$, compute the kink charge $Q = [\phi(+\infty) - \phi(-\infty)]/(2\pi)$, energy and lower-index momentum integral $\int \phi_t \phi_x \mathrm{d}x$. This line solution is not a periodic field for the monodromy theorem.

Solution. Differentiation of the two first-order equations gives

$$(\alpha_B)_{\xi\eta} = \cos\beta_B \sin\alpha_B, \qquad (\beta_B)_{\xi\eta} = \cos\alpha_B \sin\beta_B.$$

Adding and subtracting proves

$$\phi_{\xi\eta} = \sin(\alpha_B + \beta_B) = \sin\phi, \qquad \widetilde{\phi}_{\xi\eta} = \sin(\alpha_B - \beta_B) = \sin\widetilde{\phi}.$$

The physical equation is $\phi_{tt} - \phi_{xx} + \sin\phi = 0$ because $\partial_\xi \partial_\eta = \partial_x^2 - \partial_t^2$.

For $\widetilde{\phi} = 0$ both half fields equal $\phi/2$. The first-order system reads $\phi_\xi = 2a_B \sin(\phi/2)$ and $\phi_\eta = 2a_B^{-1}\sin(\phi/2)$. On the specified open branch, $\mathrm{d}\log\tan(\phi/4)/\mathrm{d}\phi = [2\sin(\phi/2)]^{-1}$. Thus

$$\log\tan(\phi/4) = a_B\xi + a_B^{-1}\eta + \delta.$$

Let $\gamma = (a_B + a_B^{-1})/2$ and $v_{\mathrm{tr}} = (1 - a_B^2)/(1 + a_B^2)$. Then $\gamma > 0$, $|v_{\mathrm{tr}}| < 1$, $\gamma^2(1 - v_{\mathrm{tr}}^2) = 1$, and choosing $\delta = -\gamma x_0$ gives

$$\phi(t,x) = 4\arctan e^{\gamma(x - v_{\mathrm{tr}}t - x_0)}.$$

It remains strictly between zero and 2π at finite points and has those two limits at the spatial ends, so $Q = 1$.

Write $b_{\mathrm{SG}}(z) = 4\arctan e^z$ for its rest profile. Direct differentiation and the half-angle identities give $b'_{\mathrm{SG}} = 2\,\mathrm{sech}\, z$ and $1 - \cos b_{\mathrm{SG}} = 2\,\mathrm{sech}^2 z = (b'_{\mathrm{SG}})^2/2$. The derivative-squared integral is $\int (b'_{\mathrm{SG}})^2 \mathrm{d}z = 4\int \mathrm{sech}^2 z \mathrm{d}z = 8$. For $z = \gamma(x - v_{\mathrm{tr}}t - x_0)$, $\phi_x = \gamma b'_{\mathrm{SG}}$ and $\phi_t = -\gamma v_{\mathrm{tr}} b'_{\mathrm{SG}}$. Consequently

$$E_{\mathrm{en}} = \frac{\gamma^2(1 + v_{\mathrm{tr}}^2) + 1}{2\gamma} \int (b'_{\mathrm{SG}})^2 \mathrm{d}z = 8\gamma.$$

The equality uses $1 = \gamma^2(1 - v_{\mathrm{tr}}^2)$. The lower-index momentum integral is

$$\int \phi_t \phi_x \mathrm{d}x = -\gamma v_{\mathrm{tr}} \int (b'_{\mathrm{SG}})^2 \mathrm{d}z = -8\gamma v_{\mathrm{tr}}.$$

Its sign differs from the raised spatial momentum. The sine-Gordon potential and energy are different from the ϕ^4 kink's potential and energy. The solution is on the line and connects distinct vacua, so the periodic monodromy conclusion has not been applied to it.

Classical Field Theory — Exercise Sheet 28

Tensor variation, gravitational waves and boundary action

Solutions

Use Lecture 28's curvature, inverse-metric and action conventions. All metrics, variations and coordinate maps are smooth. Repeated spacetime indices run from zero to three. Write $\eta = \mathrm{diag}(1,-1,-1,-1)$ only for the flat background. None of the problems asserts global existence for Einstein evolution.

Exercise 28.1 (The full Palatini tensor calculation). Let g_s be a smooth family of Lorentz metrics in fixed coordinates and put

$$k^{\mu\nu} = \partial_s g_s^{\mu\nu}\big|_0, \qquad h_{\mu\nu} = \partial_s (g_s)_{\mu\nu}\big|_0 = -g_{\mu\alpha}g_{\nu\beta}k^{\alpha\beta}, \qquad C^{\rho}_{\mu\nu} = \delta\Gamma^{\rho}_{\mu\nu}.$$

Here ∇ is the background Levi–Civita connection.

(a) Prove that C is a tensor of type $(1,2)$ and derive its expression in terms of ∇h. Do not assume that connection coefficients themselves are tensor components.

(b) Vary the full coordinate Riemann formula. Expand both covariant derivatives of C, including all its index terms, to prove $\delta R^{\rho}{}_{\sigma\mu\nu} = \nabla_\mu C^{\rho}_{\nu\sigma} - \nabla_\nu C^{\rho}_{\mu\sigma}$. Contract to obtain the Palatini identity without setting $\Gamma = 0$.

(c) Contract again to derive the inverse-metric boundary vector $v^\mu = \nabla^\mu(g_{\alpha\beta}k^{\alpha\beta}) - \nabla_\nu k^{\mu\nu}$. This is the complete tensor argument promised by Lecture 28.

Solution. For two connections, their difference evaluated on X, Y is linear over smooth functions in X. It is also linear over such functions in Y: the two Leibniz terms $X(f)Y$ cancel when replacing Y by fY. Thus $\nabla^{(s)} - \nabla$ is a $(1,2)$ tensor. Its parameter derivative at zero, C, is a tensor as well. Torsion-freeness gives $C^{\rho}_{\mu\nu} = C^{\rho}_{\nu\mu}$.

Differentiating metric compatibility gives

$$\nabla_\mu h_{\nu\lambda} = C^{\rho}_{\mu\nu}g_{\rho\lambda} + C^{\rho}_{\mu\lambda}g_{\nu\rho}.$$

Add its (μ,λ,ν) and (ν,λ,μ) forms and subtract its (λ,μ,ν) form. The lower-index symmetry of C leaves

$$C^{\rho}_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\nabla_\mu h_{\lambda\nu} + \nabla_\nu h_{\lambda\mu} - \nabla_\lambda h_{\mu\nu}).$$

The full coordinate variation of Riemann is

$$\delta R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu C^{\rho}_{\nu\sigma} - \partial_\nu C^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}C^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}C^{\lambda}_{\mu\sigma} + C^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - C^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}.$$

On the other hand, the tensor covariant derivative is

$$\nabla_\mu C^{\rho}_{\nu\sigma} = \partial_\mu C^{\rho}_{\nu\sigma} + \Gamma^{\rho}_{\mu\lambda}C^{\lambda}_{\nu\sigma} - \Gamma^{\lambda}_{\mu\nu}C^{\rho}_{\lambda\sigma} - \Gamma^{\lambda}_{\mu\sigma}C^{\rho}_{\nu\lambda}.$$

Subtract its version with μ,ν interchanged. The terms with $\Gamma^{\lambda}_{\mu\nu}$ cancel by torsion-freeness. The remaining lower-σ terms and upper-index terms are precisely the four connection-product terms in the coordinate variation. This proves $\delta R^{\rho}{}_{\sigma\mu\nu} = \nabla_\mu C^{\rho}_{\nu\sigma} - \nabla_\nu C^{\rho}_{\mu\sigma}$ in arbitrary coordinates.

Using $\mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$ now gives

$$\delta\,\mathrm{Ric}_{\mu\nu} = \nabla_\rho C^\rho_{\nu\mu} - \nabla_\nu C^\rho_{\rho\mu}.$$

Metric compatibility allows the next contraction to be written as a divergence:

$$g^{\mu\nu}\delta\,\mathrm{Ric}_{\mu\nu} = \nabla_\rho\big(g^{\mu\nu}C^\rho_{\mu\nu} - g^{\rho\mu}C^\nu_{\mu\nu}\big).$$

Let $\tau = g_{\mu\nu}k^{\mu\nu}$. Substitution of $h_{\mu\nu} = -g_{\mu\alpha}g_{\nu\beta}k^{\alpha\beta}$ into the formula for C yields

$$g^{\alpha\beta}C^\mu_{\alpha\beta} = -\nabla_\nu k^{\mu\nu} + \frac{1}{2}\nabla^\mu\tau, \qquad g^{\mu\alpha}C^\beta_{\alpha\beta} = -\frac{1}{2}\nabla^\mu\tau.$$

Their difference is $v^\mu = \nabla^\mu\tau - \nabla_\nu k^{\mu\nu}$, proving the claimed boundary-vector formula and its divergence identity.

Exercise 28.2 (Vacuum waves, gauge and the two polarisations). Set $g_{\mu\nu} = \eta_{\mu\nu} + sh_{\mu\nu} + O(s^2)$. Raise perturbation indices with η, write $h = \eta^{\mu\nu}h_{\mu\nu}$ and define $\bar h_{\mu\nu} = h_{\mu\nu} - \eta_{\mu\nu}h/2$.

(a) Derive $\Gamma^{(1)}$, $\mathrm{Ric}^{(1)}$ and $G^{(1)}$. Show explicitly that $G^{(1)}_{00}$ and $G^{(1)}_{0i}$ contain no second time derivatives of the perturbation. These are the linearized vacuum constraint equations on a slice.

(b) Under the covariant-metric Lie variation $h_{\mu\nu} \mapsto h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, derive the trace-reversed change and its divergence. In harmonic gauge $\partial^\mu\bar h_{\mu\nu} = 0$, derive the vacuum wave equation. No analytic theorem establishing that gauge globally is assumed or requested.

(c) For a mode $h_{\mu\nu} = \epsilon_{\mu\nu}e^{-\mathrm{i}\nu t + \mathrm{i}\nu x^3}$ with $\nu > 0$, impose harmonic gauge. Use residual null-wave gauge parameters to set the four temporal amplitudes to zero and derive the complete transverse traceless matrix. Count the independent polarisations. The real field is obtained by taking real parts; the plane mode itself is not claimed to have finite total energy on $\mathbb{R}^3$.

Solution. The first-order connection is

$$(\Gamma^{(1)})^\rho_{\mu\nu} = \frac{1}{2}\eta^{\rho\lambda}(\partial_\mu h_{\lambda\nu} + \partial_\nu h_{\lambda\mu} - \partial_\lambda h_{\mu\nu}).$$

The background connection is zero, so differentiating this formula gives

$$\mathrm{Ric}^{(1)}_{\mu\nu} = \frac{1}{2}(\partial_\rho\partial_\mu h^\rho{}_\nu + \partial_\rho\partial_\nu h^\rho{}_\mu - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h).$$

Its trace is $R^{(1)} = \partial_\rho\partial_\sigma h^{\rho\sigma} - \Box h$, and $G^{(1)} = \mathrm{Ric}^{(1)} - \eta R^{(1)}/2$. Expanding the time-index components, with spatial repeated indices summed, gives

$$\begin{aligned} G^{(1)}_{00} &= \frac{1}{2}(\Delta h_{jj} - \partial_i\partial_j h_{ij}),\\ G^{(1)}_{0i} &= \frac{1}{2}\left[\Delta h_{0i} - \partial_i\partial_j h_{0j} + \partial_t(\partial_i h_{jj} - \partial_j h_{ji})\right]. \end{aligned}$$

All second time derivatives have cancelled. The vacuum equations setting these expressions to zero constrain slice fields and their first time derivatives, rather than determining second time derivatives.

Trace reversal rewrites the full linearized tensor as

$$G^{(1)}_{\mu\nu} = \frac{1}{2}\left[\partial_\mu\partial_\rho\bar h^\rho{}_\nu + \partial_\nu\partial_\rho\bar h^\rho{}_\mu - \Box\bar h_{\mu\nu} - \eta_{\mu\nu}\partial_\rho\partial_\sigma\bar h^{\rho\sigma}\right].$$

The Lie variation has trace $2\partial_\rho\xi^\rho$ and hence

$$\delta\bar{h}_{\mu\nu} = \partial_\mu\xi_\nu + \partial_\nu\xi_\mu - \eta_{\mu\nu}\partial_\rho\xi^\rho, \qquad \partial^\mu\delta\bar{h}_{\mu\nu} = \Box\xi_\nu.$$

Thus harmonic gauge reduces vacuum to $\Box\bar{h}_{\mu\nu} = 0$. Parameters solving $\Box\xi_\nu = 0$ preserve that gauge. This algebra does not assert global solvability of an arbitrary gauge-fixing problem. Pure gauge perturbations give zero linearized curvature because their connection variation is $\partial_\mu\partial_\nu\xi^\rho$, whose antisymmetrised derivative is zero.

For the specified mode, harmonic gauge is $\ell^\mu\bar{\epsilon}_{\mu\nu} = 0$, where $\ell^\mu = (\nu, 0, 0, \nu)$ and $\ell_\mu\ell^\mu = 0$. A parameter with the same null phase changes the covariant amplitude by

$$\delta\epsilon_{\mu\nu} = -\mathrm{i}(\ell_\mu\alpha_\nu + \ell_\nu\alpha_\mu).$$

It preserves harmonic gauge. Set

$$\alpha_0 = \frac{\epsilon_{00}}{2\mathrm{i}\nu}, \quad \alpha_1 = \frac{\epsilon_{01}}{\mathrm{i}\nu}, \quad \alpha_2 = \frac{\epsilon_{02}}{\mathrm{i}\nu}, \quad \alpha_3 = \alpha_0 + \frac{\epsilon_{03}}{\mathrm{i}\nu}.$$

Since $\nu > 0$, these are defined and make every $\epsilon'_{0\mu} = 0$. Let $s_{\mathrm{sp}} = \epsilon'_{11}+\epsilon'_{22}+\epsilon'_{33}$. The remaining harmonic equations are $\bar{\epsilon}'_{0\nu} + \bar{\epsilon}'_{3\nu} = 0$. Their $\nu = 0$ component gives $s_{\mathrm{sp}} = 0$. Their $\nu = 1, 2$ components give $\epsilon'_{31} = \epsilon'_{32} = 0$. The $\nu = 3$ component then gives $\epsilon'_{33} = 0$. Thus $\epsilon'_{22} = -\epsilon'_{11}$ and the full amplitude is

$$\epsilon'_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The two free positive-frequency amplitudes encode two polarisations. For a real solution their negative-frequency partners are conjugate, rather than additional independent polarisations.

Exercise 28.3 (The non-null boundary action). Let Σ be a smooth non-null boundary patch of a region D. All variations have compact support in a collar of the interior of this patch and vanish near its edges and every other boundary piece. A Gaussian collar for the background may be used as a geometric given:

$$g = \epsilon_n\,\mathrm{d}n^2 + \gamma_{ab}(n, y)\mathrm{d}y^a\mathrm{d}y^b, \qquad \Sigma = \{n = 0\}, \quad D \text{ locally has } n \le 0, \quad \epsilon_n = \pm 1.$$

The outward unit normal is $N = \partial_n$. Write $\mathrm{vol}_\Sigma = \iota_N\mathrm{vol}_g$ with outward-first orientation, $K = \nabla_\mu N^\mu$, and fix only the induced metric: $h_{ab}|_\Sigma = 0$ for $h_{\mu\nu} = \delta g_{\mu\nu}$. Normal components of h are initially unrestricted.

Gaussian normal coordinates for a smooth non-null hypersurface are the local construction described in Wald, *General Relativity*, §3.3, pp. 42–43. The boundary-action discussion is Appendix E.1, pp. 458–459. Use the explicit conventions in this question, including ϵ_n.

(a) Construct a vector field ξ vanishing on Σ such that $\widetilde{h} = h + \mathcal{L}_\xi g$ has $\widetilde{h}_{nn} = \widetilde{h}_{na} = 0$ near the patch. Derive and solve the required first-order normal equations. Explain why the induced variation is still zero.

(b) In this representative compute the normal flux of the Palatini vector and the variation of $2\epsilon_n\int_\Sigma K\mathrm{vol}_\Sigma$. Prove their cancellation.

(c) Use diffeomorphism invariance and Lecture 28's Bianchi identity to extend that result to the original arbitrary normal components. State the resulting gravitational action with the frozen overall sign. Specify why this is not a null-boundary or corner/joint formula.

Solution. For the background collar the normal Lie components are

$$(\mathcal{L}_\xi g)_{nn} = 2\epsilon_n\partial_n\xi^n, \qquad (\mathcal{L}_\xi g)_{na} = \gamma_{ab}\partial_n\xi^b + \epsilon_n\partial_a\xi^n.$$

Define ξ by the normal initial-value equations

$$\partial_n \xi^n = -\frac{\epsilon_n}{2} h_{nn}, \qquad \xi^n(0,y) = 0,$$
$$\partial_n \xi^b = -\gamma^{ba}(h_{na} + \epsilon_n \partial_a \xi^n), \quad \xi^b(0,y) = 0.$$

First integrate the first line from zero to n. Then substitute that smooth solution and integrate the second line. The resulting ξ is smooth and has the desired zero boundary value. A cutoff away from the boundary extends it into D without changing these equations near the patch. Tangential compact support is preserved. On Σ, both ξ and its tangential derivatives vanish, so $(\mathcal{L}_\xi g)_{ab} = 0$ there. Thus $\widetilde{h}_{ab}|_\Sigma = 0$ as well.

Now use the Gaussian-preserving representative and write it simply as h. The inverse variation is $k^{\mu\nu} = -h^{\mu\nu}$, with the indices of h raised by the background. The first problem gives the equivalent covariant-variation vector

$$v^\mu = \nabla_\nu h^{\mu\nu} - \nabla^\mu (g^{\alpha\beta} h_{\alpha\beta}).$$

The normal components $h_{n\mu}$ vanish throughout a collar, and all components vanish on the boundary because the induced variation is fixed. Hence connection-times-h terms vanish at the boundary, and

$$v^n = -\epsilon_n \gamma^{ab} \partial_n h_{ab}.$$

The boundary density is $\epsilon_n (N_\mu v^\mu) \mathrm{vol}_\Sigma = v^n \mathrm{vol}_\Sigma$, because $N_\mu = \epsilon_n (\mathrm{d}n)_\mu$. This flux formula follows here directly from $\nabla_\mu v^\mu \mathrm{vol}_g = \partial_\mu (\sqrt{|\det g|} v^\mu) \mathrm{d}^4x$: normal-coordinate FTC leaves $v^n \mathrm{vol}_\Sigma$ at $n = 0$, while tangential integrals vanish by compact support.

In the same collar, $K = \frac{1}{2} \gamma^{ab} \partial_n \gamma_{ab}$. At the boundary $\delta\gamma^{ab} = 0$ and $\delta \mathrm{vol}_\Sigma = 0$. The normal vector remains ∂_n for this representative. Therefore

$$\delta K = \frac{1}{2} \gamma^{ab} \partial_n h_{ab}, \qquad \delta \left(2\epsilon_n \int_\Sigma K \mathrm{vol}_\Sigma \right) = \epsilon_n \int_\Sigma \gamma^{ab} \partial_n h_{ab} \mathrm{vol}_\Sigma.$$

This cancels the full normal-derivative contribution from the curvature integral, for either value of ϵ_n.

It remains to justify using this representative. The local flow of ξ fixes the boundary pointwise. The curvature integral and the oriented normal/extrinsic-curvature boundary integral are geometric integrals over the same region and patch; change of variables leaves each invariant under this boundary-fixing flow. Thus adding $\mathcal{L}_\xi g$ changes neither first variation of their sum. The bulk Einstein pairing is also unchanged. Its Lie contribution is

$$-2 \int_D G_{\mu\nu} \nabla^\mu \xi^\nu \, \mathrm{vol}_g = 0 :$$

integration by parts uses $\nabla^\mu G_{\mu\nu} = 0$ and a boundary current containing $\xi = 0$. Consequently the cancellation proves, for the original fixed-induced-metric variation too,

$$\delta \left[\int_D R_{\mathrm{sc}} \mathrm{vol}_g + 2\epsilon_n \int_\Sigma K \mathrm{vol}_\Sigma \right] = \int_D G_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g.$$

Multiplying by $-(16\pi G_{\mathrm{N}})^{-1}$ gives the stated gravitational action with boundary term $-(8\pi G_{\mathrm{N}})^{-1} \epsilon_n \int_\Sigma K \mathrm{vol}_\Sigma$. For compactly supported metric tests, adding Lecture 28's Hilbert matter variation gives the same interior equation $G_{\mu\nu} = 8\pi G_{\mathrm{N}} T_{\mu\nu}$.

The argument needs a smooth non-null patch and an invertible induced metric for its normal coordinates and inverse γ^{ab}. The support restriction removes edge and other-boundary contributions. It proves no formula for null boundaries or nonvanishing variations at corners/joints; their geometric data and boundary terms require separate analysis.